Broadcasting Education at Scale: Long-Term Labor Market Impacts of Television-Based Schools*

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Abstract

This paper examines the long-term impacts of using information and communication technologies to scale up last-mile educational services. We exploit geographic variation and cohort exposure from 1980 to 2000 to Mexico's TV-schools–lower secondary schools that substitute on-site specialized teachers with televised lectures, serving over 1.4 million children annually. Cohorts in high TV-school construction areas are 8 percentage points more likely to graduate lower secondary, with increases of 0.4 years of education and 8% in hourly earnings. Labor market returns are comparable to those from standard schools. Impacts are primarily driven by out-of-school children, with higher earning gains in urbanized areas.

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I Introduction

Over the past several decades, unprecedented school expansion efforts have taken place in most low and middle-income countries (LMICs) and broadening access to basic education has emerged as a fundamental policy tool to reduce poverty, facilitate workers' movement towards higher-paying jobs, and accelerate structural transformation (World Bank, 2018; Duflo, 2001; Porzio et al., 2022). However, countries with weak fiscal and state capacity have persistently struggled to ensure universal access to education services, leading to the disproportionate exclusion of vulnerable populations, particularly in hard-to-reach and rural areas.

For a variety of public services, Information and Communication Technologies (ICT) have opened new opportunities for the expansion of remote delivery models, allowing increased service access in rural or remote communities (Aker, 2017; World Bank, 2016). ICT-based schooling models might be particularly promising for these contexts, addressing challenges such as staffing shortages, high unit costs from reaching dispersed populations, teacher absenteeism, and low instructional quality (Evans and Acosta, 2023; Chaudhury et al., 2006). These approaches might be especially beneficial at the post-primary level, where the advanced curriculum requires higher-skilled subject-specialist teachers (Glewwe and Kremer, 2006; Banerjee et al., 2013; Echazarra and Radinger, 2019) and where 1 out of 7 children of relevant school age remain out of school (UNESCO, 2023). However, despite the potential for ICTs to universalize education access, there is limited evidence on whether these approaches can replicate the labor market benefits typically associated with traditional schooling models, particularly when operated at scale by governments.

This paper investigates the long-term labor market effects of building school infrastructure reliant on a basic technology –television– to expand access to education services. We study this question in the context of a twenty-year nationwide expansion of television-based schools, *telesecundarias* or TV-schools, in Mexico. Mexico's TV-schools are lower secondary schools (grades 7 to 9) where lessons are delivered through television broadcasts in a classroom setting, with a single support teacher per grade, instead of one teacher per subject like in standard secondary schools. The model, therefore, substantially reduces personnel costs associated with delivering education while also standardizing instruction delivery. Today, approximately 20% of all lower secondary-aged Mexican children, around 1.4 million, attend a TV-school.

Recent policy and academic discussions have primarily centered on the potential of 'high-tech' educational technologies, particularly those reliant on online or computer-based programs (Goodman et al., 2019; Muralidharan et al., 2019; Banerjee et al., 2007; Cristia et al., 2017). However, current technological and implementation constraints in LMICs make it unlikely that these approaches will sufficiently scale beyond regions with strong digital infrastructure. Alternatively, 'low-tech' solutions, such as television, offer a pragmatic alternative for delivering widespread remote instruction to marginalized areas in LMICs.[1] However, the impersonal and uniform nature of television as an instructional medium raises concerns about its effectiveness in adequately training students and equipping them with the necessary skills for the labor market. Given the evidence on the importance of 'teaching at the right level' of each student (Banerjee et al., 2007), it is unclear whether these one-size-fits-all approaches offer any value, especially since televised lectures are not tailored to each classroom or school, and even less so to individual students.

To understand how access to TV-schools affected children's educational and labor market outcomes as adults, we rely on the variation in TV-school availability at the municipal level generated by their gradual expansion between 1980 and 2000. Using census data from 2010 and employing a difference-in-differences approach, we compare the long-term outcomes of individuals with access to different TV-school densities during their school-age years, net of cohort and municipality averages and municipality-specific linear trends.

We first estimate the average effects of the nationwide program roll-out. Cohorts of adults who resided in areas with high TV-school density experienced an eight percentage point (p.p.) increase in the likelihood of graduating from lower secondary education. Increases in graduation primarily come from the enrollment of out-of-school children, rather than students switching from standard schools to TV-schools. Construction of TV-schools also increased the likelihood of graduating from primary school; however, we do not find evidence of increases in upper-secondary graduation. Overall, cohorts exposed to high TV-school density attained an additional 0.4 years of education on average.

In the labor market, affected cohorts experienced an average increase of 364 pesos ($30 USD)

[1]Case in point, the global school closures prompted by the COVID-19 pandemic dramatically increased the need for ICTs to deliver educational content remotely. While high-income countries turned to online education platforms, over 60 LMICs had to provide at-home remote educational options via television due to the widespread unavailability of more advanced technologies and internet access (World Bank, 2020; Barron Rodriguez et al., 2021).

in monthly earnings, corresponding to an 8% increase in hourly earnings. These income gains are accompanied by a 3 p.p. increase in the probability of being employed, particularly in the tertiary or services sector. These findings support the idea that, on average, expanding post-primary education services through a low-tech approach, such as TV-schools, resulted in sizable education and labor market gains.

Our conclusions remain robust to employing alternative estimators that address potential biases in staggered difference-in-differences estimates due to heterogeneity in treatment effects (Sun and Abraham, 2021). The estimated effects of TV-schools are similar when we use a matched sample of municipalities and alternative sources of treatment variation and control for the concurrent growth of other types of schools. We also find generally consistent, albeit noisier, results using an alternative source of identification by leveraging the timing of the 1993 law mandating lower secondary school access. As our primary analysis relies on census data that only provide information on the municipality of residence in adulthood, not birth, a potential threat to our identification strategy is selective migration. However, additional analyses indicate that across-municipal or out-migration are unlikely to explain our results.

To benchmark the impacts of TV-schools, we compute back-of-the-envelope estimates of the returns to education. We show that an additional year of education through TV-schools is associated with an increase in adult monthly income of 607 to 1,141 pesos ($\sim$ \$48 to \$90 US), or 10 to 19% in hourly income. These figures align with estimated returns to education in similar settings. We also compare the returns from TV-schools to those from standard schools by using a similar identification strategy using the expansion of standard schools in the country. While we estimate greater reduced form effects on both educational attainment and adult income from the roll-out of standard schools, the labor market returns per additional year of education are relatively similar for both types of schools.

The high fixed costs of establishing a TV-school model and the low marginal costs of operating it make it inherently designed to operate at scale. Originally, TV-schools were conceived as a low-cost alternative for providing educational access for rural areas with no other schooling options. However, the model was increasingly adopted in more connected and urban areas due to its lower cost and teacher requirements. This widespread expansion raises two important questions: first, whether TV-schools truly benefit their target rural areas, and second, whether their presence in more urban areas might be detrimental. For instance, in urban municipalities, individuals might have substituted away from higher quality schooling options (Kline and Walters, 2016; Mountjoy, 2022), or the human capital acquired through TV schools may be inadequate for the needs of a more service-oriented and developed labor market.

We find that TV-school openings led to educational and labor market increases in both agricultural and more urban, sectorally diversified municipalities. This underscores the labor market value of lower secondary education, even in rural settings. However, while there is no evidence of differences in graduation rates or educational achievement across regions, cohorts in areas with more diversified labor markets experienced larger income increases compared to those in more rural areas. This implies that the ability of a local economy to better absorb and utilize new skills is likely a critical component of the returns to education. Furthermore, this result mitigates concerns that the construction of TV-schools in more urban areas, where other school alternatives existed, was detrimental to affected cohorts. In sum, TV-schools appear to be an effective model for offering schooling options to marginalized students across various contexts.

This paper complements the body of work that has found that pre-school and primary school construction programs can increase school enrollment and attainment (Berlinski and Galiani, 2007; Burde and Linden, 2013; Aaronson and Mazumder, 2011), as well as improve long-term labor market outcomes (Duflo, 2001; Akresh et al., 2023). More recent studies have started documenting the medium- and long-run effects of secondary education in developing countries. This literature, however, is sparse, with existing causal studies mostly exploiting variation from admission cutoffs or scholarship assignments (Spohr, 2003; Ozier, 2016; Duflo et al., 2021). While this work has found positive effects on educational attainment and some labor market improvements, there is limited evidence to suggest that individuals experience increases in income and earnings. Relative to this literature, this paper focuses on a distinct, low-cost delivery modality at the lower secondary level designed to reach the last mile. We show that a blended learning model combining TV-based instruction with in-person school attendance can substantially benefit out-of-school children. Moreover, with large sample sizes and variation over 20 years, we identify sizable effects on income.

Additionally, the results of this paper relate to the growing literature studying the impacts of technology in education (see Bulman and Fairlie (2016) and Escueta et al. (2017) for summaries).

Most of the research on remote lessons evaluates them as complements to formal schooling and face-to-face instruction in developed countries, or in the context of Massive Open Online Courses and online college classes (Banerjee and Duflo, 2014; Alpert et al., 2016; Bettinger et al., 2017; Goodman et al., 2019). While evidence from high-income countries suggests that online programs may be a relatively poor substitute for in-person education (Figlio et al., 2013; Heissel, 2016; Bettinger et al., 2017; Bueno, 2020), recent work from developing countries, where the counterfactual might be no schooling or low-quality schooling options, shows that remote lessons can improve student achievement. For example, interactive videoconferencing in Ghana improved literacy and numeracy (Johnston and Ksoll, 2017), a large-scale program in China that connected urban teachers with rural students through satellite internet improved academic achievement and earnings (Bianchi et al., 2022), and expert-led video integration increased test scores in Pakistan (Beg et al., 2022). However, unlike these interventions that support interactive or adaptive learning, this paper examines a simpler, one-way 'low-tech' approach, much less customizable but potentially easier to scale in LMICs.

Finally, we complement the literature that has studied the short-term educational impacts of low-tech technologies to increase access to education, like radio (Jamison et al., 1981) or cell phones (Aker et al., 2012). Similar to our focus on TV-schools in Mexico, Borghesan and Vasey (2024) examine the impact of TV-schools on 7th-grade standardized scores compared to attending traditional schools. Using a marginal treatment effects framework with distance to school as an instrumental variable, the authors find that attending a TV-school benefits students in terms of learning scores one year later. This suggests that, contrary to common perceptions in Mexico (Santos, 2001), TV-schools are not necessarily a lower-quality education option relative to the counterfactual, at least for some contexts or students. Our study complements this work by not only documenting longer-term labor market gains from increased access to education through these 'low-tech' approaches but also by analyzing nationwide effects, encompassing children in areas both with and without access to standard school alternatives and with access to different labor markets.

The rest of the paper is organized as follows. Section II describes the institutional background of lower secondary education in Mexico and provides details on Mexico's TV-schools and their nationwide rollout. Section III describes the data sources and Section IV presents the empirical strategy. Section V discusses results and investigates the sensitivity of the results to alternative econometric and sample specifications. Section VII presents impacts by types of municipalities affected. Section VIII concludes.

II Context

MEXICO'S TV-SCHOOLS. Basic education in Mexico comprises primary education (grades 1 through 6, ages 6 to 12), lower secondary education (grades 7 to 9, ages 12 to 15), and upper secondary education (grades 10 to 12, ages 16 to 18). At the lower secondary level, there are three main school modalities: general schools *(secundaria general)*, technical schools *(secundaria tecnica)*, and TV-schools or *telesecundarias*. All lower secondary modalities cover the same core curriculum and issue an equally recognized certificate of completion necessary for enrollment in upper secondary.

In 2016, there were 6.7 million lower secondary students in Mexico: 51% and 27% attended general and technical schools, respectively, and 1.4 million attended TV-schools, representing 21% of the total lower secondary enrollment. Moreover, out of the 39,265 lower secondary schools in the country, 48% were TV-schools (INEE, 2017).

General and technical lower secondary schools operate like any other ordinary school, with specialized teachers for each subject in the lower secondary curriculum.[2] We denote both these school modalities as 'standard' schools throughout this article. TV-schools are small, usually with only one class per grade and between 15 and 30 students per class. The key feature of the TV-school model is that lessons are delivered in a classroom setting through centrally-produced television broadcasts. In the classroom, there is typically a single teacher per grade (or even per school), the monitor teacher (*maestro monitor*), who provides support for all subjects.[3] In contrast, standard schools have, on average, 11 or 12 subject-specialized teachers.

Lessons are simultaneously broadcasted to all TV-schools in the country following a pre-established schedule.[4] Each school hour is dedicated to a different subject, and there is a com-

[2]The main difference between general schools and technical schools is that in addition to the academic common core, technical schools offer technical subjects or workshops. Lower secondary diploma students can graduate with a certificate on the workshop they took.

[3]In 2008, 20% of TV-schools had only one or two teachers managing the three grades (SEP, 2014).

[4]When the program was first introduced, the transmission was through microwaves and TV antennas and, later, satellite technology.

bination of remote instruction and in-class work: typically, students watch a 15-minute televised lesson, followed by 35 minutes of class discussion and classwork, guided by the monitor teacher and with the help of learning guides (INEE, 2005). Lessons for different grades are staggered throughout the hour so that one grade will begin their televised lesson when the preceding grade concludes theirs. Children receive 2 to 6 hours of televised instruction per day (INEE, 2016).

The televised lessons are crafted by pedagogical experts and recorded in television studios by teachers chosen for their effective communication skills, known as TV-teachers (*telemaestros*). The monitor teachers oversee classroom activities, answer student questions, and grade homework and exams. They rely on teaching guides for all the subjects covered in the televised lessons.

As is often the case with public services operated at scale, there is considerable variability in the adequacy of infrastructure and quality of services. For instance, in 2001 it was estimated that about 60% of monitor teachers were fully trained in the model, while 40% were university graduates with no other teacher training (Creed and Perraton, 2001). Moreover, some TV-schools might lack essential resources as electricity, television signal reception or textbooks.[5] Therefore, our estimates reflect intent-to-treat effects, which account for implementation failures common in LMICs.

The fixed costs of setting up a TV-based school system–which includes building physical facilities, transmission infrastructure, creation of television programs, and designing and printing workbooks– are substantial. For instance, in 1997, the annual investment and recurrent costs of operating the system were US $ 425,750,474 (Calderoni, 1998). However, the marginal costs of operating the schools are much lower than that of operating standard schools. The average administrative cost per TV-school student is half that of standard schools. In 2002, TV-schools cost 6,811 pesos per student, compared to 12,460 pesos for general lower secondary schools and 14,572 for lower secondary technical schools (Martinez Rizo, 2005).

TV-SCHOOL ROLLOUT AND PLACEMENT. Mexico introduced TV-schools in 1968 as a response to the shortage of qualified secondary school teachers willing to work in remote rural areas and the dispersed distribution of primary education graduates seeking to continue their studies

[5] In 2001, a survey revealed that 10% of TV-schools didn't have electricity, 25% had low reception signal, and 22% didn't have the introductory textbooks (Martinez Rizo, 2005). Supervisor teachers and students had to adapt the lessons and classes to these precarious circumstances.

(Calderoni, 1998).[6]

This paper examines the rollout of TV-schools from 1980 to 2000, a period during which approximately 13,000 TV-schools opened across the country. We exclude the operations during the 1970s from our analysis due to the smaller scale and distinct operational model of that era, coupled with the lack of reliable data before the 1980s, when Mexico began to systematically track school openings. Moreover, since our interest is in the long-term effects of these schools, we only consider the impact of their construction up to 2000.

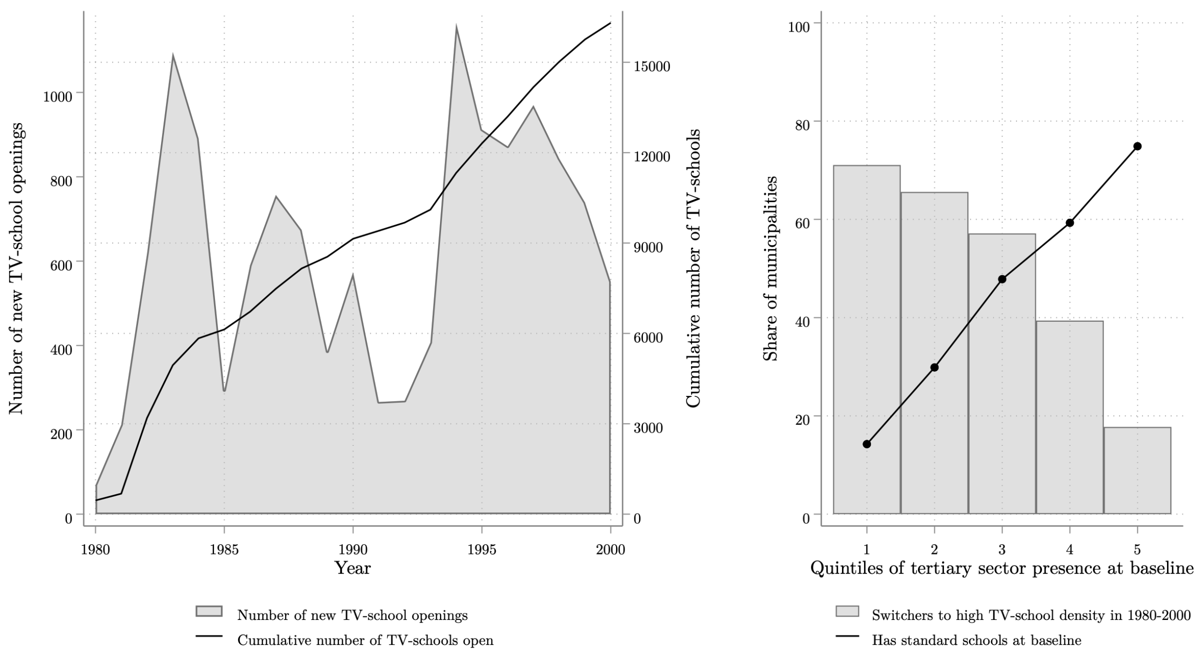
The left-side graph in Figure 1 shows the number of new TV-school openings by year and the cumulative number of school openings during this period (Appendix Figure A1 shows geographical changes in municipal TV-school density over time). Before 1980, a few hundred TV-schools were in operation, but by 2000, they had reached around 15,000. Moreover, while these schools were gradually constructed during this period, there were two major waves of openings. First, a spike in 1981, after several agreements with state-level governments were signed to start the remote schooling service increased their number from 694 to 3,279 (Martinez Rizo, 2005). Second, in 1993, when lower secondary education became compulsory, and states were constitutionally required to offer access to this education level, TV schools —cheaper and requiring fewer teachers than standard schools—became an attractive option for places in need of expanding lower secondary services (Jiménez et al., 2010).

Conceptually, TV-schools were designed as a solution to offering education services in rural and remote areas where standard schools were not available and where operating them would be too costly.[7] In fact, we corroborate that high-poverty, predominantly rural municipalities with lower tertiary sector presence in 1980 had a higher likelihood of experiencing high TV-school construction density over subsequent periods (Appendix Table A2, columns 1 to 4). The right-side

[6]In 1965, the number of primary school graduates unable to enter secondary school in Mexico was about 37% of the number of previous year's 6th graders (Mayo et al., 1975). Moreover, in 1970, 41% of the Mexican population lived in remote and small communities of no more than 2,500 inhabitants. While this fraction was only approximately 21% by 2017, significant challenges for the effective provision of public services for these communities remain (OECD., 2019).

[7]In the early days of TV-schools, government agencies nationally planned school allocations based on vague criteria such as "geographical and urban conditions, economic, cultural, social and hygienic factors" (SEP, 1967). Later on, official guidelines advised opening a TV-school in areas with a minimum distance of 4.5 km from other standard schools and where at least 30 students could benefit from the service. While we corroborate that TV-schools were more likely to open in rural areas with a low density of other lower secondary schools, we find that the distance and student minimum rules were implemented too loosely to exploit them as part of our empirical design.

Figure 1: TV-school Rollout and Placement



Notes: This figure shows the number of TV-school openings over time and the share of municipalities with high TV-school density construction by tertiary sector prevalence. The left graph shows the annual (area) and cumulative (solid line) number of new TV-school from 1980 to 2000. The right graph shows the share of municipalities with high TV-school density, categorized by quintiles of tertiary sector prevalence in 1980. The solid black line indicates the proportion of municipalities with a standard school in 1980 in each quintile. See Section III for details on variable construction. The data is sourced from the Mexican Secretariat of Education (SEP), and the 1980 Mexican census (10% IPUMS subsample).

graph in Figure 1 shows the share of municipalities with high-intensity TV-school construction by quintile of the share of adult workers in the tertiary sector at baseline in 1980, a variable highly correlated with various dimensions of poverty and rurality (Appendix Figure A2).[8] Overlaid in a black line is the proportion of municipalities with at least one standard secondary school at baseline (section III describes the variable construction).

Over 70% of the most rural and less economically diversified municipalities experienced high-density TV-school construction, and of these, fewer than 20%had a standard lower secondary school at baseline. Consequently, TV-school students tend to come from families with a lower socioeconomic background compared to those attending standard schools.[9] Nonetheless, TV-schools were also constructed in more economically diverse areas. Nearly 20% and 40% of the municipal-

[8]Using measures of rurality or poverty to construct Figure 1 leads to very similar conclusions. We consistently show results using tertiary sector presence rather than these other variables to align with the results from Section VII.

[9]For example, in 2016-2017, only 37% of TV-schools students had mothers with secondary education or higher, and almost 60% benefited from the Prospera/Oportunidades conditional cash transfer (CCT) program. In contrast, these proportions were 63% and 23%, respectively, for students in standard lower secondary schools (INEE, 2017).

ities in the 5th and 4th quintiles of tertiary sector presence, many of which already had a standard lower secondary option available, also experienced high-intensity TV-school construction. The construction program targeted less rural municipalities over the years, though they reached out to increasingly poorer municipalities (Appendix Table A2, columns (5) to (8)). The variation in contexts where TV-schools opened raises questions on the extent of impact heterogeneity, particularly as the availability of alternative educational services and the characteristics of local labor markets significantly varied. We explore this heterogeneity in Section VII.

III Data and Sample

III.1 Data

CENSUS DATA. Outcome data for our primary specifications comes from the 10 percent subsample of the 2010 Mexican census accessed through the Integrated Public Use Microdata Series (IPUMS) project (IPUMS, 2013) and collected by the National Statistics Office (INEGI). The census data has information on different dimensions of an individual's educational, labor market, and other socio-demographic outcomes in adulthood. In particular, we report the effects on years of education and the probability of completing primary, lower, and upper secondary school. To capture impacts related to labor market experiences, we estimate effects on the likelihood of working (relative to being inactive or unemployed), an individuals' reported monthly labor income (winsorzised at the 0.1% level to address extreme top outliers), individuals' hourly wages, the (unconditional) probability of working in the primary, secondary or tertiary sectors, and the probability of reporting enrollment in social security, a proxy for formal employment.[10]

The census includes questions about an individual's municipality of residence and the state of birth but not about their municipality of birth.[11] In Section V.1 we discuss the tests we conduct to ensure that our estimates are not simply driven by selective migration. In addition, we use the 2000 census to estimate the short-term effects of TV-schools, which might have been less affected by migration as one can deduce respondents' locations in 1995.

[10] Municipal boundaries are harmonized by IPUMS and account for changes in census geography. For instance, if the boundaries between two census units change from one census sample year to another, units are combined to create a single unit. This adjustment explains the discrepancy between the number of municipalities reported in this paper and the total number of existing municipalities in Mexico.

[11] A question asks municipality of residence 5 years prior, in 2005. Our results are robust to assigning people to this municipality instead of the current one.

Finally, we use the 1980 census to obtain baseline information about municipal-level characteristics, including population size, school enrollment rates, the fraction of people living in rural localities, and poverty levels.[12]

Administrative School Data. The census data is merged with administrative data on the number of new TV-schools opened in each municipality during the 1980-2000 period obtained from the Mexican Secretariat of Education (SEP). The school database contains the address and opening date for all schools by type of service offered in the country. We collapse the number of schools-shifts by school type at the municipal level to merge with the census data.[13]

The SEP did not maintain a consolidated database of schools until the 1980s. The quality of data for early years is questionable, with some TV-schools reportedly opening before 1968 or many appearing for the first time when official reporting began, which might suggest that some schools listed as newly constructed were simply newly reported. To mitigate this issue, we only consider new school constructions post-1980. We further discuss this issue of school data quality and variable construction in Appendix B.

Additionally, we use the yearly student enrollment numbers at the school-shift-grade level from the SEP to investigate enrollment changes by school type after TV-schools were introduced. These enrollment counts are reported by each school to the SEP and only available post-1990.

III.2 Sample and Descriptive Statistics

Sample. We consider the TV-school opening period from 1980 to 2000, excluding the 191 municipalities that reportedly experienced high-intensity openings during 1968-1979. We also exclude Mexico City from the sample because its municipalities correspond to relatively close neighborhoods with high within-area migration. Our final sample includes 2,134 municipalities, of which 1,084 experienced high TV-school construction density.

The sample includes individuals born between 1945 and 1990 (aged 20 to 65 in 2010), and it is

[12]The poverty index was constructed using primary component analysis and includes the following municipal variables for 1980: the share of illiterate people and people with incomplete primary over the age of 15, the share of households without piped water, with dirt floors, without electricity, and the share of the working population with income of up to 2 minimum salaries.

[13]In Mexico, many school buildings operate a morning and an afternoon shift. We count these as two separate schools. However, almost all TV-schools operate in a single shift since they tend to serve communities with very few students.

restricted to non-migrants from their state of birth. As discussed in the next section, this restriction assumes that the municipality of residence in adulthood is a good proxy for the municipality of school exposure. Including migrants from their state of birth would introduce noise and confound program impacts with other improvements in local opportunities. However, this restriction implies that the estimates only represent those who remained in their state of birth into adulthood, constituting 92% of the included cohorts for the relevant sample. In Section V.1, we show the robustness of our findings under different assumptions regarding the treatment of these migrants.

Descriptive Statistics. Appendix Table A1 reports summary statistics for the analysis sample, showing separately averages for the entire sample, for municipalities by treatment status (switchers and non-switchers, see details of the treatment construction in Section IV), by sectoral composition at baseline, and by average density of standard schools presence. We display weighted means and standard deviations to ensure they are representative of the population. Individuals in treated areas tend to have lower average years of education, lower incomes, and substantially more likely to work in the agricultural sector.

IV Empirical strategy

Our empirical strategy combines two sources of identifying variation: whether an area eventually experienced high-intensity TV-school construction and the staggered timing of this event over the 1980 to 2000 period. We measure the intensity of exposure through a variable identifying the TV-school density at the cohort-municipality level:

$$TS_{mc} = \frac{\text{Number of TV-schools}_{mc}}{\text{Population ages 12 to 14}_{m}} \times 50$$

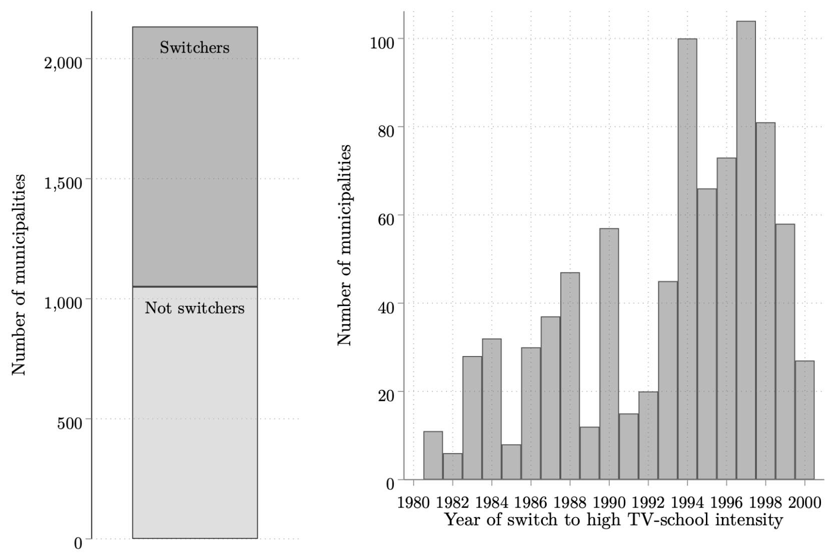
where TS_{mc} is the number of TV-schools available in municipality m per 50 eligible children when individuals from cohort c were 15 years old.[14] The normalization of the number of schools with the size of the targeted population mitigates the imprecision in the measurement of the intensity of TV-school exposure, and the scaling factor of 50 approximates the number of seats available in a newly created TV-school. Our primary treatment variable is a binarized measure $AboveTS_{mc} \equiv$

[14] Students transition to lower secondary school when they are around 12 years of age. However, grade repetition is relatively high, and transition ages to lower secondary school are imperfect, particularly for populations in marginal and rural areas.

$1[TS_{mc}$ above median$]$, identifying individuals with access to a density of TV-schools above the sample median.[15] We also report effects using the continuous measure of exposure TS_{mc} as a complementary analysis.[16]

Figure 2 shows the variation exploited.[17] About half of the municipalities in the sample never switch to high-intensity TV-school construction, and the frequency of switchers varies across different years. Unlike study designs that rely on a single-year event, this staggered implementation makes it less likely that other simultaneous programs or policies confound the estimated effects.

Figure 2: Source of Variation: Municipalities Switching to High TV-School Intensity



Notes: This figure illustrates the data structure. Our identifying variation comes from a 'hybrid' data structure (Miller, 2023). The left panel shows the number of municipalities in the analysis data that eventually switch to high TV-school intensity status (n=1,084) and those that do not (n=1,050). The right panel presents a histogram of the switching event dates, with one observation per municipality, restricting to the municipalities that experienced high-density construction. The data is sourced from the Mexican Secretariat of Education (SEP).

For computational ease, we collapse our data into birth-year by municipality-of-residence cells and implement a dynamic two-way fixed-effects difference-in-differences regression (TWFE), for-

[15]The median density of TV-schools in areas with openings is one TV-school per 357 eligible children (ages 12 to 14). Among individuals with access to a high density of TV-schools ($AboveTS_{mc} = 1$), the average density is one TV-school per 125 children. In comparison areas, there was one TV-school per 625 children.

[16]Differences-in-differences with continuous treatment require the so-called "strong" parallel trend assumption (Callaway et al., 2024), which intuitively stipulates that changes in outcomes for municipalities with small changes in TV-schools access provide a good counterfactual for the changes in outcomes that would have been observed for municipalities with larger changes in TV-school construction.

[17]Figure A3 presents the spatial and temporal evolution of the variation exploited, highlighting the municipalities that transition to high TV-school intensity every five years.

mally estimating the following equation for cohort c living in municipality m:

$$Y_{mc} = \alpha + \sum_{\tau \in [25,5], \tau \neq 16} \beta_\tau (\text{AboveTS})_m \times 1[\text{Age at switch}_m = \tau] + \gamma_m + \lambda_c + \gamma_m \cdot t_c + \varepsilon_{mc} \tag{1}$$

Where Y_{mc} is the outcome of interest, and γ_m and λ_c are municipality and cohort fixed effects. We also include a municipality-specific linear time trend ($\gamma_m \cdot t_c$) to control for underlying linear trends in the outcome variables that differ across municipalities. β_τ is the average treatment on the treated (ATT) difference-in-differences effect estimate of the TV-school exposure intensity at age 15, as a function of the individual's age τ during the high-intensity TV-school openings in their municipality. ε_{mc} is the error term. To account for the presence of heteroskedasticity and serial correlation, robust standard errors are clustered at the municipal level.

To reduce differential compositional concerns in the sample, we restrict our window of interest from -10 to +10 years centered around the year of switch to high TV-school intensity, i.e., limiting our analysis to individuals aged 5 to 25 at the time of this switch.[18] For control municipalities, we impose similar cohort restrictions by creating a "synthetic switching year", which is randomly selected from the distribution of actual switching years in treated municipalities and assigned to a control municipality.

Staggered Treatment. The TWFE estimator does not result in consistent estimates if treatment effects are heterogeneous across groups or time (Borusyak et al., 2024; de Chaisemartin, 2018; Sun and Abraham, 2021; Callaway et al., 2024). To address this concern, we show that our results are generally robust to re-estimating them using the heterogeneity-robust estimators proposed by Sun and Abraham (2021). This estimator is appropriate and interpretable in the context of staggered treatment timing and treatment effect heterogeneity. It involves estimating a fully parametric saturated regression of municipality- and relative time-specific treatment effects through interaction terms, allowing for treatment effect heterogeneity, and using the never-treated units as controls. Then, the estimates are aggregated as a weighted average using the sample shares of each cohort in each period. We consider the Sun and Abraham (2021) estimators as our primary heterogeneity-robust method and compare the full dynamic effects and the aggregated average treatment effects

[18] Indeed, Appendix Figure A4 shows the distribution of population weights in the sample, providing evidence of a steep drop ten years after the switch to high TV-school intensity.

to the estimates from Equation (1) and the DiD_{ATT} above.

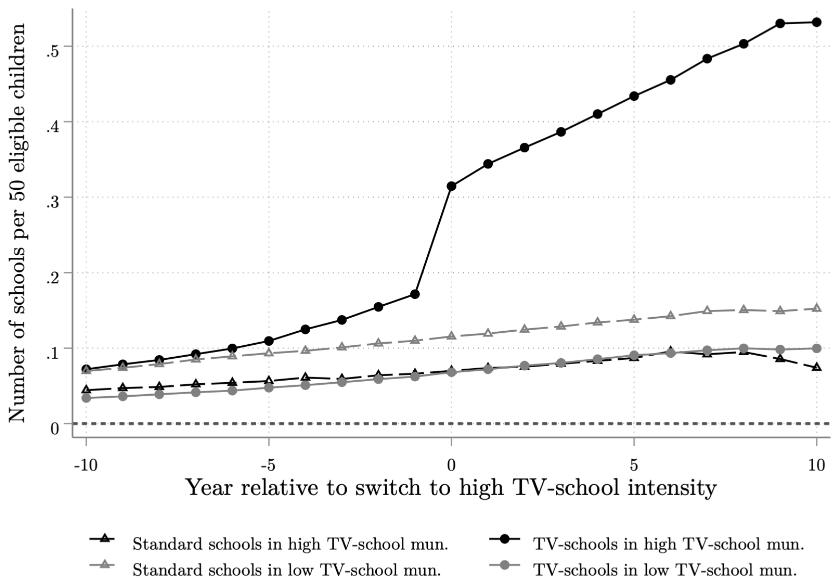
Single Estimates. We complement the dynamic specification with single coefficient estimates. The single pooled coefficient of a standard TWFE regression provides a variance-weighted average of the underlying dynamic coefficients, with weights based on the residual variance of the treatment variable after residualizing with respect to the controls, which, in our case, include municipality-specific trends. Given that our dynamic treatment effects increase over time, the TWFE pooled coefficient estimate mechanically will put more weight on the coefficients closer to the switching year than on longer-term coefficients. Therefore, this pooled coefficient is difficult to interpret and will be less policy-relevant. Instead, we calculate the average treatment on the treated (ATT) effect as the equally weighted mean of the dynamic treatment coefficients β_τ, considering only the fully treated periods $\tau \in [3, 10]$. This is our preferred specification and is reported as DiD_{ATT}. The standard TWFE single-coefficient estimates, also excluding the partially treated periods, are reported as supplementary results and denoted as DiD_{TWFE}.

Parallel Trends and concurrent standard lower secondary school expansion. The lead coefficients relative to the time of switch $\tau = 16$ in Equation (1) provide an empirical test of pre-treatment differential trends, which supports our identifying assumption.

Additionally, the phased rollout of TV-schools lessens the concerns that other programs or public investments drive the observed effects, as such programs would have needed to precisely coincide with the year and municipality of TV-school construction. A plausible confounder, however, could arise if the government simultaneously built standard lower secondary schools alongside TV-schools. If this were the case, our estimates would reflect the impact of broader school access rather than the specific effects attributable to TV-schools.

Figure 3 shows the variation in TV-schools and standard lower secondary school density over time, in high and low-intensity TV-school municipalities. The figure also highlights the variation we use to identify effects, showing a discrete increase in the number of TV-schools in municipalities classified as switching to high-density TV school construction at the time of switch, as well as a gradual increase in the years after the switch. We do not find evidence that the transition to high-intensity TV-schools is correlated with simultaneous shifts to a high density of standard schools.

Figure 3: Density of Schools Relative to the High TV-School Intensity Switching Year



Notes: This figure shows the TV-school density and the standard school density in municipalities that are ever classified as having high TV-school intensity from 1980 to 2000, and those that never get this classification (low intensity). The solid black circles show the treatment variation we exploit in the analysis. The data is sourced from the Mexican Secretariat of Education (SEP).

V Main Results

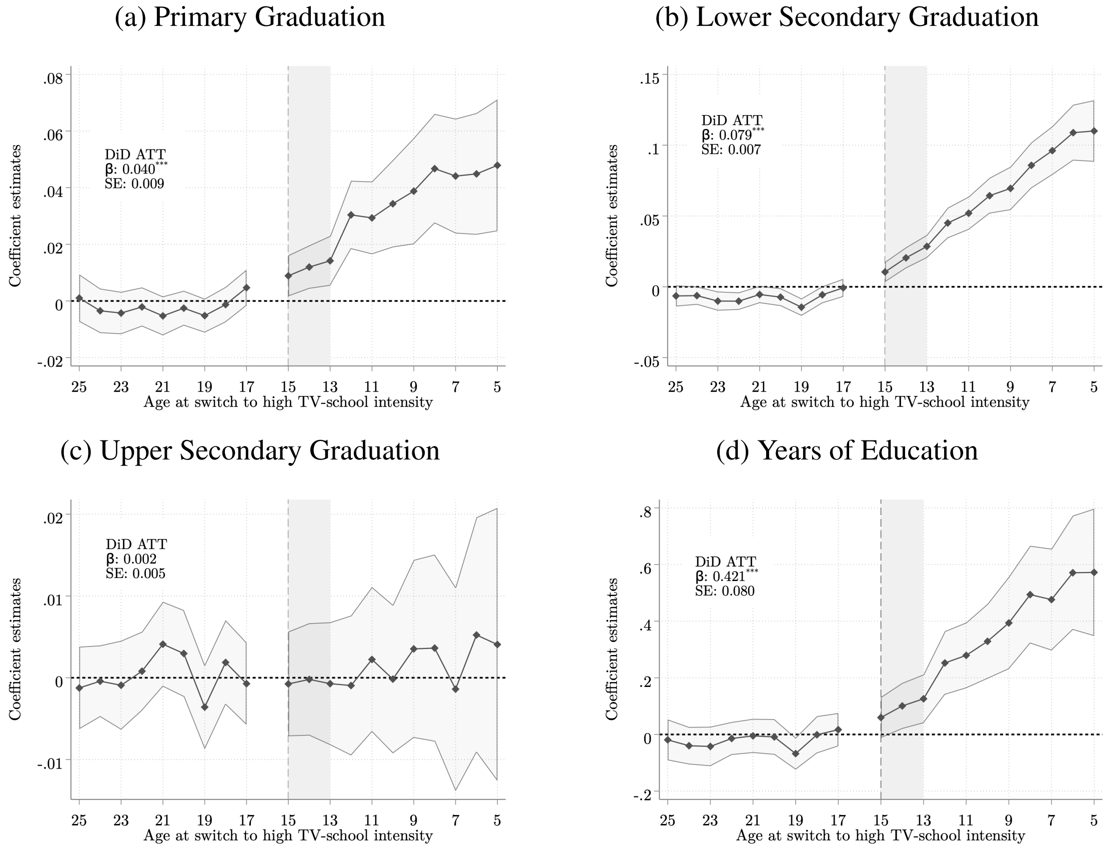
Education Outcomes. Figure 4 presents the main event studies on educational outcomes resulting from the nationwide construction of TV-schools. We document positive and statistically significant effects for primary (Figure 4a) and lower secondary graduation (Figure 4b) for cohorts aged 15 or younger when their municipality experienced intense TV-school construction, which result in overall increases in years of education (Figure 4d). Consistent with the assumption of parallel trends, we find estimates close to zero for older cohorts, who were likely too old to benefit from this construction. We also observe smaller effects for partially treated cohorts and evidence of increases over time.

Aggregating the event study estimates for lower secondary school graduation, we calculate an overall increase of 8 p.p., corresponding to a 12.6% increase over the baseline rate of 63%. The probability of graduating from primary schools shows an increase of 4 percentage points from a mean baseline graduation rate of 85%.[19] This result aligns with emerging research in LMICs that

[19]The DiD_{ATT}, DiD_{TWFE}, and Sun and Abraham (2021) coefficient estimates are reported in Table A3.

finds that access to higher levels of education can encourage completion at lower levels (Jagnani and Khanna, 2020; Sandholtz, 2022). However, we find little evidence of meaningful gains in upper secondary graduation (Figure 4c). This absence of effects for subsequent educational levels is perhaps unsurprising, given the limited availability of upper-lower secondaries in many of these areas at the time.

Figure 4: Event Studies of Impacts of Access to High TV-School Intensity on Education Outcomes



Notes: This figure shows the effect estimates of access to high TV-schools density by age at the year of switch to a high TV-school intensity construction on different outcomes. The econometric specification is a dynamic TWFE event study model that includes municipality and cohort fixed effects, and municipality-specific linear time trends. See Equation (1) for details. Event study coefficient estimates are shown with a solid line, and 95% confidence intervals are shown with a shaded area. All effects are computed with respect to age 16. The dynamic coefficient estimates are summarized with an equally weighted average treatment on the treated single point-estimate DiD_{ATT} superposed in the graph alongside its standard error. See Section IV for details. Outcomes are: indicators for graduating from primary, lower secondary, and upper secondary education, and years of education. Data is from the Mexican Secretariat of Education (SEP) for 1980–2000, and the 2010 Mexican census (10% IPUMS subsample).

Overall, cohorts in municipalities with intensive TV-school construction gained an additional 0.4 years of education from a baseline of 9.2 (Figure 4d), representing a 4.6% increase in the average total educational attainment. Re-estimating the effects using the continuous treatment measure indicates that constructing an additional TV-School per 50 children in the municipality led to an increase in educational attainment of 0.4 years and increased the primary and lower secondary school graduation shares by 5 and 13 percentage points, respectively (Appendix Table A5).[20] These results join an existing literature showing that improving schooling access, either by increasing school construction or by reducing travel time, can have an important effect on educational attainment (Duflo, 2001; Burde and Linden, 2013; Muralidharan and Prakash, 2017).

To understand why TV-schools significantly increased educational attainment, it is important to consider the counterfactual schooling options available to children. We use the yearly school enrollment data, available after 1990, to estimate the impacts on enrollment by school type following a high-intensity construction of TV-schools in the municipality.[21] Appendix Figure A5 shows a significant increase in TV-school enrollment at the onset of the TV-school construction, which grows steadily over time. The point estimates for average school enrollment in standard schools are negative and decrease over time, although they are noisy and remain statistically insignificant for all post-TV-school construction periods. Taking these point estimates at face value, they suggest that over time, about two-thirds of affected children would not have enrolled in school at all had the TV schools not been constructed, while roughly one-third might have transferred from standard schools to a TV-school. Therefore, the main gains in human capital are likely driven by enrolling previously out-of-school children, with less evidence of students switching between school types. We investigate this heterogeneity further in Section VII.

Labor Market Outcomes. Figure 5a shows consistent gains in the monthly labor earnings for affected cohorts. The pooled estimate corresponds to overall gains of 365 pesos (approximately $29 USD at the time), corresponding to a 12% increase over the mean labor earnings of 3,110 pesos. We further estimate an increase of 8% in the exposed individual's hourly income (Figure 5b).

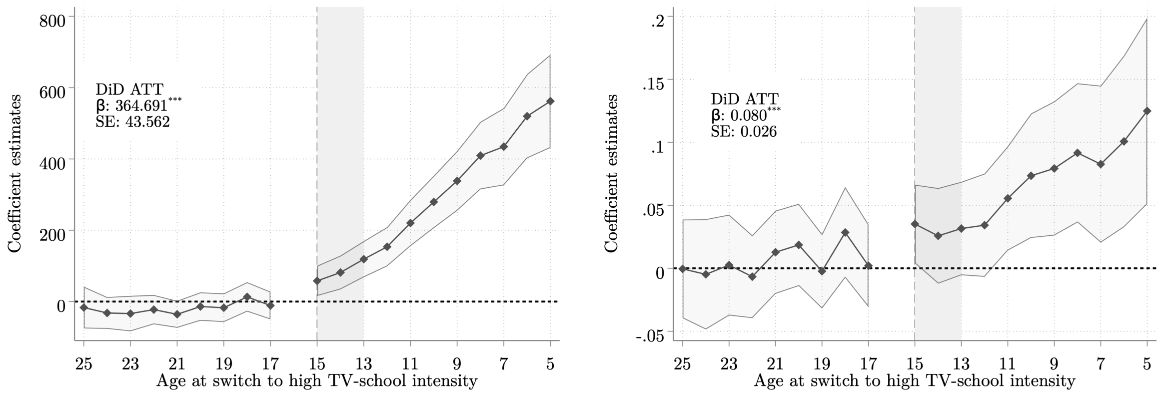
We then investigate the impacts on outcomes related to the changes in labor market composition, which could be one of the mechanisms contributing to the positive income effects. First, we find a 3 p.p. increase in the probability of working in adulthood (Figure 6a), a magnitude that corresponds to the increase in the probability of reporting a non-zero income (Figure 6b). These

[20] Appendix Figures A9 and A10 present the dynamic treatment effects estimated using the continuous treatment—the number of TV-schools opened in the municipality, scaled by the eligible population—centered around the age at the year of the first TV-school construction. Similar to the results with the binary treatment, these figures indicate no pre-treatment trends.

[21] In 1990, 29% of municipalities did not have any standard lower secondary schools.

Figure 5: Event Studies of Impacts of Access to High TV-School Intensity on Income

(a) Monthly Income (pesos) (b) Hourly Income (log)



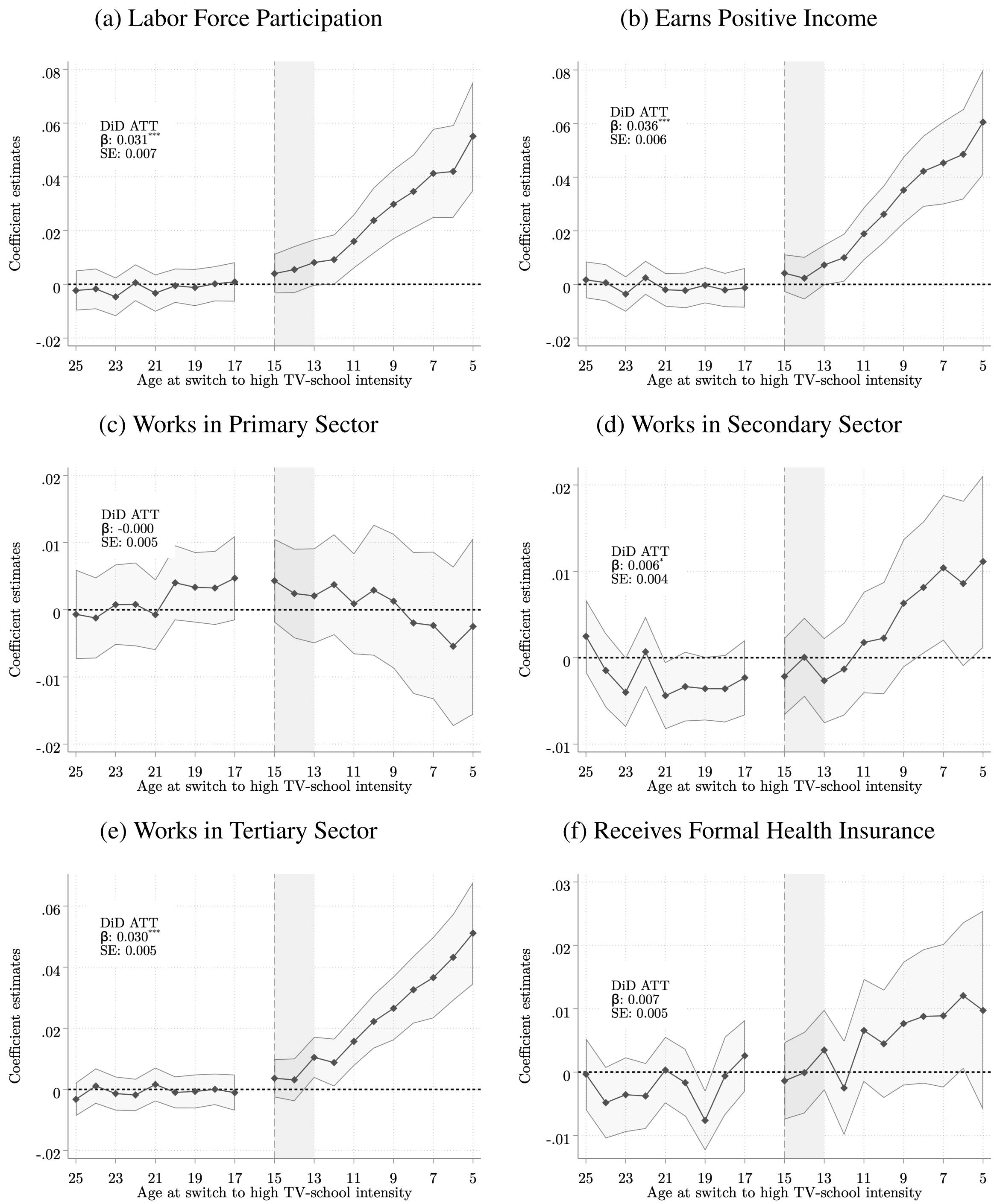
Notes: This figure shows the effect estimates of access to high TV-schools density by age at the year of switch to a high TV-school intensity construction on different outcomes. The econometric specification is a dynamic TWFE event study model that includes municipality and cohort fixed effects, and municipality-specific linear time trends. See Equation (1) for details. Event study coefficient estimates are shown with a solid line, and 95% confidence intervals are shown with a shaded area. All effects are computed with respect to age 16. The dynamic coefficient estimates are summarized with an equally weighted average treatment on the treated single point-estimate DiD_{ATT} superposed in the graph alongside its standard error. See Section IV for details. Outcomes are: monthly adult income in pesos, and the logarithmic transformation of hourly income. We do not condition on whether individuals are working. Data is from the Mexican Secretariat of Education (SEP) for 1980–2000, and the 2010 Mexican census (10% IPUMS subsample).

represent a 4.6% and 6.4% increase in the shares from baseline values of 66% and 55%, respectively.

Notably, there is a significant 3.3 p.p. rise in the probability of working in the service or tertiary sector (Figure 6e), corresponding to the increase in labor market participation and representing an 8% extensive-margin increase in the overall proportion of individuals within the exposed cohorts working in tertiary sector activities. The rise in tertiary sector participation predominantly stems from a 10% increase in wholesale and retail trade, a 20% in financial services and insurance, a 26% increase in public administration and defense, a 6.6% increase in business services and real state, and a 17.5% in education, and a 10% in health and social work (all statistically significant at conventional levels), whereas there are no significant increases in the probability of working in hotels and restaurants, transportation, storage and communications, or private household services (Appendix Table A4).[22] However, in contrast to previous findings (Porzio et al., 2022), we do not find a significant decrease in the likelihood of employment in the primary sector (Figure 6c), with

[22]Baseline values for industry shares are the following: Wholesale and retail trade, 11%; education, 4%; hotels and restaurants, 3%; transportation, storage, and communications, 3%; public administration and defense, 3%; business services and real state, 3%; private household services, 2%; health and social work, 2%; financial services and insurance, 1%.

Figure 6: Event Studies of Impacts of Access to High TV-School Intensity on Labor Market Composition Outcomes



Notes: This figure shows the effect estimates of access to high TV-schools density by age at the year of switch to a high TV-school intensity construction on different outcomes. The econometric specification is a dynamic TWFE event study model that includes municipality and cohort fixed effects, and municipality-specific linear time trends. See Equation (1) for details. Event study coefficient estimates are shown with a solid line, and 95% confidence intervals are shown with a shaded area. All effects are computed with respect to age 16. The dynamic coefficient estimates are summarized with an equally-weighted average treatment on the treated single point-estimate DiD_{ATT} superposed in the graph alongside its standard error. See Section IV for details. Outcomes are: indicators for participation in the labor market and unconditional probability of earning positive income, unconditionally participating in the primary sector, secondary sector, and tertiary sector, and for receiving formal health insurance. Data is from the Mexican Secretariat of Education (SEP) for 1980–2000, and the 2010 Mexican census (10% IPUMS subsample).

a precise zero estimated, and only a marginal and barely statistically significant 0.6 p.p. effect in the probability of employment in manufacturing (Figure 6d). Although point estimates suggest that affected cohorts are more likely to be enrolled in employment-related social security, a common proxy for formal sector participation, the effects are not statistically significant (Figure 6f).

Overall, our findings suggest that cohorts exposed to a high-intensity TV-school availability exhibit economically and statistically significant increases in adult labor market earnings. This can be attributed, in part, to an overall increase in the probability of employment in the services sector. However, given that the estimated percentage point sectoral compositional changes are small, it is unlikely that the income effects are entirely explained by changes in sectorial composition.

V.1 Validity and Robustness

This section discusses several approaches to address three potential threats to the identification strategy: differential trends, differential migration, and concurrent expansion of other programs. Finally, we discuss a battery of sensitivity analyses.

Alternative estimators and the parallel trend assumption. The evidence presented in the previous section, indicating the absence of effects for older cohorts for our preferred specifications, aligns with the identifying assumption of parallel trends. We next investigate in Figure A6a and Figure A6b whether excluding municipality-specific linear trends and using the estimator suggested by Sun and Abraham (2021), which is robust to treatment effect heterogeneity, affects our conclusions.

Comparing the effects for primary school graduation, the figures indicate the presence of pre-trends in primary graduation rates in the years leading up to the analysis in TWFE specifications that exclude municipality-specific linear trends. This pattern is not unexpected, as TV-school construction targeted areas with increasing demand from primary school students. Moreover, this is one of the reasons motivating the inclusion of municipality-specific trends in our preferred specification. While directly controlling for the number of primary schools in the pre-period interacted with cohort dummies somewhat reduces the pre-treatment coefficients, it does not entirely eliminate them.

In contrast, we do not find evidence that pre-trends in primary graduation affected lower secondary graduation. Estimates with and without linear-specific trends are very similar, indicating little evidence of pre-treatment impacts before the introduction of TV-schools. The estimators from Sun and Abraham (2021), which do not include unit-specific trends, are also very close to our TWFE estimates, suggesting that in this setting, concerns regarding staggered treatment are less of a concern.

In the labor market, there is little evidence of pre-trends in the probability of working and having a positive income. The effects are similar to that of Sun and Abraham (2021). There is some evidence of monthly income and hourly income show a similar pattern to that of years of education, with some evidence of existing pre-trends in the specifications that exclude municipal linear trends, though there is little evidence of pre-trends in closer relative years.

Overall, we conclude that different estimators lead to similar conclusions, though the inclusion of unit-specific trends helps address some underlying evidence of pre-trends in primary education, years of education, monthly income, and the likelihood of working in agriculture. We find no evidence that the inclusion of linear trends affects the estimates for other outcomes.

THE 1993 SHOCK AS AN ALTERNATIVE SOURCE OF VARIATION. In 1993, the Mexican constitution was amended to mandate lower secondary school education. This legislative change was meant to guarantee universal access to lower secondary schooling nationwide and prompted the rapid establishment of TV-schools. We leverage the variation created by this change to identify effects. Given the swift implementation following the legislation, this provides a relatively 'clean' source of variation, less susceptible to the potential of municipalities' endogenous decisions to set up these schools. Effects are estimated using Equation 1, except we only use the treatment variation from the 145 municipalities that experienced high-intensity TV-school construction in 1993 or 1994, using the rest as a comparison group. Appendix Figure A12 shows the main impacts on education and labor market outcomes. The estimated effects on education are of similar magnitude and significance compared to the main specifications. The effects on income and hourly wages, while positive, are noisier and statistically insignificant, which may be attributed to the reduced variation used. Nonetheless, the magnitudes suggest effects similar to those observed in the main analysis, which we take as conservative evidence pointing in a similar direction.

Migration. Migration, both within the country and abroad, poses a threat to identification. As discussed in Section III, the census data only reports the municipality of residence in adulthood, but not at birth. If migration decisions are uncorrelated with access to secondary schools, this assignment would just introduce measurement error in the estimates, attenuating the effects towards zero. However, if there is selective migration in response to the treatment, the estimates would capture the effects of this regional sorting instead of the causal effect of the TV-school expansion. For instance, if individuals with the highest returns to education permanently migrate from poorer municipalities –with high density of TV-schools– to urban municipalities—with relatively low density of TV-schools—they would be incorrectly classified as 'untreated' and we would underestimate the effects of the TV-schools. On the other hand, if areas with TV-schools attract particularly motivated students or families to settle, one could overestimate the impact of these schools. We take several steps to investigate the extent to which our estimates might be affected by these issues.

First, as discussed, our main specifications exclude interstate migrants, since the census does report the state of birth. This corresponds to excluding 8% of the sample. While this implies that our estimates are only relevant for the sample of individuals who remained in the state in which they were born, this reduces concerns about selective interstate in-migration.

To assess the extent to which out-migration affects our estimates, we follow others in the literature (Parker and Vogl, 2023; Atkin, 2016) to test whether the size of cohorts of non-state migrants are differentially affected by the treatment. For instance, if people are less likely to migrate in response to the introduction of TV-schools, the size of the cohorts should increase differentially across municipalities. Figure A11a shows the coefficients of a regression analogous to equation 1 where the dependent variable is the log of population size for each municipality-cohort cell. We find no evidence of significant changes in cohort sizes following the introduction of TV-schools. Considering that Mexican men are more likely to migrate to the U.S., we also test whether there is an effect on the gender ratio by cohort. Again, we do not find significant changes (Figure A11b). We also re-run these estimates including inter-state migrants, and find similar results, suggesting that there was no large inflow of people to treated municipalities.[23]

[23] Even if there was an inflow of people from different states, the main estimates would not be directly affected as interstate migrants are not included them in the regression. However, they might indirectly affect returns to education, for instance, if these flows had general equilibrium effects. We also note that this approach of studying cohort sizes helps us better understand the potential for international migration affecting our estimates since these individuals would disappear from any sample in Mexico.

As a second approach to directly test the effects of TV-school construction on short-term migration, we use data from the 2000 census, which indicates the municipality of residence five years prior, in 1995. This allows us to assign respondents to their original locations in 1995 closer to the time they would have attended lower secondary school. We focus on cohorts born between 1970 and 1985 (aged 10 to 25 in 1995) and only use variation from TV-school construction during the 1990s, excluding municipalities that experienced high construction in earlier years. This approach helps ensure that the location of individuals in 1995 more closely matches their municipality of secondary school attendance. Appendix Figure A13a shows no effects on short-term migration for affected cohorts, by estimating whether they had a different municipality of residence in 2000. We also estimate effects on lower secondary graduation using this sample and find effects of similar magnitude to our main estimates (Appendix Figure A13b).

Overall, we conclude that while there are data limitations intrinsic to using the census to measure long-term outcomes due to the possibility of inter-municipal migration, any possible migration effects due to these schools are unlikely to be large enough to substantially affect our results.

Alternative Samples and Controls. To address concerns about potential sources of bias contaminating our results, we conduct several robustness tests, reported in Appendix Table A7. Column (1) shows our main estimates for comparison. Column (2) presents estimates controlling for the simultaneous growth of standard secondary school density at the municipal level, which could confound the impacts of TV-school growth. Adding this control does not affect the magnitudes of the main estimates.

To address concerns that municipalities receiving TV-schools might have very distinct growth trajectories compared to other municipalities, we re-estimate effects using a sample that only includes municipalities that received at least one TV-school during the study period. We find similar effects for this group (Column 3). Column (4) shows the effects when excluding municipalities that received at least one TV-school but never "switched" to high-intensity construction. These municipalities tend to be more urban and populated, meaning the TV-density threshold per person is not met. Again, the conclusions remain similar.

Finally, Column (5) combines the difference-in-difference approach with propensity score matching. This sample is restricted to municipalities that experienced construction and matched control municipalities with similar 1980 characteristics.[24] Again, the effects are positive and significant, and if anything, they are somewhat larger than the original estimates.

VI Benchmarking the Effects of TV-Schools

Next, we benchmark the impacts of TV-school construction on labor market outcomes. To do this, we proceed in two ways. First, we estimate the implied returns to education, leveraging the TV-school construction as a source of exogenous variation. Second, we compare the impacts of TV-schools to that of standard school constructions during the same period, using a similar identification strategy to evaluate both modalities.

IMPLIED RETURNS TO EDUCATION. We provide back-of-the-envelope estimates of the implied returns to education using Wald estimates. We note, however, that strictly speaking the exclusion restriction assumption that underpins the validity of these estimates is likely to be violated since it requires assuming that there are no externalities or general equilibrium effects stemming from TV-school construction. However, it is helpful to provide a reference point for evaluating the impact of these schools relative to the existing literature.

We estimate that an additional year of education following the high-intensity access to TV-schools is associated with a 19% increase in adult hourly income and an increase in monthly income of 866 pesos (Panel A, Table 1). For comparison, the OLS estimates of the returns to education for the same sample, are also 19%. Using the continuous TV-school density treatment variable reported in Panel B, we find that an additional TV-school per 50 children is associated with an increase of 1,141 pesos in monthly income and a 10% increase in hourly income when using our preferred DiD_{ATT} estimates. Using the DiD_{TWFE} estimates, this corresponds to an increase of 607 pesos, and a 15% increase in hourly income.

As a comparison, the worldwide average return to schooling is around 10%, and they tend to be higher in low or middle-income economies and lower for secondary education (7.2%) than for tertiary education (15.2%) (Montenegro and Patrinos, 2014). Other OLS estimates using Mince-

[24] We matched based on the following characteristics in 1980: primary school enrollment, secondary school enrollment, the share of adults with incomplete primary education, the share of households without piped water, electricity, and with dirt floors, share of adults earning less than two minimum wages, the share of illiterate adults, total population, the share of people living in rural localities, and share of people in the primary sector

Table 1: Impacts of Access to Schooling on Educational Achievement and Income by Type of School

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Lower Sec. Grad.		Years of Education		Monthly Income (pesos)		Hourly Income (log)	
	DiD_{TWFE}	DiD_{ATT}	DiD_{TWFE}	DiD_{ATT}	DiD_{TWFE}	DiD_{ATT}	DiD_{TWFE}	DiD_{ATT}
Panel A: High-intensity School Construction (Binary Treatment)								
School type: TV-Schools								
High TV-School Intensity	0.038***	0.079***	0.216***	0.421***	133.861***	364.691***	0.023	0.080***
	(0.004)	(0.007)	(0.036)	(0.080)	(22.857)	(43.562)	(0.015)	(0.026)
Dependent Mean	0.64	0.63	9.23	9.18	3151.06	3110.25	2.77	2.75
Observations	42823	45393	42823	45393	42819	45389	42824	45394
Panel B: Number of Schools per 50 Children (Continuous Treatment)								
School type: TV-Schools								
TV-School Intensity	0.159***	0.128***	0.770***	0.410***	412.587***	468.063***	0.114**	0.041
	(0.011)	(0.014)	(0.101)	(0.108)	(63.285)	(87.321)	(0.045)	(0.070)
Dependent Mean	0.61	0.60	8.93	8.93	3168.45	3214.74	2.77	2.78
Observations	43026	47511	43026	47511	43026	47509	43027	47512
School type: Standard Schools								
Std. School Intensity	0.177***	0.113***	1.082**	1.215***	580.563	1045.685***	0.174	0.066
	(0.053)	(0.041)	(0.486)	(0.435)	(360.005)	(333.187)	(0.109)	(0.240)
Dependent Mean	0.60	0.60	8.91	8.89	3018.04	3009.43	2.73	2.73
Observations	42518	44651	42518	44651	42518	44649	42519	44652

Notes: This table shows the impact of access to schooling by school type (standard schools and TV-schools). Odd columns present single-coefficient estimates from a two-way fixed effects (TWFE) model, DiD_{TWFE}, which includes municipality and cohort fixed effects, and municipality-specific linear time trends. Even columns present our preferred specification, the equally-weighted average treatment on the treated (ATT) of post-treatment estimates from the dynamic TWFE event study model, DiD_{ATT}, also including municipality and cohort fixed effects, and municipality-specific linear time trends. See Equation (1) and Section IV for details on the econometric specifications. Panel A reports coefficient estimates for the binary treatment–an indicator identifying individuals with access to a high density of schools in the municipality at age 15, defined as being above the sample median density. This binary treatment effect is only reported for TV-schools. Panel B reports coefficient estimates for continuous treatment–school density in the municipality at age 15, measured as the number of TV-schools per 50 eligible children. This effect is reported both for standard schools and TV-schools. The outcomes are: an indicator for graduating from lower secondary education (col. 1-2), years of education (col. 3-4), the unconditional monthly adult income in pesos (col. 5-6), and logarithmic transformation of hourly income (col. 7-8). Regressions use administrative school construction data from the Mexican Secretariat of Education for 1980–2000, and outcome data from the 10% subsample of the 2010 Mexican census from IPUMS, restricting it to non inter-state migrants. Data is aggregated into birth-year by municipality-of-residence cells, and standard errors are clustered at the municipal level. Statistical significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

rian regressions in Mexico report a return to an extra year of primary education of around 8%, whereas the returns for an extra year of secondary education and of college are around 10% and 11% respectively (Morales-Ramos, 2011).

COMPARISONS TO STANDARD SCHOOLS. We take advantage of the expansion of standard schools from 1980 to 2000 to assess how estimates differ from those of the TV-school roll-out. It is worth noting that this should not be taken as a direct comparison of the effects of different schooling approaches for the same population, as TV schools and standard schools operate in different types of areas. Standard schools are more likely to open in more urban and sectorally diversified municipalities, enrolling children with different demographic profiles. However, this comparison can provide a sense of the magnitude of nation-wide impacts from two different lower secondary schooling modalities.

For this exercise, we focus on the effects using the continuous measures of school density to enable better comparability across school alternatives.[25] Figures A14 and A15 present the event studies for the standard school treatment, centered around the first standard school construction in the municipality, showing no pre-treatment trends.

The impacts of constructing one standard school and one TV-school per 50 children on lower secondary graduation rates are nearly identical (Table 1, Panel B). However, the overall educational attainment, measured by overall years of education, is almost three times higher for standard schools than for TV-schools (1.21 additional years compared to 0.41 additional years of education), with both results significant at the 1% level. This difference in total educational attainment might be explained by the additional educational gains in higher levels of education in areas that experienced the construction of standard schools (Appendix A14c), whereas, as discussed, we do not find those corresponding effects for high TV-school density construction (Appendix A9c).

The average monthly income increase from an additional standard school is 1,046 pesos, more than two times as much as the 468 pesos increase from an additional TV-school. The increases in hourly income are positive for both types of schools, but are not significant at conventional levels using the continuous treatment measure. Back-of-the-envelope calculations of the returns of both types of schools indicate that TV-schools are associated with a monthly income increase of 1,141 pesos per additional year of education. In contrast, standard schools are associated with a return of 864 pesos per additional year of education. Therefore, despite higher overall impacts on overall educational attainment and adult income for standard schools compared to TV-schools, the labor market returns per additional year of education are relatively similar across both modalities.

The differences in the reduced form point estimates between standard schools and TV-schools may arise from several factors. First, standard schools are located in regions with greater access to upper secondary or tertiary education. Second, standard schools may be located in more developed

[25] In addition, using a binary measure for standard school construction, analogous to the high-intensity TV-schools access measure, does not yield significant effects. This is likely because standard schools were built gradually, preventing a distinct school density increase that could be used to identify impacts with the binary measure. Table A8 reports the binary treatment estimates for both school types.

areas with more diversified labor markets, potentially yielding higher overall economic returns to similar levels of human capital. Finally, standard schools may provide more relevant human capital for continuing education and for the labor market. In the next section, we investigate this further by analyzing how the impacts of TV-schools differ based on the baseline characteristics of the region where they operate.

VII Scale-up Considerations: Do Effects Differ by Region?

An often-cited concern related to scaling up programs is the potential for context-dependent impacts that may not align with those estimated in small-scale, tightly controlled experiments (Banerjee et al., 2017; Al-Ubaydli et al., 2019). Recent work has documented significant heterogeneity in educational program effectiveness following large-scale implementation, highlighting the importance of considering how variations in local context influence outcomes.[26]

A particular feature of the TV-school model is that it was originally conceived to operate in rural areas with thin teacher labor pools and limited schooling alternatives. However, as the model was scaled, TV-schools were established in municipalities with a variety of labor market opportunities, degrees of market integration, and alternative schooling options. This raises the question of whether the positive nationwide average effects estimated in the previous section mask important differences in program impacts by area characteristics. In particular, examining whether TV-schools benefited the originally intended rural areas is important, given the uncertain demand for and potential returns on education in places with few off-farm employment opportunities. Additionally, the TV-school model might not be suitable for more urban, sectorally diversified municipalities with access to standard schools and labor markets that potentially require skills that are hard to impart through ICTs.

To understand this dimension of effect heterogeneity, we categorize municipalities by their baseline prevalence of tertiary sector jobs. Specifically, we analyze the heterogeneity impacts by municipalities with above or below the median share of tertiary sector jobs at baseline (10%). There was substantial within- and across-state variation in the sectoral composition of municipalities in

[26]For instance, in education, several aspects have been shown to substantially affect program effectiveness, ranging from differences in design or inputs (Kerwin and Thornton, 2021; Ganimian, 2020), state capacity (Bold et al., 2018), and population targeted (Al-Ubaydli et al., 2017), to supply-side constraints (Davis et al., 2017), cultural practices (Ashraf et al., 2020), general equilibrium effects (Khanna, 2023) and what constitutes the closest substitutes to the program (Mountjoy, 2022; Fort et al., 2020).

1980, and as discussed in Section II, sectoral composition correlates with the prevalence of agricultural jobs, rurality, and poverty, allowing us to distinguish between the originally intended more marginalized locations and more developed areas.[27]

Table 2: Heterogenous Effects of a High TV-school Intensity by Type of Region

	Full sample	By services jobs		By standard school presence				Services jobs + standard school presence		
		Low	High	None at baseline	Low density	Has at baseline	High density	Low serv. + low density	High serv. + low dens.	High serv. + high dens.
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Panel A: Education Outcomes										
Lower Sec. Graduation	0.079***	0.060***	0.063***	0.076***	0.074***	0.074***	0.066***	0.068***	0.062***	0.053***
	(0.007)	(0.009)	(0.012)	(0.008)	(0.008)	(0.013)	(0.013)	(0.010)	(0.015)	(0.018)
Mean	0.46	0.38	0.55	0.41	0.42	0.53	0.56	0.37	0.50	0.62
Observations	45393	23730	21663	25743	32020	19650	13373	20064	11956	9707
Years of Education	0.421***	0.265***	0.330**	0.297***	0.330***	0.478***	0.445***	0.279***	0.321***	0.225
	(0.080)	(0.079)	(0.150)	(0.070)	(0.079)	(0.146)	(0.168)	(0.092)	(0.124)	(0.282)
Mean	7.48	6.72	8.30	7.00	7.09	8.10	8.40	6.66	7.81	8.91
Observations	45393	23730	21663	25743	32020	19650	13373	20064	11956	9707
Panel B: Income										
Monthly Income	364.691***	121.131***	403.200***	182.028***	190.840***	475.079***	544.608***	127.646***	220.292***	483.657***
	(43.562)	(36.884)	(81.575)	(41.590)	(38.541)	(69.983)	(92.779)	(40.820)	(81.165)	(145.672)
Mean	1807.08	1249.48	2417.72	1430.51	1513.25	2300.33	2510.51	1201.87	2035.63	2888.38
Observations	45389	23725	21664	25739	32016	19650	13373	20059	11957	9707
Hourly Income (log)	0.080***	0.018	0.102**	0.062	0.065**	0.084***	0.073	0.029	0.126**	0.038
	(0.026)	(0.035)	(0.041)	(0.040)	(0.031)	(0.032)	(0.048)	(0.038)	(0.058)	(0.051)
Mean	2.23	1.92	2.56	2.00	2.08	2.52	2.59	1.88	2.40	2.75
Observations	45394	23730	21664	25744	32021	19650	13373	20064	11957	9707
Panel C: Labor market composition										
Labor Market Participation	0.031***	0.005	0.031***	0.015**	0.012**	0.042***	0.056***	0.002	0.020	0.030*
	(0.007)	(0.007)	(0.011)	(0.007)	(0.006)	(0.011)	(0.016)	(0.007)	(0.013)	(0.018)
Mean	0.58	0.55	0.62	0.56	0.56	0.61	0.63	0.54	0.60	0.65
Observations	45393	23729	21664	25743	32020	19650	13373	20063	11957	9707
Services Sector	0.030***	0.007	0.033***	0.020***	0.017***	0.036***	0.044***	0.006	0.029***	0.032
	(0.005)	(0.005)	(0.011)	(0.005)	(0.005)	(0.009)	(0.013)	(0.005)	(0.010)	(0.023)
Mean	0.23	0.16	0.30	0.19	0.19	0.28	0.30	0.16	0.26	0.34
Observations	45394	23730	21664	25744	32021	19650	13373	20064	11957	9707

Notes: This table shows the impact of access to a high density of TV-schools on education and labor market outcomes by different municipality characteristics using split-sample regressions. The samples of analysis are: Full sample (col. 1), below and above the median of the share of services sector in 1980 (col. 2-3), without and with presence of standard schools in 1980 (col. 4,6) below and above the density of standard schools in 2000 (col. 5,7), low share of services sector in 1980 and low density of standard schools in 2000 (col. 8), high share of services sector in 1980 and low density of standard schools in 2000 (col. 9), high share of services sector and high density of standard schools in 2000 (col. 10). The treatment is an indicator identifying individuals with access to a high density of schools in the municipality at age 15, defined as being above the sample median density of schools. The coefficient estimates are the equally-weighted average treatment on the treated (ATT) of post-treatment estimates from the dynamic TWFE event study model, DiD_{ATT}, which includes municipality and cohort fixed effects, and municipality-specific linear time trends. See Equation (1) and Section IV for details on the econometric specification. The outcomes are: an indicator for graduating from lower secondary education, years of education, the unconditional monthly adult income in pesos, the logarithmic transformation of hourly income, an indicator for participating in the labor market, and an indicator for participating in the services sector. Regressions use administrative school construction data from the Mexican Secretariat of Education for 1980–2000, and outcome data from the 10% subsample of the 2010 Mexican census from IPUMS. Data is aggregated into birth-year by municipality-of-residence cells, and standard errors are clustered at the municipal level. Statistical significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2 reports the estimated pooled average treatment effects on the treated, DiD_{ATT}, calculated by running separate split-sample regressions for each group of municipalities with a given characteristic. Figure 8 displays the full set of estimated dynamic coefficients for the main out-

[27] Figure A17 shows the evolution of constructions over time by sectoral composition. Figures A1 and A16 show that despite the Southern regions of Mexico having fewer tertiary sector jobs compared to the North, there is substantial spatial heterogeneity even within states. This within- and across-state variation in this labor market characteristic makes it conducive for investigating the heterogeneity of the treatment effects. It is also a policy-relevant dimension observable in most censuses.

comes, showing an absence of pre-trends for each of the heterogeneity groups.[28]

EDUCATIONAL OUTCOMES. We find no evidence that the availability of jobs with potentially higher returns to education influences enrollment in TV-schools and subsequent educational achievement. The effects on years of education and lower secondary education graduation rates are similar across areas with low and high tertiary sector presence at baseline, with a 6 p.p. increase in lower secondary education for both and an overall increase of 0.27 and 0.33 years of education, respectively (see Columns 2 and 3 of Table 2 and Figures 8a and 8b). Municipalities with fewer services jobs at baseline start from a lower educational achievement baseline, resulting in a 15% increase in lower secondary graduation rates compared to a 11% in high services regions.[29]

Regions with high and low service jobs may also have different access to alternative schooling options, resulting in different counterfactual scenarios for attending TV schools. Therefore, the average treatment effects on educational attainment may reflect changes in overall enrollment rates (extensive margin) as well as shifts in enrollment across school types (intensive margin). To investigate whether the relative importance of each of these margins differs by region, we examine school enrollment patterns in both standard schools and TV-schools following the onset of high TV-school construction in municipalities with varying shares of services jobs.

Figure 7 shows that in regions predominantly involved in agriculture, TV-school enrollment increases at the extensive margin without affecting standard school enrollment, as conventional schooling alternatives were largely unavailable.[30] In contrast, more developed areas exhibit evidence of a decline in standard school enrollment. Although we cannot reject that there was no change in enrollment in standard schools, the point estimates suggest a potential yet statistically imprecise substitution effect at the intensive margin.

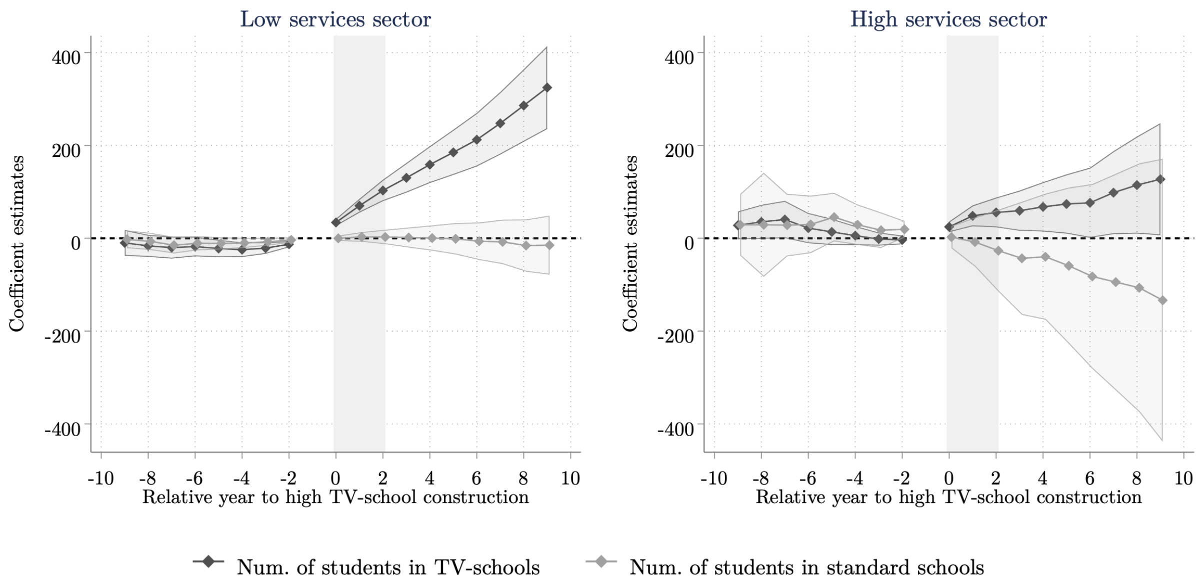
Table 1 showed larger estimated effects on educational attainment and labor market outcomes from an additional standard school compared to an additional TV-school. If this difference arises because standard schools provide a higher quality of education more aligned with labor market de-

[28] We also estimate the differential effects between municipalities with low and high service presence by running a single regression similar to Equation (1) interacting all coefficients with indicators for their heterogeneity group. The differential point-estimate effects and the p-value for the joint hypothesis of equality between groups are presented in Tables A9 and A10. The estimated dynamic coefficients for the rest of the outcomes are displayed in Figure A18.

[29] We do not detect differences in effects for primary and upper secondary education graduation either. See Appendix Figures A18a and A18b and Table A9.

[30] As with Figure A5, the yearly enrollment data at the school-level is only available post-1990.

Figure 7: Event Studies of Impacts of Access to High TV-School Intensity on Enrollment by Type of School and Sectoral Composition (post-1990)



Notes: This figure shows the effect estimates of access to high density of TV-schools on TV-school enrollment on 7th grade (1st grade of lower secondary) and standard school enrollment by sectoral composition at baseline, calculated by academic year relative to the year of switch to a high TV-school intensity construction. The outcome is measured as the number of students enrolled in first grade in a given secondary school type in the municipality. The econometric specification is a dynamic TWFE event study model that includes municipality and cohort fixed effects, and municipality-specific linear time trends. See equation (1) for details. Event study coefficient estimates are represented by a solid line, and 95% confidence intervals with a vertical line. All effects are computed relative to the period before the high TV-school density switch. Data comes from the Mexican Secretariat of Education (SEP) for the years 1990–2000.

mands, then the observed student shifts from standard to TV-schools would be concerning. Alternatively, while both standard schools and TV-schools might offer comparable educational content, standard schools could be serving populations with better access to higher educational opportunities or more robust labor markets, thereby yielding different effects on similar educational inputs.

To investigate these possibilities, Table 2 examines the differential impacts of TV-schools based on the presence of standard schools in the municipality. We show results using a measure of standard schools at baseline (an indicator for the presence of at least one standard school in 1984–Columns 4 and 6) and a contemporaneous measure of standard school presence (an indicator variable for high-density of standard schools at the end of our sample period–Columns 5 and 7). Additionally, we examine the effects of TV-schools across different combinations of standard school presence and municipality types, where sample size permits: municipalities with low services presence at baseline and low density of standard schools (Column 8), those with high services presence and low density of standard schools (Column 9), and those with both high services presence and high density of standard schools (Column 10).

Across municipalities with different characteristics, we find that lower secondary education graduation rates consistently increase between 5 p.p. and 8 p.p., all significant at the 1% level. The presence of standard schools does not seem to differentially change the impact of TV-school construction on these graduation rates. However, regions with a high density of standard schools exhibit slightly lower graduation rates following intense TV-school construction (Column 7) compared to regions with lower densities of standard schools (Columns 4 and 5). This pattern is consistent with the switch in standard to TV-school enrollment shown in Figure 7. Columns 9 and 10 further support this, suggesting that in richer municipalities, the impact of TV-school construction on graduation is smaller if there is a high density of standard schools.

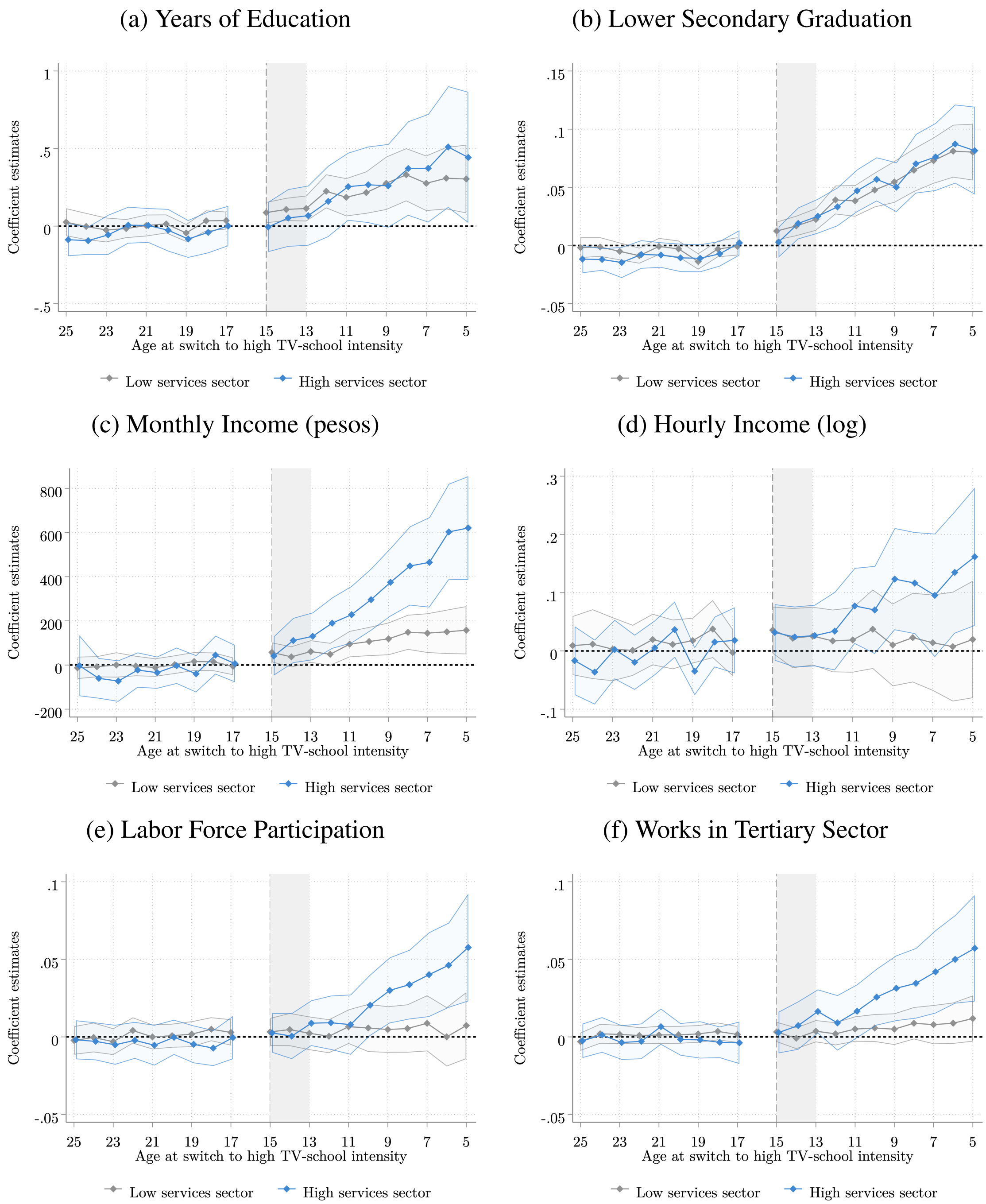
In terms of years of education, the impacts are also consistently positive. They also appear to be higher in areas with a higher share of service jobs. Interestingly, regions with a higher density of standard schools show larger increases (0.44–0.48 years) than those with low school density (0.30–0.33 years). This might be explained by the increased presence of higher educational institutions in areas with standard school presence. Overall, we conclude that the introduction of TV-schools led to comparable increases in lower secondary graduation rates across regions, despite the potential student transfers from standard to TV-schools in some areas.

LABOR MARKET OUTCOMES. We observe consistently positive effects on labor income across different types of municipalities, although the magnitude and statistical significance of these effects vary. The most substantial impacts are in regions with a higher share of service jobs at baseline, where income increases by 403 pesos, representing a 17% increase from the average income of 2,418 pesos. Conversely, more agriculturally intensive regions experience a rise of 121 pesos, a 10% increase from an average income of 1,250 pesos. Both increases are statistically significant at the 1% level, and the difference between these impacts is also significant at the same level, as shown in Figure 8c.[31]

Additionally, regions with low standard school density see income increases ranging from 182 to 191 pesos (13%), while areas with high standard school densities experience larger increases, between 475 and 544 pesos (2122%). Collectively, these findings suggest not only that income

[31] The differences in heterogeneity estimates and the p-value for the joint significance tests are in Table A10.

Figure 8: Event Studies of Impacts of Access to High TV-School Intensity on Outcomes by Sectoral Composition



Notes: This figure shows the effect estimates of access to high TV-schools density by age at the year of switch to a high TV-school intensity construction on different outcomes. This is decomposed by the presence of services sector at baseline, defined as an indicator for whether the municipality had an above-median share of services sector jobs in 1980. The econometric specification is a dynamic TWFE event study model that includes municipality and cohort fixed effects, and municipality-specific linear time trends. See Equation (1) for details. Event study coefficient estimates are shown with a solid line, and 95% confidence intervals are shown with a shaded area. All effects are computed with respect to age 16. See Section IV for details. Outcomes are: years of education, an indicator for graduating from lower secondary education, unconditional monthly adult income in pesos, and the logarithmic transformation of hourly income, indicators for participation in the labor market, and for unconditionally participating in the tertiary sector. Data is from the Mexican Secretariat of Education (SEP) for 1980–2000, and the 2010 Mexican census (10% IPUMS subsample).

gains are considerably larger in more developed areas but also that regions with a high density of standard schools may have other unobserved characteristics associated with higher impacts in the labor market.

The estimates for hourly income are less precise, and using this outcome variable, we cannot rule out that there are no effects in more rural areas. The main gains seem to be concentrated in municipalities with a high prevalence of service jobs and limited access to standard schools (Column 9).

In terms of changes in the labor market composition, the probability of employment increases by 3 p.p. (a 5% increase over the 62% average) in municipalities with higher sectoral diversification. There are no significant changes at the extensive margin of labor supply in municipalities with fewer service jobs, and this difference is significant at the 5% level.[32] Additionally, there is a significant 3 p.p. increase in the share of individuals working in the tertiary sector in those sectorally-diversified regions, representing an 11% increase in the overall share. We find no corresponding effect in low-service sector municipalities, raising questions about whether increased education can effectively drive labor reallocation in predominantly poor and agriculturally based areas if there are limited off-farm opportunities.

In summary, while we do not find evidence that local labor market conditions influenced investments in lower secondary education, we find that the labor market value of educational gains varies based on area characteristics. Although not conclusive, the results suggest that an area's capacity to absorb post-primary skills might be an critical factor in determining the economic value of a secondary degree. Importantly, we do not find evidence of any long-term detrimental effects from TV-school constructions in areas where alternative schooling options were available. We interpret these findings optimistically regarding the role of TV-schools in expanding access for children across diverse socioeconomic contexts.

VIII Conclusion

Improving access to public services for traditionally marginalized populations remains a critical concern for policymakers in LMICs. ICT-based programs, which can facilitate information

[32] Appendix Table A10 shows that the likelihood of earning positive income unconditionally increases by 3 p.p. and 1.1 p.p. in high and low service job areas, respectively, with the second increase and the difference being significant at the 10% level.

dissemination, expand coverage, and bypass agency issues among public sector employees, present a promising solution to extend access in remote locations at scale (Aker, 2017; World Bank, 2016; Jepsen and Rivkin, 2009). In the context of remote and rural education, the appeal of ICT lies not only in its ability to reduce marginal costs of delivery but also in its reduced reliance on the quality of labor inputs.

The challenge of attracting and retaining qualified teachers in rural and marginalized areas is significant, and teachers who take these posts also tend to be less qualified and more likely to lack formal teacher training , with women particularly underrepresented (Evans and Acosta, 2023; Bobba et al., 2021; Elacqua et al., 2018). Despite several LMICs implementing policies to attract and retain teachers to these locations through financial incentives, subsidized housing, and behavioral nudges (Pugatch and Schroeder, 2018; Castro and Esposito, 2022; Kamere et al., 2019; Ajzenman et al., 2022), these efforts are insufficient at current expenditure levels to close the existing gaps (Evans and Acosta, 2023).

This paper documents the long-term effects of the expansion of television-based schools in Mexico, a model that could solve two prevalent barriers to expanding post-primary education in developing countries: the supply constraint of trained secondary education teachers and the high rates of teacher absenteeism (Banerjee and Duflo, 2006; Duflo et al., 2012). Employing a difference-in-difference identification strategy, we find consistent evidence of increases in educational attainment for cohorts who were young enough to benefit from access to these schools, with individuals exposed to high-intensity TV-school openings by age 16 completing an 0.4 additional years of education and obtaining significantly higher earnings in adulthood.

These results provide much-needed evidence of the long-run impacts of one of the most basic forms of technology-aided instruction. Recent surveys report mixed results on the effectiveness of the use of technology in education (Bulman and Fairlie, 2016; Escueta et al., 2017), with the most substantial returns observed in programs that use technology to personalize instruction to each student (e.g., Banerjee et al., 2007; Muralidharan et al., 2019). In contrast, the mode of content delivery in TV-schools, a one-size-fits-all lesson taught by remote teachers, differs significantly from these programs. The success of TV-schools, however, may derive from their 'blended environment', where the benefits of superstar teachers delivering the content (Acemoglu et al., 2014) are combined with in-class support, guided activities, and peer interactions.

In settings where low-tech remote educational interventions may be the only possibility to provide education, access to TV-schools can offer sizable gains for out-of-school children. Additionally, if the success of interventions using technology to personalize instruction replicates to TV-schools-like settings, the estimated impacts in this paper could be a lower bound for future programs using more sophisticated and personalized educational technologies.

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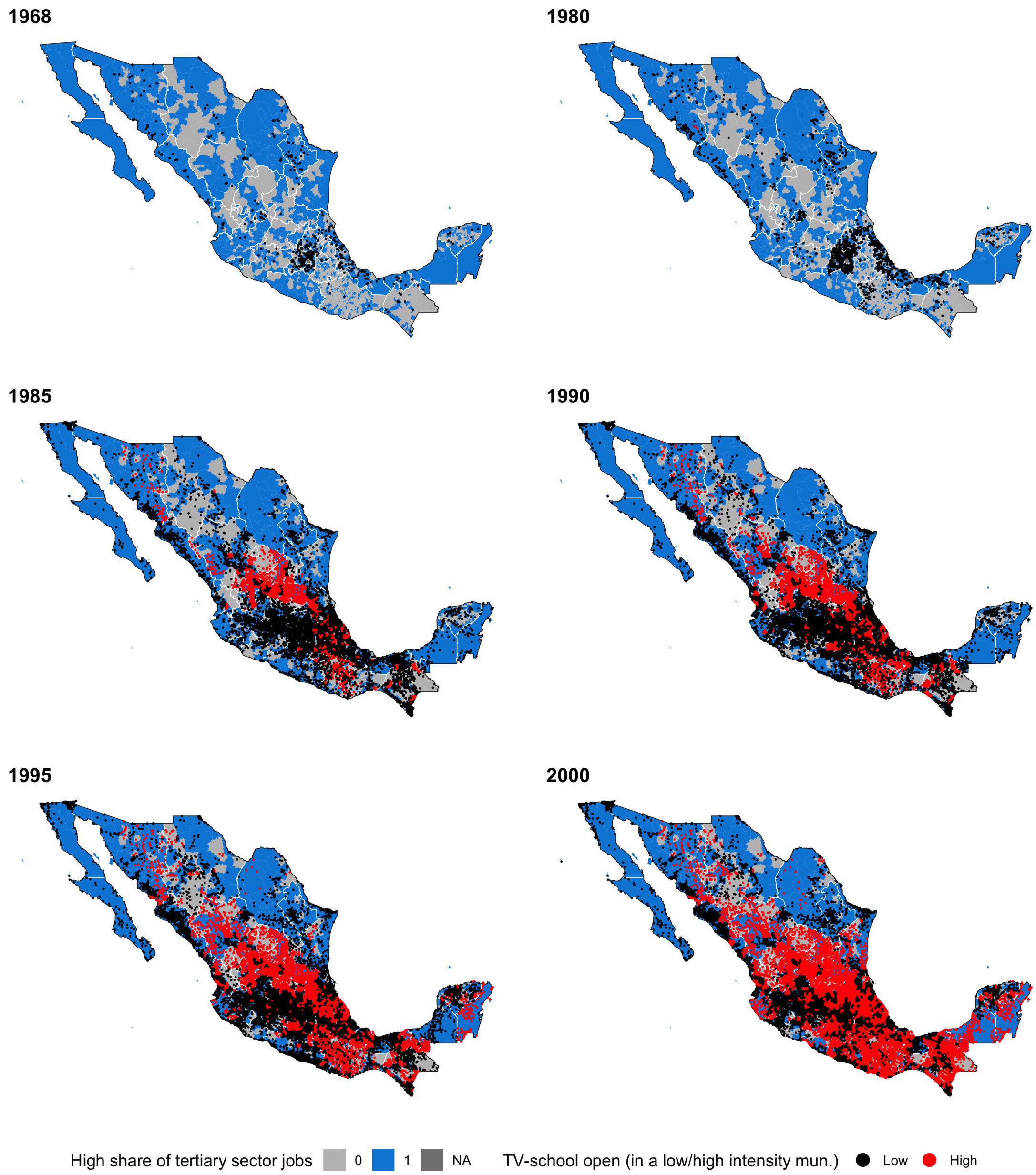
Additional Tables and Figures

Figure A1: Expansion of TV-Schools (1968–2000)

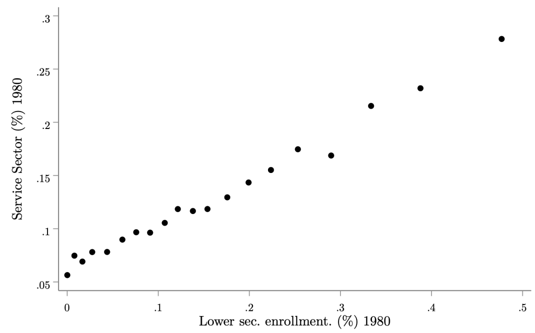


Notes: The maps show the open TV-schools in each year, colored by whether their municipality is above or below the median TV-school density in a given year. TV-school density is defined as the number of schools per 50 children aged 11 to 14 in each municipality in 1980. Municipalities are also categorized by whether they have an above-median or below-median share of tertiary sector jobs in 1980. Geographical frontiers correspond to municipalities. Data comes from the school registry data from the Mexican Secretariat of Education (SEP).

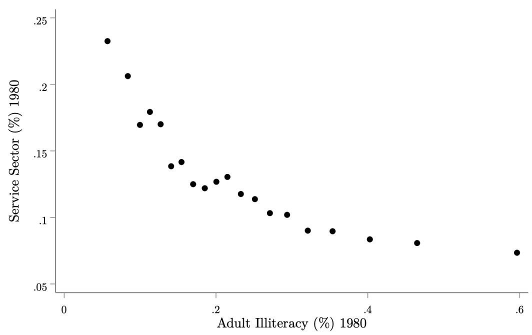
Figure A2: Correlation between Share of Workers in the Tertiary Sector and Municipal Characteristics in 1980

(a) Poverty Index

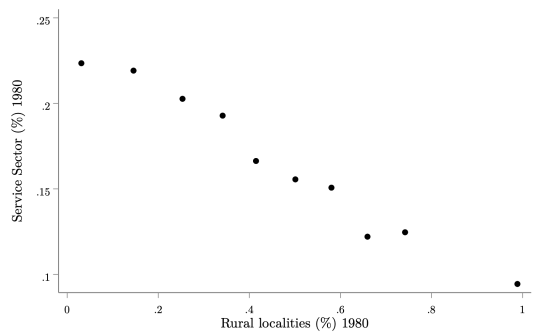
(b) Lower Secondary Graduation



(c) Illiteracy Rate

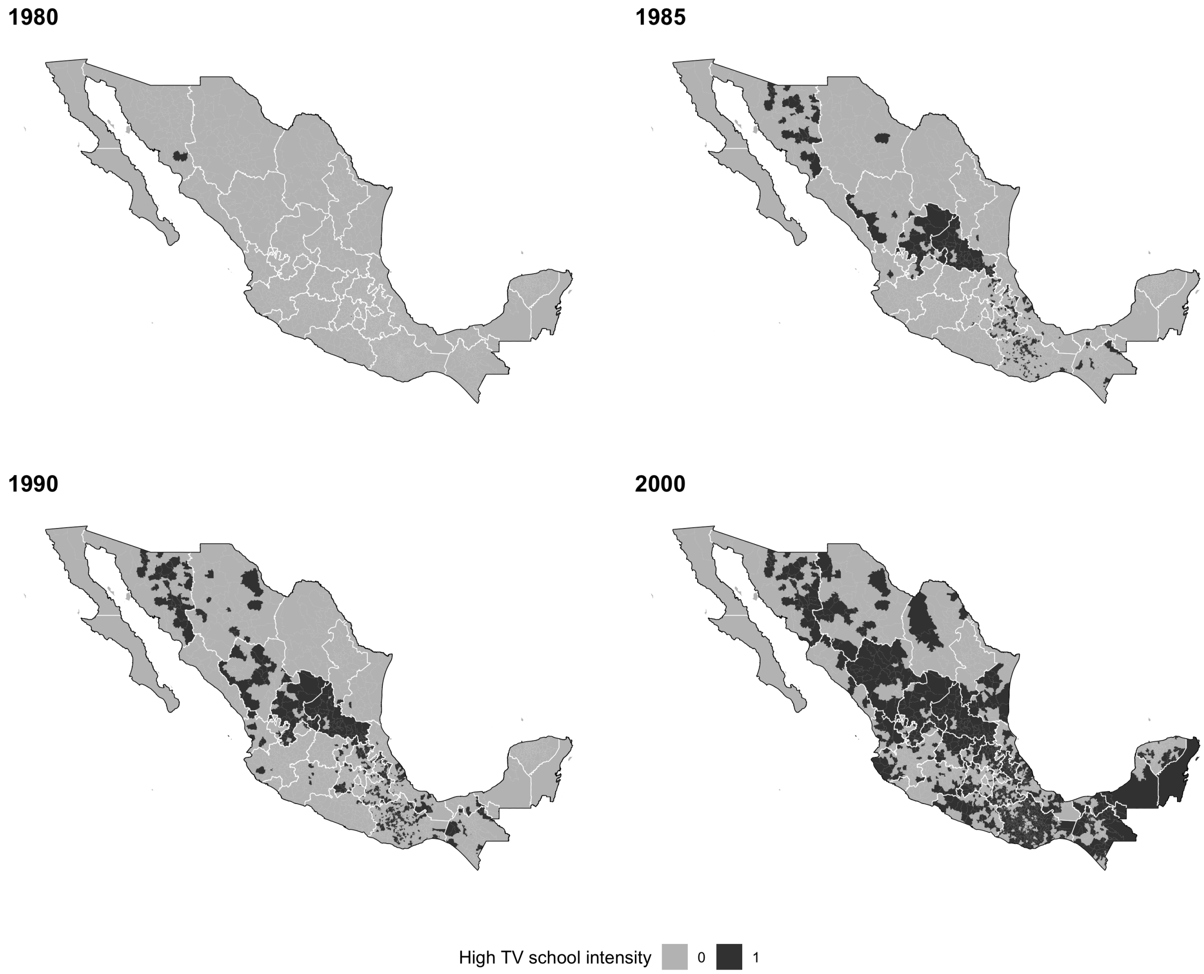


(d) Share of Population in Rural Localities



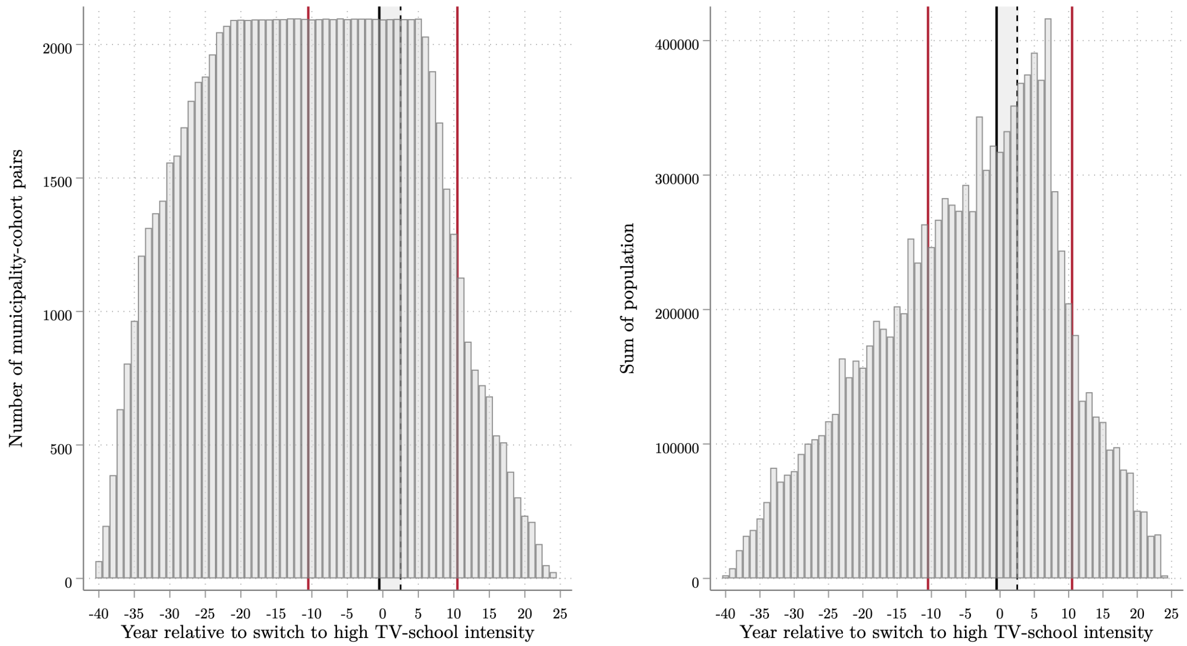
Notes: The figures above show binscatter plots of the relationship between the share of the adult population working in the service sector in a given municipality and (a) the poverty index of that municipality (constructed using a primary component analysis including the share of illiterate people and people with incomplete primary over the age of 15, the share of households without piped water, with dirt floors, without electricity, and the share of the working population with income of up to 2 minimum salaries), (b) lower secondary school graduation, (c) the adult illiteracy rate, and (c) the share of the municipal population living in rural localities, defined as those with less than 2,500 people. All data is from the 1980 census (IPUMS subsample).

Figure A3: Switchers to High TV-school Intensity (1980–2000)



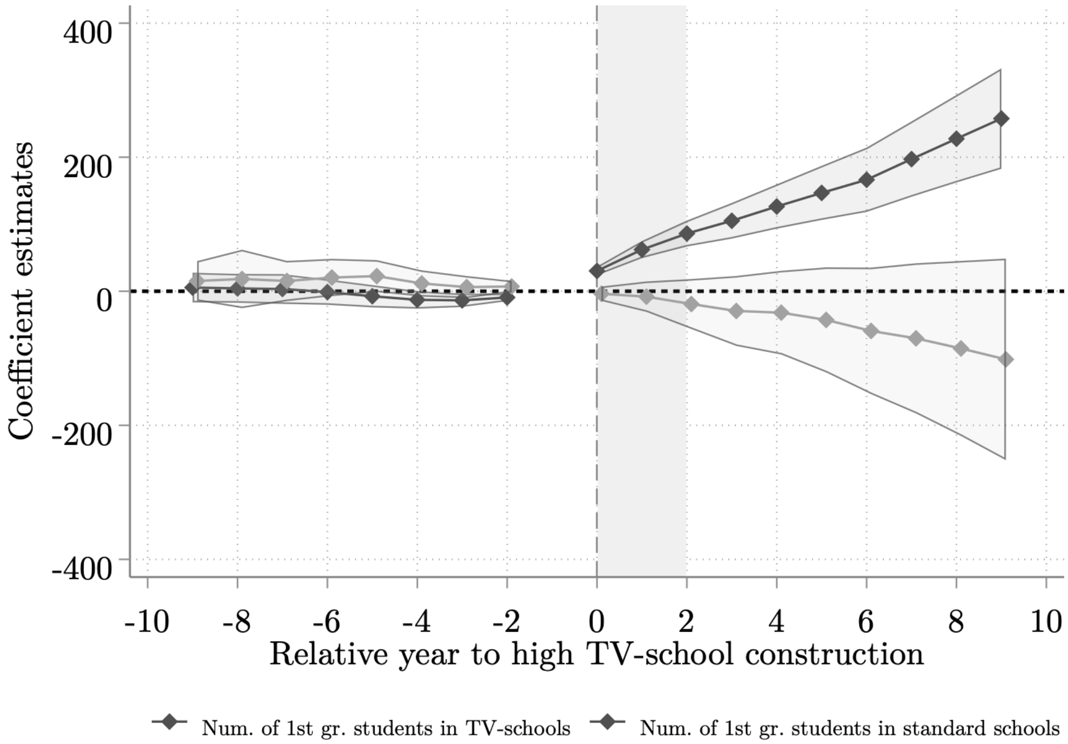
Notes: The maps show the spatial and temporal identifying variation used in the empirical strategy. It highlights the municipalities that switch to having a high TV-school intensity during the 1980-2000 period. Data comes from the school registry data from the Mexican Secretariat of Education (SEP).

Figure A4: Distribution of Sample Population Weights



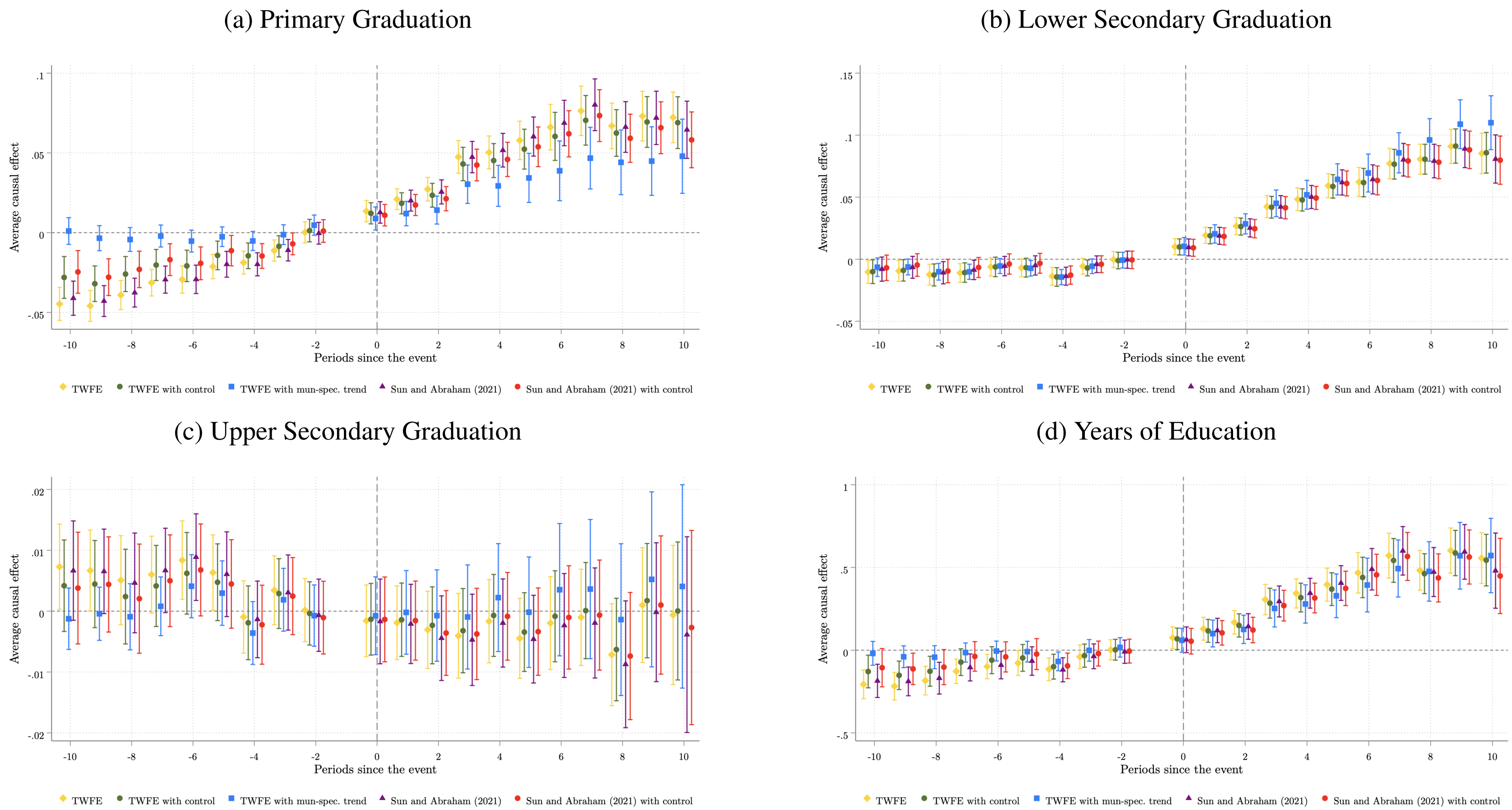
Notes: This graph shows the distribution of the population weights by cohort relative to the switch year to high TV-school intensity. The left graph plots the number of municipality-cohort pairs at each relative year. The right graph plots the sum of the weighted average population at each relative year. The red lines indicate the relative year periods in the analysis. Data comes from the 2010 Mexican census (10% IPUMS subsample).

Figure A5: Event Studies of Impacts of Access to High TV-School Intensity on Enrollment by Type of School (post-1990)



Notes: This figure shows the effect estimates of access to high density of TV-schools on TV-school enrollment and standard school enrollment by academic year relative to the year of switch to a high TV-school intensity construction. The outcome is measured as the number of students enrolled in first grade in a given secondary school type in the municipality. The econometric specification is a dynamic TWFE event study model that includes municipality and cohort fixed effects, and municipality-specific linear time trends. See Equation (1) for details. Event study coefficient estimates are represented by a solid line, and 95% confidence intervals with a vertical line. All effects are computed relative to the period before the high TV-school density switch. Data comes from the Mexican Secretariat of Education (SEP) for the years 1990–2000.

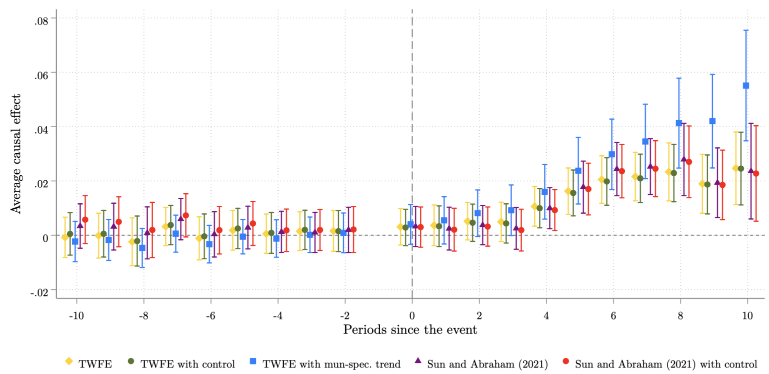
Figure A6: Event Studies of Impacts of Access to High TV-School Intensity using Different Econometric Specifications on Education Outcomes



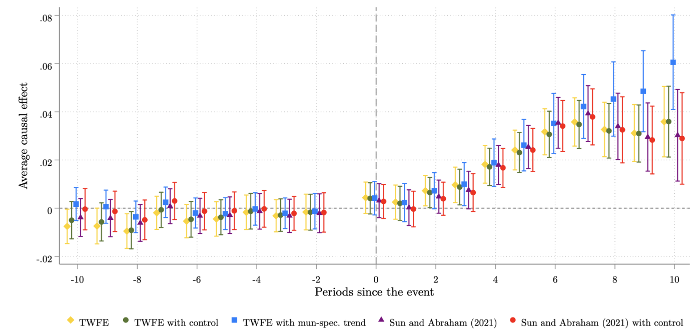
Notes: This figure shows the estimated effects of access to high TV-schools density on educational outcomes, by periods relative to the year of the switch to a high TV-school intensity. The econometric specifications are: A standard dynamic two-way fixed effects model only including municipality and cohort fixed effects (TWFE), a dynamic TWFE with municipality and cohort fixed effects with primary school density as control (TWFE with control), a dynamic TWFE with municipality and cohort fixed effects and municipality-specific linear time trends (TWFE with mun-spec. trend), the Sun and Abraham (2021)'s heterogeneity-robust estimator without (Sun and Abraham) and with primary school density as a control (Sun and Abraham with control). Outcomes are: indicators for participation in the labor market and unconditional probability of earning positive income, unconditional monthly adult income in pesos, and the logarithmic transformation of hourly income. Event study coefficient estimates are shown with dots, and 95% confidence intervals are shown with a shaded area. All effects are calculated with respect to period -1. See Section IV for details. Data is from the Mexican Secretariat of Education (SEP) for 1980–2000, and the 2010 Mexican census (10% IPUMS subsample).

Figure A7: Event Studies of Impacts of Access to High TV-School Intensity using Different Econometric Specifications on Labor Outcomes

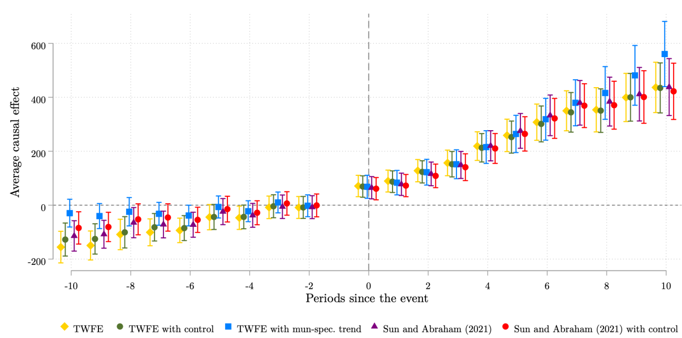
(a) Labor Force Participation



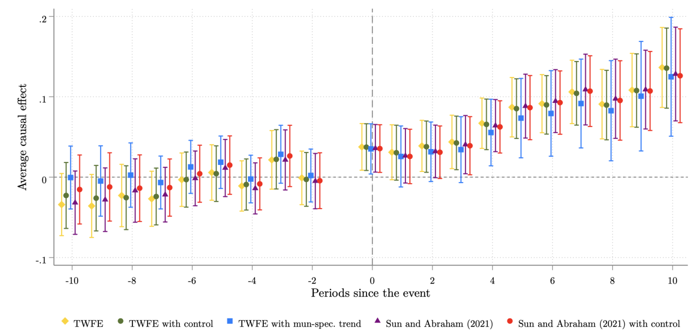
(b) Earns Positive Income



(c) Monthly income (pesos)

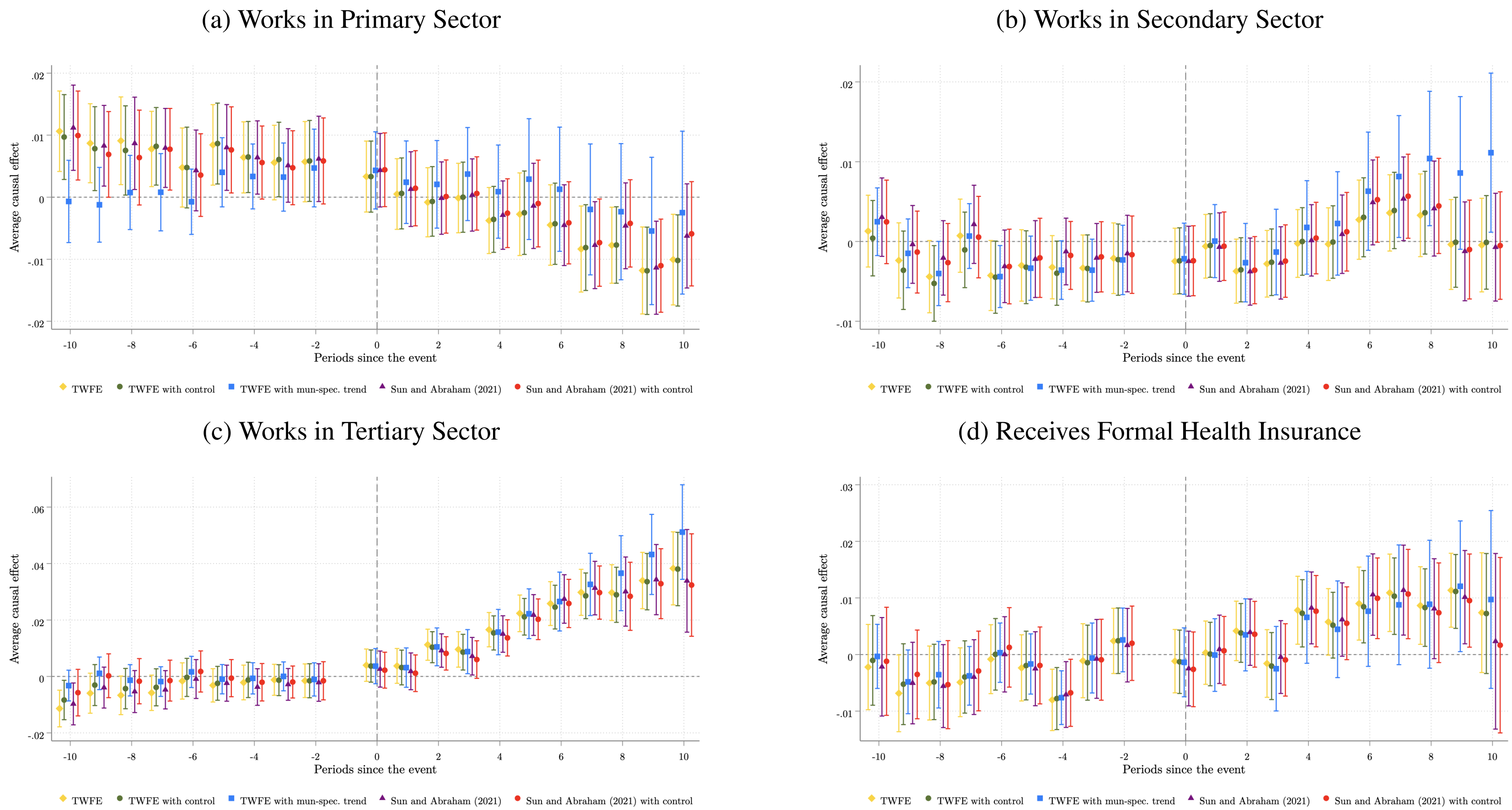


(d) Hourly income (log)



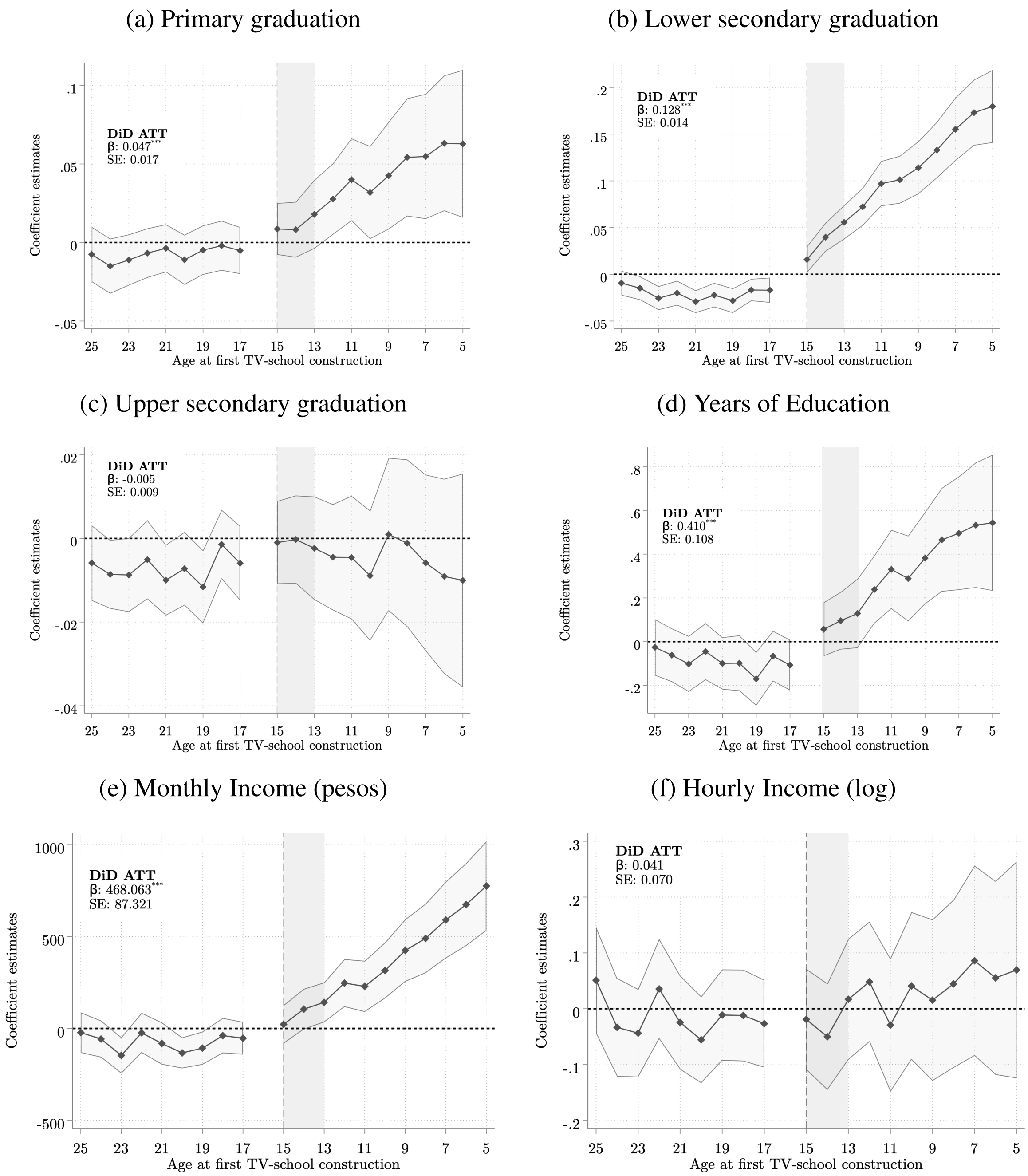
Notes: This figure shows the estimated effects of access to high TV-schools density on educational outcomes, by periods relative to the year of the switch to a high TV-school intensity. The econometric specifications are: A standard dynamic two-way fixed effects model only including municipality and cohort fixed effects (TWFE), a dynamic TWFE with municipality and cohort fixed effects with primary school density as control (TWFE with control), a dynamic TWFE with municipality and cohort fixed effects and municipality-specific linear time trends (TWFE with mun-spec. trend), the Sun and Abraham (2021)'s heterogeneity-robust estimator without (Sun and Abraham) and with primary school density as a control (Sun and Abraham with control). Outcomes are: indicators for graduating from primary, lower secondary, and upper secondary education, and years of education. Event study coefficient estimates are shown with dots, and 95% confidence intervals are shown with a shaded area. All effects are calculated with respect to period -1. See Section IV for details. Data is from the Mexican Secretariat of Education (SEP) for 1980–2000, and the 2010 Mexican census (10% IPUMS subsample).

Figure A8: Event Studies of Impacts of Access to High TV-School Intensity on using Different Econometric Specifications on Labor Market Composition Outcomes



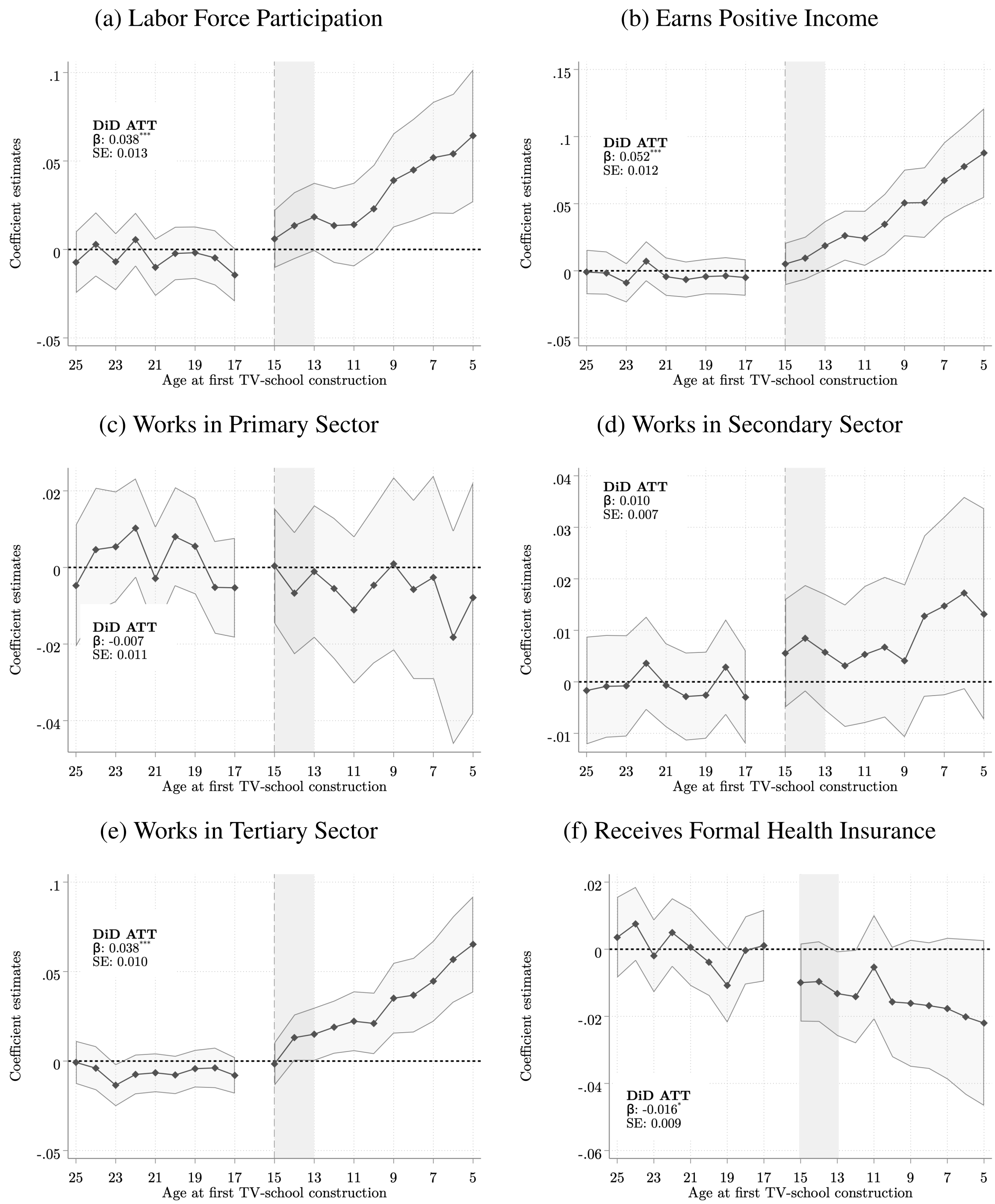
Notes: This figure shows the estimated effects of access to high TV-schools density on educational outcomes, by periods relative to the year of the switch to a high TV-school intensity. The econometric specifications are: A standard dynamic two-way fixed effects model only including municipality and cohort fixed effects (TWFE), a dynamic TWFE with municipality and cohort fixed effects with primary school density as control (TWFE with control), a dynamic TWFE with municipality and cohort fixed effects and municipality-specific linear time trends (TWFE with mun-spec. trend), the Sun and Abraham (2021)'s heterogeneity-robust estimator without (Sun and Abraham) and with primary school density as a control (Sun and Abraham with control). Outcomes are: indicators for unconditionally participating in the primary sector, secondary sector, and tertiary sector, and for receiving formal health insurance. Event study coefficient estimates are shown with dots, and 95% confidence intervals are shown with a shaded area. All effects are calculated with respect to period -1. See Section IV for details. Data is from the Mexican Secretariat of Education (SEP) for 1980–2000, and the 2010 Mexican census (10% IPUMS subsample).

Figure A9: Event Studies of Impacts of TV-School Intensity on Education Outcomes and Income (Continuous Treatment Variable)



Notes: This figure shows the effect estimates of the TV-schools density by age at the first TV-school construction on different outcomes, using the continuous treatment variable. The econometric specification is a dynamic TWFE event study model that includes municipality and cohort fixed effects, and municipality-specific linear time trends. See Equation (1) for details. Event study coefficient estimates are shown with a solid line, and 95% confidence intervals are shown with a shaded area. All effects are computed with respect to age 16. The dynamic coefficient estimates are summarized with an equally-weighted average treatment on the treated single point-estimate DiD_{ATT} superposed in the graph alongside its standard error. See Section IV for details. Outcomes are: indicators for graduating from primary, lower secondary, and upper secondary education, years of education, unconditional monthly income in pesos, and the logarithmic transformation of hourly income. Data is from the Mexican Secretariat of Education (SEP) for 1980–2000, and the 2010 Mexican census (10% IPUMS subsample).

Figure A10: Event Studies of Impacts of TV-School Intensity on
Labor Market Composition Outcomes (Continuous Treatment Variable)

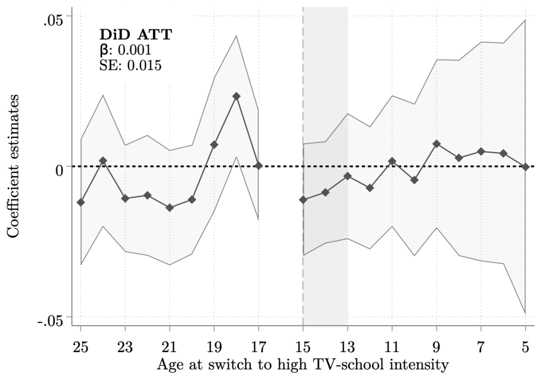


Notes: This figure shows the effect estimates of the TV-schools density by age at the first TV-school construction on different outcomes, using the continuous treatment variable. The econometric specification is a dynamic TWFE event study model that includes municipality and cohort fixed effects, and municipality-specific linear time trends. See Equation (1) for details. Event study coefficient estimates are shown with a solid line, and 95% confidence intervals are shown with a shaded area. All effects are computed with respect to age 16. The dynamic coefficient estimates are summarized with an equally-weighted average treatment on the treated single point-estimate DiD_{ATT} superposed in the graph alongside its standard error. See Section IV for details. Outcomes are: indicators for participation in the labor market and unconditional probability of earning positive income, unconditionally participating in the primary sector, secondary sector, and tertiary sector, and for receiving formal health insurance. Data is from the Mexican Secretariat of Education (SEP) for 1980–2000, and the 2010 Mexican census (10% IPUMS subsample).

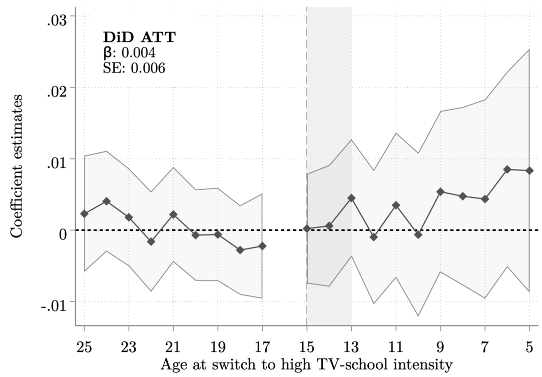
Figure A11: Event Studies of Impacts of TV-School Intensity on Cohort Population Size and Share of Females

Panel A. Excluding inter-state migrants (main sample)

(a) Log (Municipal Population)

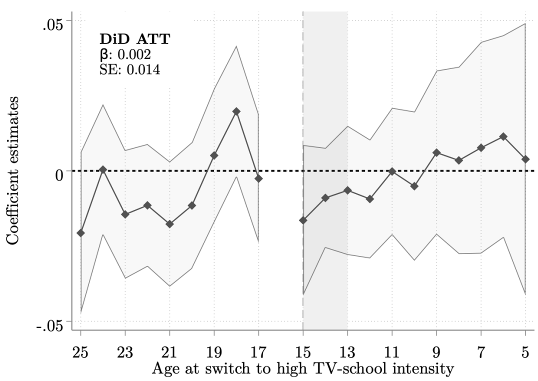


(b) Share of Females

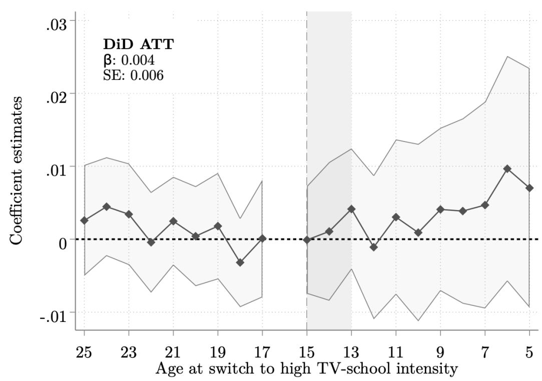


Panel B. Including inter-state migrants

(c) Log (Municipal Population)



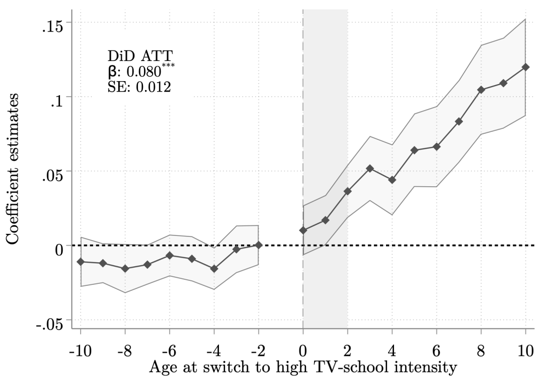
(d) Share of Females



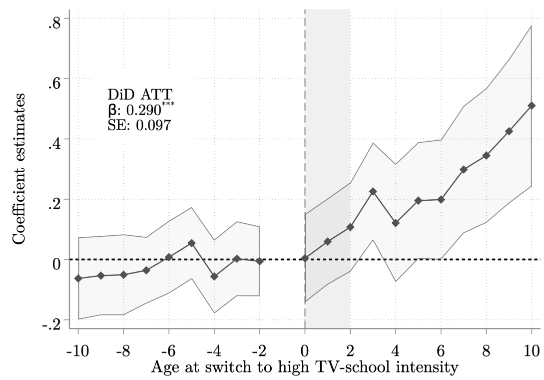
Notes: This figure shows the effect estimates of the TV-schools density by age at the first TV-school construction on different outcomes. The econometric specification is a dynamic TWFE event study model that includes municipality and cohort fixed effects, and municipality-specific linear time trends. See Equation (1) for details. Event study coefficient estimates are shown with a solid line, and 95% confidence intervals are shown with a shaded area. All effects are computed with respect to age 16. The dynamic coefficient estimates are summarized with an equally-weighted average treatment on the treated single point-estimate DiD_{ATT} superposed in the graph alongside its standard error. See Section IV for details. Outcomes are the logarithm of the population in a cohort, and the share of females in the cohort. Data is from the Mexican Secretariat of Education (SEP) for 1980–2000, and the 2010 Mexican census (10% IPUMS subsample).

Figure A12: Event Studies of Impacts of TV-School Intensity on Education and Labor Market Outcomes (1993 construction shock)

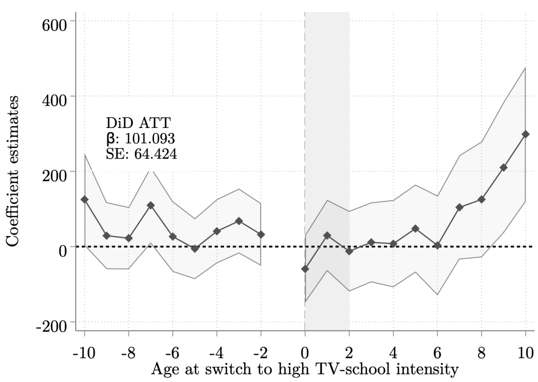
(a) Lower Secondary Education



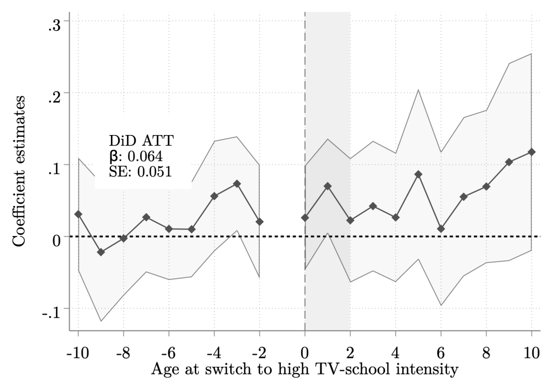
(b) Years of Education



(c) Monthly Income (pesos)



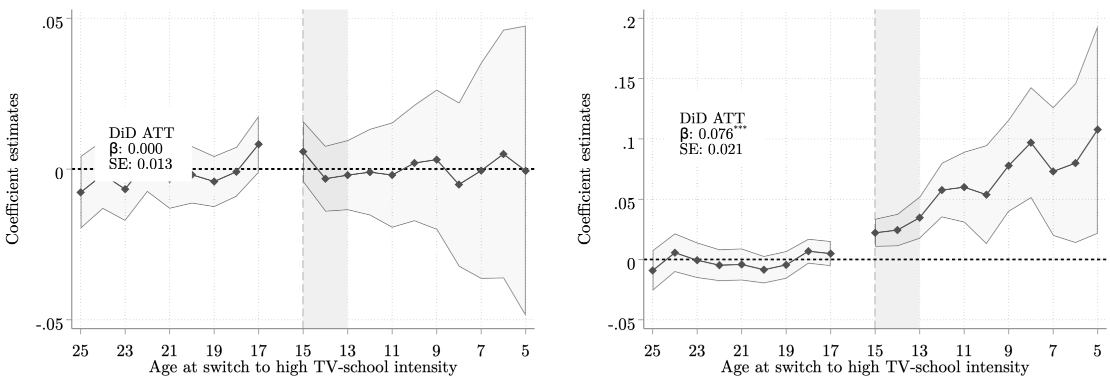
(d) Hourly Income (log)



Notes: This figure shows the effect of the TV-schools density by age from high-density construction in 1993 or 1994. The econometric specification is a dynamic TWFE event study model that includes municipality and cohort fixed effects, and municipality-specific linear time trends. See Equation (1) for details. Event study coefficient estimates are shown with a solid line, and 95% confidence intervals are shown with a shaded area. All effects are computed with respect to age 16. The dynamic coefficient estimates are summarized with an equally-weighted average treatment on the treated single point-estimate DiD_{ATT} superposed in the graph alongside its standard error. See Section IV for details. Outcomes are migration and lower secondary graduation. Data is from the Mexican Secretariat of Education (SEP), and the 2010 Mexican census (10% IPUMS subsample).

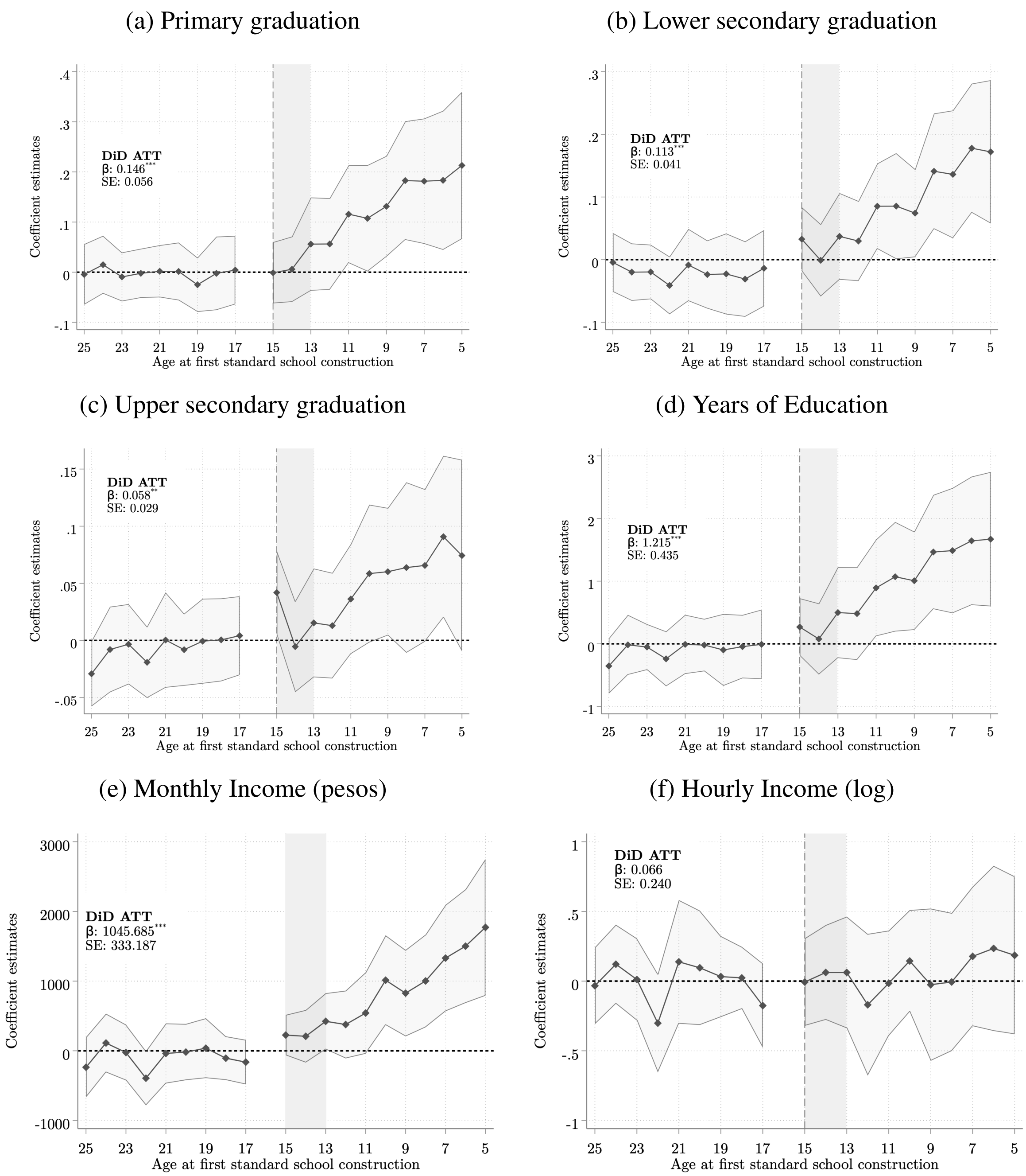
Figure A13: Event Studies of Impacts of TV-School Intensity on Migration (1995-2000) and Short-Term Effects on Education (2000 Census)

(a) Migration (1995 to 2000) (b) Lower Secondary Graduation



Notes: This figure shows the short-term effect estimates of the TV-schools density by age as measured in 2000. The econometric specification is a dynamic TWFE event study model that includes municipality and cohort fixed effects, and municipality-specific linear time trends. See Equation (1) for details. Event study coefficient estimates are shown with a solid line, and 95% confidence intervals are shown with a shaded area. All effects are computed with respect to age 16. The dynamic coefficient estimates are summarized with an equally-weighted average treatment on the treated single point-estimate DiD_{ATT} superposed in the graph alongside its standard error. See Section IV for details. Outcomes are migration and lower secondary graduation. Data is from the Mexican Secretariat of Education (SEP) for 1990–2000, and the 2000 Mexican census (10% IPUMS subsample).

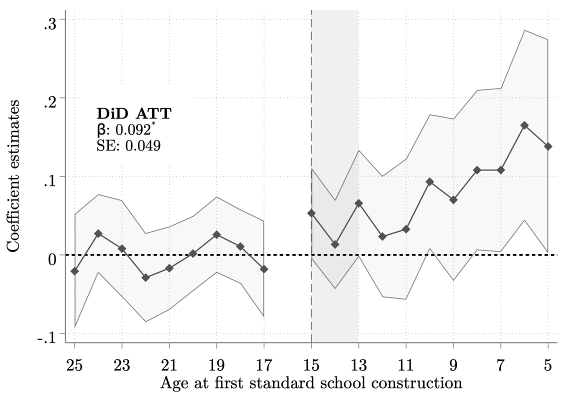
Figure A14: Event Studies of Impacts of Standard School Intensity on Education Outcomes and Income (Continuous Treatment Variable)



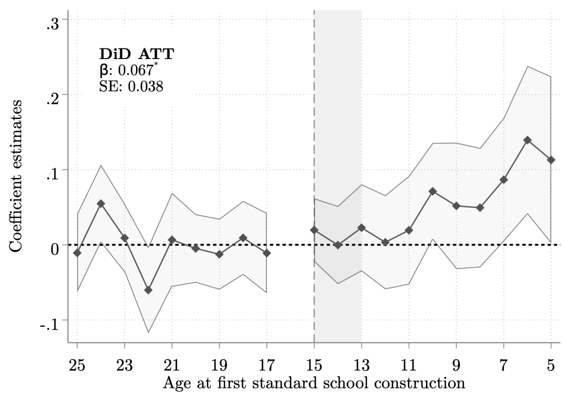
Notes: This figure shows the effect estimates of the standard school density by age at the first standard school construction on different outcomes, using the continuous treatment variable. The econometric specification is a dynamic TWFE event study model that includes municipality and cohort fixed effects, and municipality-specific linear time trends. See Equation (1) for details. Event study coefficient estimates are shown with a solid line, and 95% confidence intervals are shown with a shaded area. All effects are computed with respect to age 16. The dynamic coefficient estimates are summarized with an equally-weighted average treatment on the treated single point-estimate DiD_{ATT} superposed in the graph alongside its standard error. See Section IV for details. Outcomes are: indicators for graduating from primary, lower secondary, and upper secondary education, years of education, unconditional monthly income in pesos, and the logarithmic transformation of hourly income. Data is from the Mexican Secretariat of Education (SEP) for 1980–2000, and the 2010 Mexican census (10% IPUMS subsample).

Figure A15: Event Studies of Impacts of Standard School Intensity on Labor Market Composition Outcomes (Continuous Treatment Variable)

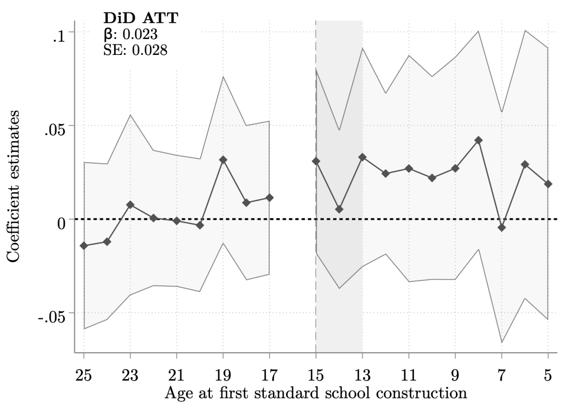
(a) Labor Force Participation



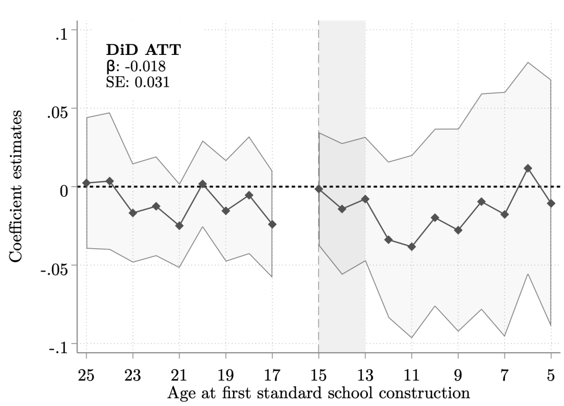
(b) Earns Positive Income



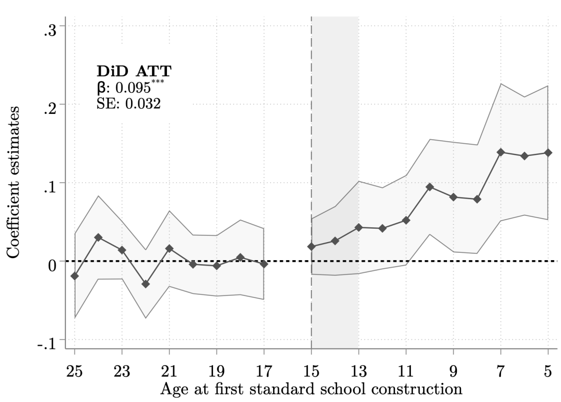
(c) Works in Primary Sector



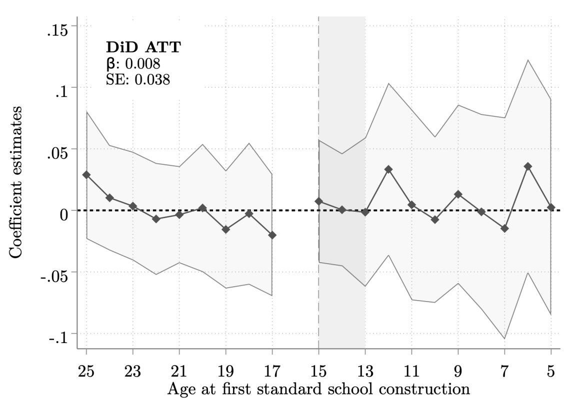
(d) Works in Secondary Sector



(e) Works in Tertiary Sector

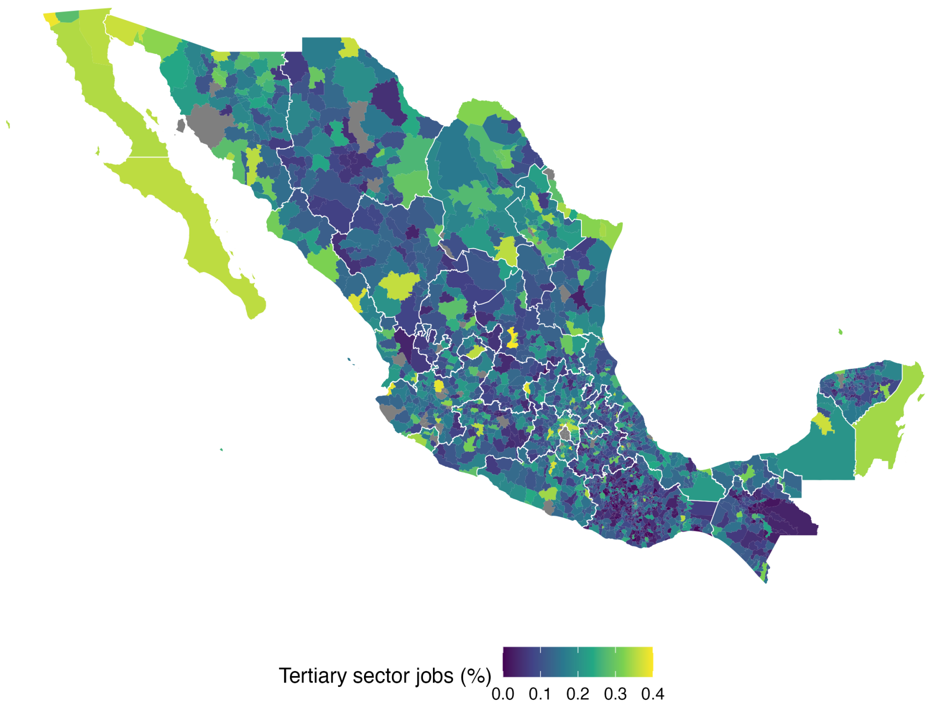


(f) Receives Formal Health Insurance



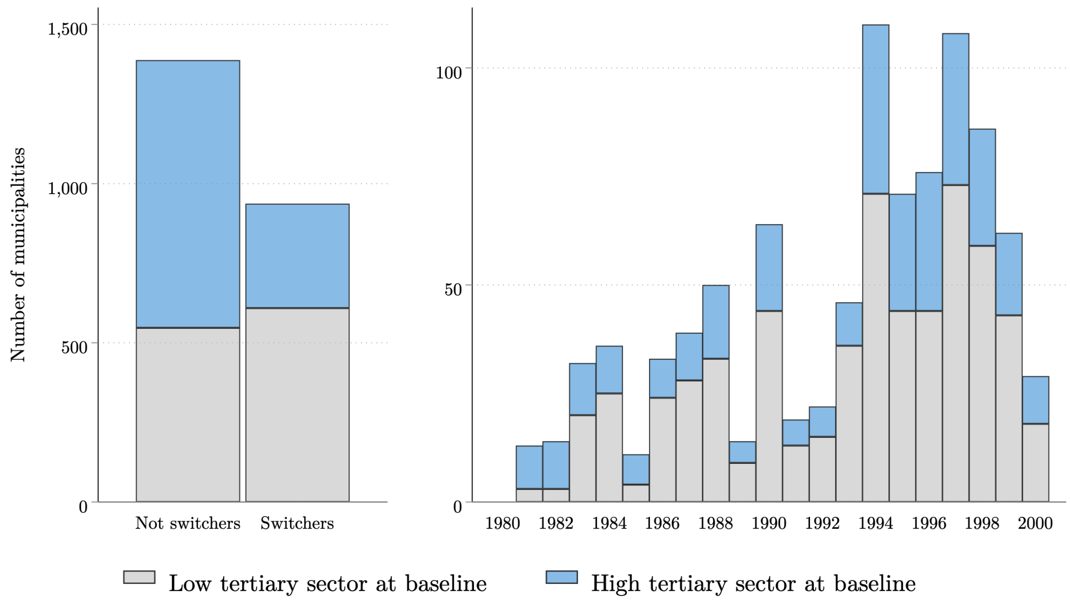
Notes: This figure shows the effect estimates of the standard school density by age at the first standard school construction on different outcomes, using the continuous treatment variable. The econometric specification is a dynamic TWFE event study model that includes municipality and cohort fixed effects, and municipality-specific linear time trends. See Equation (1) for details. Event study coefficient estimates are shown with a solid line, and 95% confidence intervals are shown with a shaded area. All effects are computed with respect to age 16. The dynamic coefficient estimates are summarized with an equally-weighted average treatment on the treated single point-estimate DiD_{ATT} superposed in the graph alongside its standard error. See Section IV for details. Outcomes are: indicators for participation in the labor market and unconditional probability of earning positive income, unconditionally participating in the primary sector, secondary sector, and tertiary sector, and for receiving formal health insurance. Data is from the Mexican Secretariat of Education (SEP) for 1980–2000, and the 2010 Mexican census (10% IPUMS subsample).

Figure A16: Prevalence of the Tertiary Sector in 1980



Notes: The maps show the share of the population working on tertiary sector jobs in 1980 in each municipality. Data comes from the 1980 Mexican census.

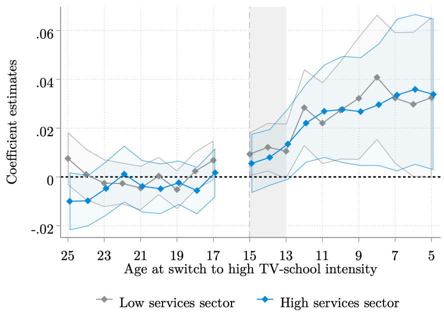
Figure A17: Evolution of TV-School Constructions by Sectoral Composition in 1980



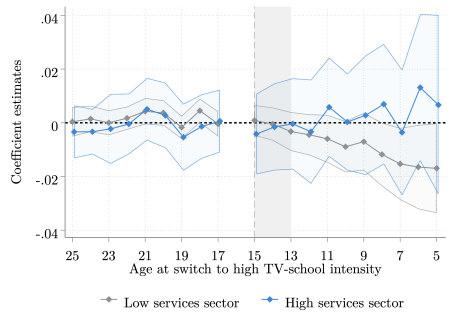
Notes: The left panel shows the number of municipalities in the analysis data that eventually switch to high TV-school intensity status (n=1,164) and those that do not (n=1,161) by prevalence of tertiary sector at baseline. The right panel presents a histogram of the switching event dates, with one observation per municipality by prevalence of tertiary sector at baseline. The data is sourced from the Mexican Secretariat of Education (SEP).

Figure A18: Event Studies of Impacts of TV-School Intensity on Outcomes by Sectoral Composition

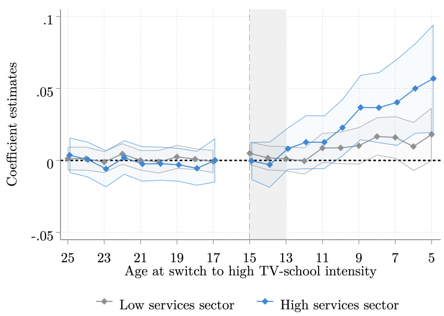
(a) Primary Graduation



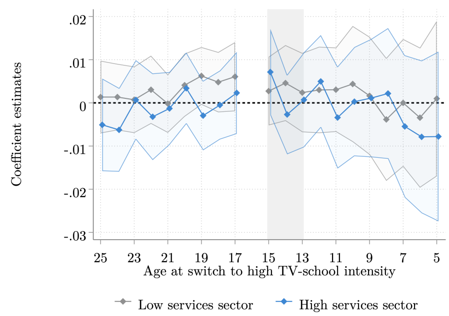
(b) Upper Secondary Graduation



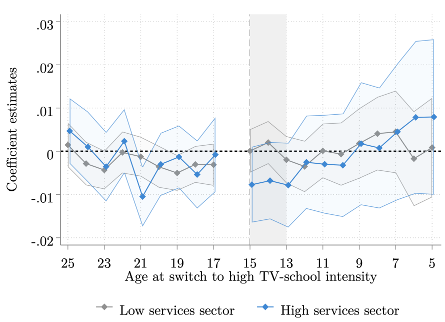
(c) Earns Positive Income



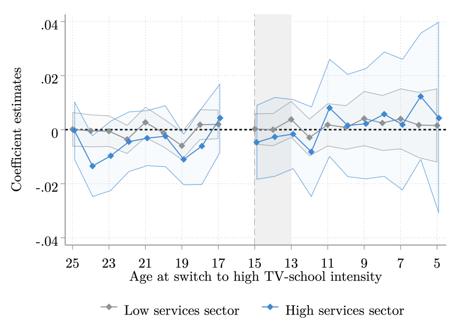
(d) Works in Primary Sector



(e) Works in Secondary Sector



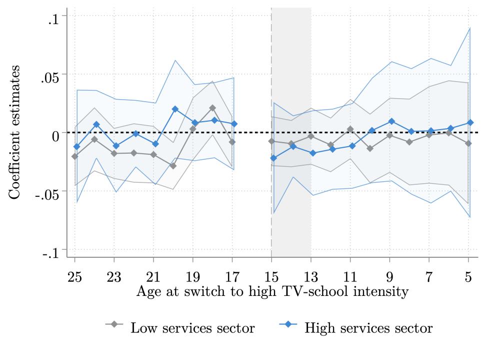
(f) Receives Formal Health Insurance



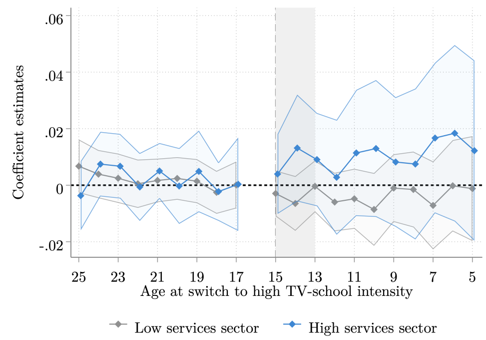
Notes: This figure shows the effect estimates of access to high TV-schools density by age at the year of switch to a high TV-school intensity construction on different outcomes. This is decomposed by the presence of services sector at baseline, defined as an indicator for whether the municipality had an above-median share of services sector jobs in 1980. The econometric specification is a dynamic TWFE event study model that includes municipality and cohort fixed effects, and municipality-specific linear time trends. See Equation (1) for details. Event study coefficient estimates are shown with a solid line, and 95% confidence intervals are shown with a shaded area. All effects are computed with respect to age 16. See Section IV for details. Outcomes are: indicators for graduating from primary and upper secondary education, for participation in the labor market and unconditional probability of earning positive income, and indicators for unconditionally participating in the primary sector and the secondary sector. Data is from the Mexican Secretariat of Education (SEP) for 1980–2000, and the 2010 Mexican census (10% IPUMS subsample).

Figure A19: Event Studies of Impacts of TV-school Intensity on Cohort Population Size and Share of Females by Sectoral Composition

(a) Log(Municipal Population)



(b) Share of Females

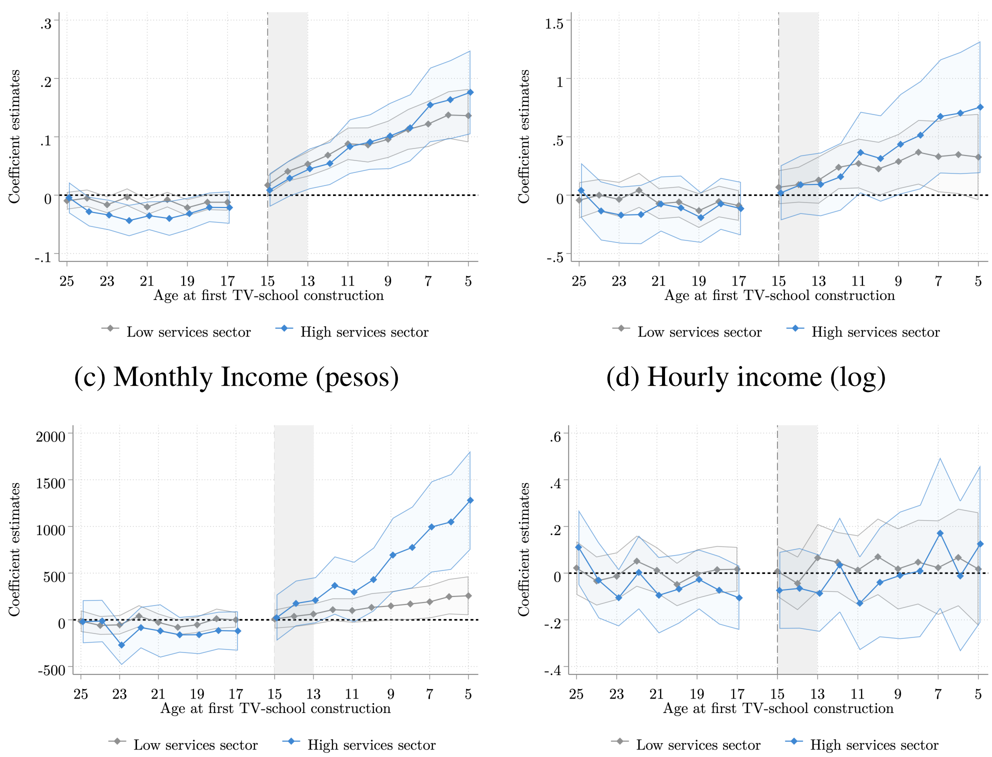


Notes: This figure shows the effect estimates of the TV-schools density by age at the first TV-school construction on different outcomes. This is decomposed by the presence of services sector at baseline. The econometric specification is a dynamic TWFE event study model that includes municipality and cohort fixed effects, and municipality-specific linear time trends. See Equation (1) for details. Event study coefficient estimates are shown with a solid line, and 95% confidence intervals are shown with a shaded area. All effects are computed with respect to age 16. The dynamic coefficient estimates are summarized with an equally-weighted average treatment on the treated single point-estimate DiD_{ATT} superposed in the graph alongside its standard error. See Section IV for details. Outcomes are the logarithm of the population in a cohort, and the share of females in the cohort. Data is from the Mexican Secretariat of Education (SEP) for 1980–2000, and the 2010 Mexican census (10% IPUMS subsample).

Figure A20: Event Studies of Impacts of TV-School Intensity on Education Outcomes and Income by Sectoral Composition

(a) Lower secondary Graduation (b) Years of Education

(c) Monthly Income (pesos) (d) Hourly income (log)



Notes: This figure shows the effect estimates of the TV-schools density by age at the first TV-school construction on different outcomes. This is decomposed by the presence of services sector at baseline, defined as an indicator for whether the municipality had an above-median share of services sector jobs in 1980. The econometric specification is a dynamic TWFE event study model that includes municipality and cohort fixed effects, and municipality-specific linear time trends. See Equation (1) for details. Event study coefficient estimates are shown with a solid line, and 95% confidence intervals are shown with a shaded area. All effects are computed with respect to age 16. See Section IV for details. Outcomes are: indicators for graduating from lower secondary education, years of education, unconditional monthly adult income in pesos, and the logarithmic transformation of hourly income. Data is from the Mexican Secretariat of Education (SEP) for 1980–2000, and the 2010 Mexican census (10% IPUMS subsample).

Table A1: Summary Statistics

	Full Sample	Treatment Status		Sectoral Composition		Standard School Density	
		Switchers	Non-Switchers	Agricultural	Services	Low Density	High Density
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Population Characteristics							
Female	0.52	0.53	0.52	0.53	0.52	0.53	0.52
	(0.06)	(0.07)	(0.06)	(0.08)	(0.06)	(0.07)	(0.06)
Age	32.25	32.55	32.16	32.43	32.22	33.02	31.96
	(7.70)	(8.88)	(7.28)	(8.61)	(7.51)	(8.63)	(7.30)
Years of Education	9.22	7.46	9.79	6.79	9.71	7.42	9.89
	(2.04)	(1.96)	(1.72)	(1.90)	(1.69)	(1.95)	(1.63)
Primary Graduation	0.85	0.74	0.89	0.69	0.89	0.74	0.90
	(0.15)	(0.19)	(0.11)	(0.20)	(0.11)	(0.19)	(0.11)
Lower Sec. Graduation	0.64	0.47	0.70	0.40	0.69	0.46	0.71
	(0.21)	(0.22)	(0.18)	(0.21)	(0.17)	(0.22)	(0.16)
Higher Sec. Graduation	0.33	0.19	0.37	0.15	0.37	0.19	0.38
	(0.17)	(0.14)	(0.15)	(0.12)	(0.15)	(0.13)	(0.15)
Labor Market Participation	0.66	0.57	0.69	0.56	0.68	0.58	0.69
	(0.11)	(0.11)	(0.10)	(0.11)	(0.10)	(0.11)	(0.10)
Earns Positive Income	0.56	0.43	0.61	0.39	0.60	0.45	0.61
	(0.15)	(0.16)	(0.12)	(0.16)	(0.13)	(0.16)	(0.13)
Monthly Income	3,091.38	1,736.57	3,530.03	1,451.23	3,422.63	1,842.32	3,558.41
	(1,603.50)	(982.03)	(1,518.16)	(915.30)	(1,506.09)	(1,036.14)	(1,526.37)
Hourly Income (log)	2.75	2.26	2.91	2.09	2.88	2.31	2.91
	(0.57)	(0.57)	(0.46)	(0.59)	(0.46)	(0.57)	(0.47)
Agricultural Sector	0.09	0.19	0.06	0.23	0.07	0.19	0.06
	(0.11)	(0.12)	(0.08)	(0.13)	(0.08)	(0.12)	(0.08)
Manufacturing Sector	0.16	0.12	0.18	0.12	0.17	0.13	0.18
	(0.08)	(0.08)	(0.08)	(0.09)	(0.08)	(0.09)	(0.08)
Services Sector	0.37	0.23	0.41	0.18	0.40	0.23	0.41
	(0.15)	(0.13)	(0.13)	(0.11)	(0.13)	(0.12)	(0.13)
Formal Health Insurance	0.41	0.21	0.47	0.16	0.46	0.22	0.48
	(0.22)	(0.15)	(0.19)	(0.14)	(0.19)	(0.16)	(0.19)
Panel B: Municipality Regional Characteristics							
Density of TV-schools	0.13	0.31	0.07	0.28	0.10	0.24	0.09
	(0.15)	(0.18)	(0.06)	(0.20)	(0.11)	(0.18)	(0.11)
Density of standard schools	0.15	0.12	0.16	0.09	0.16	0.05	0.18
	(0.10)	(0.12)	(0.09)	(0.10)	(0.10)	(0.04)	(0.10)
Rural area (1980 census)	0.38	0.97	0.19	0.91	0.27	0.76	0.23
	(0.55)	(0.50)	(0.42)	(0.43)	(0.51)	(0.50)	(0.50)
Rural area (1990 census)	0.27	0.77	0.11	0.81	0.16	0.64	0.13
	(0.44)	(0.42)	(0.31)	(0.39)	(0.36)	(0.48)	(0.34)
Lower sec. enroll. (1980 census)	0.29	0.14	0.33	0.10	0.32	0.16	0.33
	(0.14)	(0.09)	(0.12)	(0.07)	(0.12)	(0.10)	(0.12)
Poverty Index (1980 census)	-1.64	0.38	-2.30	0.74	-2.13	0.01	-2.27
	(1.79)	(1.47)	(1.34)	(1.50)	(1.42)	(1.63)	(1.42)
Municipality-cohort Count	54,450	30,091	24,359	28,406	26,044	39,205	15,245
Municipalities Count	2,325	1,164	1,161	1,156	1,169	1,614	711

Notes: This table reports the summary statistics for the population variables used in the analysis (Panel A) and for relevant municipality characteristics (Panel B) for different municipality subsamples: Full sample (col. 1), municipalities that do not or do switch to having high-density of TV-schools in 1980–2000 (col. 2-3), municipalities with below or above the median share of services sector in 1980 (col. 4-5), and municipalities with below or above the density of standard schools in 2000 (col.6-7). The population outcomes reported are an indicator for gender, age, graduation from primary, lower secondary, and higher education, participation in the labor market, unconditional probability of earning positive income, unconditional monthly adult income in pesos, logarithmic transformation of hourly income, and indicators for unconditionally participating in the primary sector, secondary sector, and tertiary sector. The region characteristics reported are the density of TV-schools, measured as the number of TV-schools per 50 eligible children, the density of standard schools, measured as the number of standard schools per 50 eligible children, indicators for being a rural municipality using the 1980 and 1990 census, municipality-level lower secondary enrollment in 1980, and the municipal poverty index in 1980. The table shows the sample mean with population weights, and the standard deviation in parentheses below. The school construction data comes from administrative records from the Mexican Secretariat of Education (SEP) for 1980–2000, and outcome data from the 10% subsample of the 2010 Mexican census from IPUMS.

Table A2: Correlates of High-Intensity TV-School Openings and Timing

	High TV-School intensity				High TV-School Intensity Switch Year			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Poverty Index	0.137***	0.117***	0.058***	0.036***	0.538***	0.616***	0.783***	0.473***
	(0.005)	(0.007)	(0.010)	(0.010)	(0.113)	(0.121)	(0.181)	(0.175)
Tertiary sector (share)		-0.693***	-0.004	-0.239		4.327	0.134	3.703
		(0.140)	(0.151)	(0.160)		(2.874)	(3.203)	(2.823)
Primary enrollment			0.183*	0.055			-6.663***	-3.977**
			(0.096)	(0.092)			(1.665)	(1.790)
Lower sec. enrollment			-0.635***	-0.876***			5.919*	6.594**
			(0.126)	(0.131)			(3.119)	(2.753)
Rural locality (share)			0.407***	0.434***			-4.713***	-2.482**
			(0.040)	(0.041)			(1.045)	(1.076)
Log(total population)				0.037***				0.425*
				(0.010)				(0.221)
State FE	No	No	No	Yes	No	No	No	Yes
R^2	0.24	0.25	0.30	0.43	0.03	0.03	0.09	0.30
Observations	2109	2109	2109	2064	850	850	850	850

Notes: This table presents associations between the characteristics of municipalities in 1980 and a high-intensity TV-school indicator (columns 1 to 4). Columns 1 to 4 regress an indicator variable denoting a high intensity of TV-school openings in the 1980-2000 period on 1980 municipal characteristics, a poverty index, the share of rural localities, primary school enrollment, tertiary sector prevalence and lower secondary school enrollment. Columns 5 to 8 restrict the sample to municipalities with high-intensity construction during this period and regress the switching year to high-intensity on those same characteristics. Standard errors in parenthesis. ***p<0.01, **p<0.05, *p<0.10.

Table A3: Comparison of the Estimates across Specifications of the Impacts of a High-Intensity of TV-Schools on Education and Labor Market Outcomes

	DiD_{TWFE}	DiD_{ATT}	DiD_{TWFE}	DiD_{ATT}	SA'21	SA'21	
	(1)	(2)	(3)	(4)	(5)	(6)	Mean
Panel A: Education Outcomes							
Lower Sec. Graduation	0.067***	0.068***	0.038***	0.079***	0.068***	0.068***	0.63
	(0.005)	(0.005)	(0.004)	(0.007)	(0.005)	(0.005)	
N	42823	45393	42823	45393	45393	43534	
Years of Education	0.543***	0.466***	0.216***	0.421***	0.460***	0.429***	9.18
	(0.045)	(0.053)	(0.036)	(0.080)	(0.060)	(0.059)	
N	42823	45393	42823	45393	45393	43534	
Primary Graduation	0.085***	0.064***	0.028***	0.040***	0.064***	0.058***	0.85
	(0.006)	(0.006)	(0.004)	(0.009)	(0.006)	(0.006)	
N	42823	45393	42823	45393	45393	43534	
Higher Sec. Graduation	-0.008***	-0.002	-0.000	0.002	-0.004	-0.002	0.32
	(0.003)	(0.003)	(0.003)	(0.005)	(0.004)	(0.004)	
N	42823	45393	42823	45393	45393	43534	
Panel B: Labor Market Outcomes							
Monthly Income	410.350***	354.612***	133.861***	364.691***	372.007***	350.873***	3110.25
	(39.938)	(36.400)	(22.857)	(43.562)	(38.557)	(38.041)	
N	42819	45389	42819	45389	45389	43530	
Hourly Income (log)	0.091***	0.091***	0.023	0.080***	0.091***	0.090***	2.75
	(0.012)	(0.016)	(0.015)	(0.026)	(0.017)	(0.017)	
N	42824	45394	42824	45394	45394	43535	
Labor Market Participation	0.016***	0.018***	0.007**	0.031***	0.019***	0.018***	0.66
	(0.003)	(0.004)	(0.003)	(0.007)	(0.004)	(0.004)	
N	42823	45393	42823	45393	45393	43534	
Earns Positive Income	0.030***	0.027***	0.011***	0.036***	0.027***	0.026***	0.56
	(0.003)	(0.004)	(0.003)	(0.006)	(0.005)	(0.005)	
N	42819	45389	42819	45389	45389	43530	
Agricultural Sector	-0.011***	-0.006**	0.000	-0.000	-0.005*	-0.004*	0.10
	(0.002)	(0.003)	(0.002)	(0.005)	(0.003)	(0.003)	
N	42824	45394	42824	45394	45394	43535	
Manufacturing Sector	0.002*	0.001	0.003*	0.006*	0.001	0.002	0.17
	(0.001)	(0.002)	(0.002)	(0.004)	(0.002)	(0.002)	
N	42824	45394	42824	45394	45394	43535	
Services Sector	0.027***	0.026***	0.006**	0.030***	0.025***	0.024***	0.36
	(0.003)	(0.004)	(0.003)	(0.005)	(0.004)	(0.004)	
N	42824	45394	42824	45394	45394	43535	
Formal Health Insurance	0.010***	0.007**	0.002	0.007	0.007*	0.006*	0.41
	(0.002)	(0.003)	(0.002)	(0.005)	(0.004)	(0.004)	
N	42823	45393	42823	45393	45393	43534	
Mun-specific trends	No	No	Yes	Yes	No	No	
Control: Prim. school density	No	No	No	No	No	Yes	

Notes: This table compares the impact of access to TV-schools on education outcomes (Panel A) and labor market outcomes (Panel B) using a binary treatment and different econometric specifications. The models are: A standard two-way fixed effects (TWFE) model, which only includes municipality and cohort fixed effects, specified as a single-coefficient estimate model (col. 1) and as an equally-weighted average treatment on the treated (ATT) of post-treatment estimates from a dynamic TWFE event study model (col. 2); a TWFE model that besides municipality and cohort fixed effects, also includes municipality-specific linear time trends, specified as a single-coefficient estimate model (col. 3) and as an equally-weighted ATT of post-treatment estimates from the dynamic TWFE event study model (col. 4), which is our preferred specification; and an heterogeneity-robust estimator proposed by Sun and Abraham (2021), without (col. 5) and with primary school density as a control (col. 6). See Equation (1) and Section IV for details on the econometric specifications. The treatment is an indicator identifying individuals with access to a high density of TV-schools in the municipality at age 15, defined as being above the sample median density, measured as the number of TV-schools per 50 eligible children. The outcomes are: an indicator for graduating from primary, lower secondary, and upper secondary education, years of education, participation in the labor market, unconditional probability of earning positive income, unconditional monthly adult income in pesos, logarithmic transformation of hourly income, and indicators for unconditionally participating in the primary sector, secondary sector, and tertiary sector, and for receiving formal health insurance. Regressions use administrative school construction data from the Mexican Secretariat of Education (SEP) for 1980–2000, and outcome data from the 10% subsample of the 2010 Mexican census from IPUMS, restricting it to non inter-state migrants. Data is aggregated into birth-year by municipality-of-residence cells, and standard errors are clustered at the municipal level. Statistical significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: Comparison of the Estimates across Specifications of the Impacts of a High-Intensity of TV-Schools on Participation in Industries

	DiD_{TWFE}	DiD_{ATT}	DiD_{TWFE}	DiD_{ATT}	SA'21	SA'21	
	(1)	(2)	(3)	(4)	(5)	(6)	Mean
Panel A: Participation in Primary Sector Industries							
Agriculture, fishing, and forestry	-0.012***	-0.006**	0.000	0.000	-0.005*	-0.004*	0.09
	(0.002)	(0.002)	(0.002)	(0.005)	(0.002)	(0.003)	
N	42824	45394	42824	45394	45394	43535	
Mining and extraction	0.001**	0.000	0.000	-0.001	0.000	-0.000	0.00
	(0.000)	(0.000)	(0.000)	(0.001)	(0.000)	(0.000)	
N	42824	45394	42824	45394	45394	43535	
Panel B: Participation in Secondary Sector Industries							
Manufacturing	-0.001	0.000	0.002*	0.007***	0.001	0.002	0.11
	(0.001)	(0.001)	(0.001)	(0.002)	(0.001)	(0.001)	
N	42824	45394	42824	45394	45394	43535	
Electricity, gas, water and waste management	0.001***	0.000	0.000	-0.000	0.000	0.000	0.00
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	
N	42824	45394	42824	45394	45394	43535	
Construction	0.002***	0.000	0.001	-0.000	-0.000	-0.000	0.06
	(0.001)	(0.001)	(0.001)	(0.003)	(0.002)	(0.002)	
N	42824	45394	42824	45394	45394	43535	
Panel C: Participation in Tertiary Sector Industries							
Wholesale and retail trade	0.004***	0.006***	0.002	0.011***	0.007***	0.006***	0.11
	(0.001)	(0.001)	(0.001)	(0.003)	(0.001)	(0.001)	
N	42824	45394	42824	45394	45394	43535	
Hotels and restaurants	0.001	0.001	-0.000	-0.001	0.001	0.000	0.03
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	
N	42824	45394	42824	45394	45394	43535	
Transportation, storage, and communications	0.003***	0.002**	0.001	0.002	0.002**	0.002**	0.03
	(0.001)	(0.001)	(0.001)	(0.002)	(0.001)	(0.001)	
N	42824	45394	42824	45394	45394	43535	
Financial services and insurance	-0.000	0.000	0.001**	0.002***	0.000	0.000*	0.01
	(0.000)	(0.000)	(0.000)	(0.001)	(0.000)	(0.000)	
N	42824	45394	42824	45394	45394	43535	
Public administration and defense	0.005***	0.006***	0.002**	0.008***	0.006***	0.006***	0.03
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	
N	42824	45394	42824	45394	45394	43535	
Business services and real state	0.002***	0.002***	0.001	0.002**	0.002***	0.002**	0.03
	(0.000)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	
N	42824	45394	42824	45394	45394	43535	
Education	0.005***	0.005***	0.001	0.007***	0.005***	0.005***	0.04
	(0.001)	(0.001)	(0.001)	(0.002)	(0.001)	(0.001)	
N	42824	45394	42824	45394	45394	43535	
Health and social work	0.002***	0.002***	0.001	0.002*	0.002***	0.002***	0.02
	(0.000)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	
N	42824	45394	42824	45394	45394	43535	
Other services	0.001*	-0.002	-0.002	-0.004	-0.003	-0.003	0.04
	(0.001)	(0.003)	(0.001)	(0.005)	(0.005)	(0.005)	
N	42824	45394	42824	45394	45394	43535	
Private household services	0.005***	0.003***	-0.000	0.000	0.003***	0.003***	0.02
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	
N	42824	45394	42824	45394	45394	43535	
Mun-specific trends	No	No	Yes	Yes	No	No	
Control: Prim. school density	No	No	No	No	No	Yes	

Notes: This table compares the impact of access to TV-schools on participation in labor market industries using a binary treatment and different econometric specifications. The models are: A standard two-way fixed effects (TWFE) model, which only includes municipality and cohort fixed effects, specified as a single-coefficient estimate model (col. 1) and as an equally-weighted average treatment on the treated (ATT) of post-treatment estimates from a dynamic TWFE event study model (col. 2); a TWFE model that besides municipality and cohort fixed effects, also includes municipality-specific linear time trends, specified as a single-coefficient estimate model (col. 3) and as an equally-weighted ATT of post-treatment estimates from the dynamic TWFE event study model (col. 4), which is our preferred specification; and an heterogeneity-robust estimator proposed by Sun and Abraham (2021), without (col. 5) and with primary school density as a control (col. 6). See Equation (1) and Section IV for details on the econometric specifications. The treatment is an indicator identifying individuals with access to a high density of TV-schools in the municipality at age 15, defined as being above the sample median density, measured as the number of TV-schools per 50 eligible children. The outcomes are indicators for the (unconditional) participation in different labor market industries, specified in the row names. Regressions use administrative school construction data from the Mexican Secretariat of Education (SEP) for 1980–2000, and outcome data from the 10% subsample of the 2010 Mexican census from IPUMS, restricting it to non inter-state migrants. Data is aggregated into birth-year by municipality-of-residence cells, and standard errors are clustered at the municipal level. Statistical significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Comparison of the Estimates across Specifications of the Impacts of the Density of TV-Schools on Education and Labor Market Outcomes

	DiD_{TWFE}	DiD_{ATT}	DiD_{TWFE}	DiD_{ATT}	Mean
	(1)	(2)	(3)	(4)	
Panel A: Education Outcomes					
Lower Sec. Graduation	0.212***	0.077***	0.159***	0.128***	0.60
	(0.014)	(0.011)	(0.011)	(0.014)	
N	43026	47511	43026	47511	
Years of Education	1.578***	0.598***	0.770***	0.410***	8.93
	(0.115)	(0.069)	(0.101)	(0.108)	
N	43026	47511	43026	47511	
Primary Graduation	0.227***	0.092***	0.091***	0.047***	0.83
	(0.016)	(0.012)	(0.014)	(0.017)	
N	43026	47511	43026	47511	
Higher Sec. Graduation	-0.006	-0.014**	-0.011*	-0.005	0.31
	(0.006)	(0.005)	(0.007)	(0.009)	
N	43026	47511	43026	47511	
Panel B: Labor Market Outcomes					
Monthly Income	1065.834***	342.939***	412.587***	468.063***	3214.74
	(92.662)	(51.983)	(63.285)	(87.321)	
N	43026	47509	43026	47509	
Hourly Income (log)	0.336***	0.099***	0.114**	0.041	2.78
	(0.028)	(0.035)	(0.045)	(0.070)	
N	43027	47512	43027	47512	
Labor Market Participation	0.036***	0.008	0.040***	0.038***	0.66
	(0.006)	(0.007)	(0.008)	(0.013)	
N	43027	47511	43027	47511	
Earns Positive Income	0.081***	0.025***	0.039***	0.052***	0.57
	(0.007)	(0.007)	(0.008)	(0.012)	
N	43026	47509	43026	47509	
Agricultural Sector	-0.037***	-0.011*	0.004	-0.007	0.10
	(0.004)	(0.006)	(0.006)	(0.011)	
N	43027	47511	43027	47511	
Manufacturing Sector	0.001	-0.007*	0.010**	0.010	0.17
	(0.003)	(0.004)	(0.005)	(0.007)	
N	43027	47511	43027	47511	
Services Sector	0.078***	0.026***	0.029***	0.038***	0.36
	(0.006)	(0.005)	(0.006)	(0.010)	
N	43027	47511	43027	47511	
Formal Health Insurance	0.032***	0.005	0.009*	-0.016*	0.41
	(0.004)	(0.005)	(0.005)	(0.009)	
N	43027	47512	43027	47512	
Mun-specific trends	No	No	Yes	Yes	

Notes: This table compares the impact of access to TV-schools on education outcomes (Panel A) and labor market outcomes (Panel B) using a continuous treatment and different econometric specifications. The models are: A standard two-way fixed effects (TWFE) model, which only includes municipality and cohort fixed effects, specified as a single-coefficient estimate model (col. 1) and as an equally-weighted average treatment on the treated (ATT) of post-treatment estimates from a dynamic TWFE event study model (col. 2); a TWFE model that besides municipality and cohort fixed effects, also includes municipality-specific linear time trends, specified as a single-coefficient estimate model (col. 3) and as an equally-weighted ATT of post-treatment estimates from the dynamic TWFE event study model (col. 4), which is our preferred specification. See Equation (1) and Section IV for details on the econometric specifications. The continuous treatment is school density in the municipality at age 15, measured as the number of TV-schools per 50 eligible children. The outcomes are: an indicator for graduating from primary, lower secondary, and upper secondary education, years of education, participation in the labor market, unconditional probability of earning positive income, unconditional monthly adult income in pesos, logarithmic transformation of hourly income, and indicators for unconditionally participating in the primary sector, secondary sector, and tertiary sector, and for receiving formal health insurance. Regressions use administrative school construction data from the Mexican Secretariat of Education (SEP) for 1980–2000, and outcome data from the 10% subsample of the 2010 Mexican census from IPUMS, restricting it to non inter-state migrants. Data is aggregated into birth-year by municipality-of-residence cells, and standard errors are clustered at the municipal level. Statistical significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Comparison of the Estimates across Specifications of the Impacts of a High-Intensity of TV-Schools on Education and Labor Market Outcomes, with the Sample Including Inter-state Migrants

	DiD_{TWFE}	DiD_{ATT}	DiD_{TWFE}	DiD_{ATT}	SA'21	SA'21	
	(1)	(2)	(3)	(4)	(5)	(6)	Mean
Panel A: Education Outcomes							
Lower Sec. Graduation	0.064***	0.064***	0.035***	0.074***	0.063***	0.062***	0.64
	(0.005)	(0.006)	(0.004)	(0.008)	(0.006)	(0.006)	
N	42831	45402	42831	45402	45402	43542	
Years of Education	0.524***	0.440***	0.203***	0.408***	0.428***	0.395***	9.27
	(0.052)	(0.070)	(0.039)	(0.095)	(0.072)	(0.072)	
N	42831	45402	42831	45402	45402	43542	
Primary Graduation	0.082***	0.061***	0.026***	0.039***	0.061***	0.055***	0.86
	(0.006)	(0.006)	(0.004)	(0.009)	(0.006)	(0.006)	
N	42831	45402	42831	45402	45402	43542	
Higher Sec. Graduation	-0.007**	-0.003	-0.000	0.002	-0.005	-0.004	0.33
	(0.003)	(0.004)	(0.003)	(0.007)	(0.005)	(0.005)	
N	42831	45402	42831	45402	45402	43542	
Panel B: Labor Market Outcomes							
Monthly Income	414.114***	355.308***	133.538***	378.144***	370.981***	348.239***	3324.97
	(48.086)	(40.543)	(24.334)	(44.622)	(43.431)	(42.667)	
N	42827	45398	42827	45398	45398	43538	
Hourly Income (log)	0.086***	0.088***	0.019	0.083***	0.089***	0.087***	2.82
	(0.012)	(0.016)	(0.015)	(0.025)	(0.016)	(0.016)	
N	42831	45402	42831	45402	45402	43542	
Labor Market Participation	0.015***	0.017***	0.006*	0.030***	0.018***	0.017***	0.67
	(0.003)	(0.004)	(0.003)	(0.007)	(0.004)	(0.004)	
N	42830	45401	42830	45401	45401	43541	
Earns Positive Income	0.029***	0.027***	0.011***	0.034***	0.027***	0.026***	0.58
	(0.003)	(0.004)	(0.003)	(0.006)	(0.004)	(0.004)	
N	42827	45398	42827	45398	45398	43538	
Agricultural Sector	-0.011***	-0.006**	-0.001	-0.000	-0.005**	-0.005**	0.09
	(0.002)	(0.002)	(0.002)	(0.004)	(0.002)	(0.002)	
N	42831	45402	42831	45402	45402	43542	
Manufacturing Sector	0.002*	0.001	0.003*	0.006*	0.001	0.002	0.17
	(0.001)	(0.002)	(0.002)	(0.004)	(0.002)	(0.002)	
N	42824	45394	42824	45394	45394	43535	
Services Sector	0.026***	0.024***	0.004	0.027***	0.024***	0.022***	0.37
	(0.003)	(0.004)	(0.003)	(0.006)	(0.004)	(0.004)	
N	42831	45402	42831	45402	45402	43542	
Formal Health Insurance	0.008***	0.006*	0.000	0.005	0.005	0.005	0.43
	(0.002)	(0.003)	(0.003)	(0.006)	(0.004)	(0.004)	
N	42830	45401	42830	45401	45401	43541	
Mun-specific trends	No	No	Yes	Yes	No	No	
Control: Prim. school density	No	No	No	No	No	Yes	

Notes: This table compares the impact of access to TV-schools on education outcomes (Panel A) and labor market outcomes (Panel B) using a binary treatment and different econometric specifications. The models are: A standard two-way fixed effects (TWFE) model, which only includes municipality and cohort fixed effects, specified as a single-coefficient estimate model (col. 1) and as an equally-weighted average treatment on the treated (ATT) of post-treatment estimates from a dynamic TWFE event study model (col. 2); a TWFE model that besides municipality and cohort fixed effects, also includes municipality-specific linear time trends, specified as a single-coefficient estimate model (col. 3) and as an equally-weighted ATT of post-treatment estimates from the dynamic TWFE event study model (col. 4), which is our preferred specification; and an heterogeneity-robust estimator proposed by Sun and Abraham (2021), without (col. 5) and with primary school density as a control (col. 6). See Equation (1) and Section IV for details on the econometric specifications. The treatment is an indicator identifying individuals with access to a high density of TV-schools in the municipality at age 15, defined as being above the sample median density, measured as the number of TV-schools per 50 eligible children. The outcomes are: an indicator for graduating from primary, lower secondary, and higher education, years of education, participation in the labor market, unconditional probability of earning positive income, unconditional monthly adult income in pesos, logarithmic transformation of hourly income, and indicators for unconditionally participating in the primary sector, secondary sector, and tertiary sector, and for receiving formal health insurance. Regressions use administrative school construction data from the Mexican Secretariat of Education (SEP) for 1980–2000, and outcome data from the 10% subsample of the 2010 Mexican census from IPUMS, with the sample including inter-state migrants. Data is aggregated into birth-year by municipality-of-residence cells, and standard errors are clustered at the municipal level. Statistical significance levels: $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$.

Table A7: Robustness to Alternative Samples and Specifications

	Main	Control std. school	Exclude Never TS	Exclude low TS	PSM sample
	(1)	(2)	(3)	(4)	(5)
Primary Graduation	0.040***	0.040***	0.037***	0.047***	0.034*
	(0.009)	(0.009)	(0.009)	(0.009)	(0.019)
Lower Sec. Graduation	0.079***	0.078***	0.075***	0.068***	0.117***
	(0.007)	(0.007)	(0.008)	(0.008)	(0.022)
Years of Education	0.421***	0.425***	0.400***	0.316***	0.673***
	(0.080)	(0.078)	(0.081)	(0.085)	(0.182)
Labor Market Participation	0.031***	0.033***	0.031***	0.014**	0.045*
	(0.007)	(0.007)	(0.007)	(0.007)	(0.027)
Monthly Income	364.691***	369.728***	359.909***	221.083***	409.475**
	(43.562)	(41.344)	(45.498)	(45.913)	(186.790)
Hourly Income (log)	0.080***	0.085***	0.076***	0.046	0.159**
	(0.026)	(0.026)	(0.026)	(0.028)	(0.074)
Mun-specific trends	Yes	Yes	Yes	Yes	Yes

Notes: This table shows the impact of high-density school construction for different samples and including controls. Column (1) shows the main estimates for reference. Column (2) controls for the concurrent growth of standard schools (standard school density interacted with year-fixed effects. Column (3) excludes municipalities that never experienced TV-School construction. Column (4) excludes municipalities that experienced construction o fat least one TV-school, but not enough to switch to high density. Column (5) uses a matched sample of municipalities to estimate effects. The table presents our preferred specification, the equally-weighted average treatment on the treated (ATT) of post-treatment estimates from the dynamic TWFE event study model, DiD_{ATT}, also including municipality and cohort fixed effects, and municipality-specific linear time trends. See Equation (1) and Section IV for details on the econometric specifications. The treatment is an indicator identifying individuals with access to a high density of schools in the municipality at age 15, defined as being above the sample median density, measured as the number of schools per 50 eligible children. The outcomes are: an indicator for graduating from lower secondary education, years of education, the unconditional monthly adult income in pesos, and logarithmic transformation of hourly income. Regressions use administrative school construction data from the Mexican Secretariat of Education for 1980–2000, and outcome data from the 10% subsample of the 2010 Mexican census from IPUMS, restricting it to non inter-state migrants. Data is aggregated into birth-year by municipality-of-residence cells, and standard errors are clustered at the municipal level. Statistical significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Impacts of Access to High-Intensity of Schools on Educational Achievement and Labor Income by Type of School

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Lower Sec. Grad.		Years of Education		Monthly Income (pesos)		Hourly Income (log)	
	DiD_{TWFE}	DiD_{ATT}	DiD_{TWFE}	DiD_{ATT}	DiD_{TWFE}	DiD_{ATT}	DiD_{TWFE}	DiD_{ATT}
School Type: TV-Schools								
High TV-School Intensity	0.038***	0.079***	0.216***	0.421***	133.861***	364.691***	0.023	0.080***
	(0.004)	(0.007)	(0.036)	(0.080)	(22.857)	(43.562)	(0.015)	(0.026)
Dependent Mean	0.64	0.63	9.23	9.18	3151.06	3110.25	2.77	2.75
Observations	42823	45393	42823	45393	42819	45389	42824	45394
School Type: Standard Schools								
High Std. School Intensity	0.013**	-0.009	0.098**	-0.004	56.548	-172.576*	0.022	0.017
	(0.006)	(0.009)	(0.047)	(0.088)	(50.018)	(88.732)	(0.016)	(0.026)
Dependent Mean	0.56	0.57	8.53	8.60	3070.94	3141.67	2.74	2.76
Observations	39864	40962	39864	40962	39861	40959	39865	40963

Notes: This table shows the impact of access to schooling by school type (standard schools and TV-schools). Odd columns present single-coefficient estimates from a two-way fixed effects (TWFE) model, DiD_{TWFE}, which includes municipality and cohort fixed effects, and municipality-specific linear time trends. Even columns present our preferred specification, the equally-weighted average treatment on the treated (ATT) of post-treatment estimates from the dynamic TWFE event study model, DiD_{ATT}, also including municipality and cohort fixed effects, and municipality-specific linear time trends. See Equation (1) and Section IV for details on the econometric specifications. The treatment is an indicator identifying individuals with access to a high density of schools in the municipality at age 15, defined as being above the sample median density, measured as the number of schools per 50 eligible children. The outcomes are: an indicator for graduating from lower secondary education (col. 1-2), years of education (col. 3-4), the unconditional monthly adult income in pesos (col. 5-6), and logarithmic transformation of hourly income (col. 7-8). Regressions use administrative school construction data from the Mexican Secretariat of Education for 1980–2000, and outcome data from the 10% subsample of the 2010 Mexican census from IPUMS, restricting it to non inter-state migrants. Data is aggregated into birth-year by municipality-of-residence cells, and standard errors are clustered at the municipal level. Statistical significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A9: Heterogenous Impacts of Access to High-Intensity of TV-Schools on Educational Outcomes by Services Sector Presence

	Difference in heterogeneity estimates				DiD_{ATT} estimates	
	DiD_{TWFE}	DiD_{ATT}	DiD_{TWFE}	DiD_{ATT}	Low services	High services
	(1)	(2)	(3)	(4)	(5)	(6)
Primary Graduation	0.016	0.011	-0.007	-0.001	0.031***	0.030***
	(0.011)	(0.012)	(0.008)	(0.017)	(0.012)	(0.012)
p-value	0.159	0.346	0.402	0.946	0.010	0.015
N	42823	45393	42823	45393	45393	45393
Lower Sec. Graduation	0.034***	0.029***	0.001	0.003	0.060***	0.063***
	(0.008)	(0.010)	(0.008)	(0.015)	(0.009)	(0.012)
p-value	0.000	0.003	0.911	0.842	0.000	0.000
N	42823	45393	42823	45393	45393	45393
Higher Sec. Graduation	0.010	0.010	0.002	0.014	-0.011**	0.004
	(0.006)	(0.008)	(0.006)	(0.012)	(0.005)	(0.011)
p-value	0.100	0.195	0.683	0.242	0.047	0.746
N	42823	45393	42823	45393	45393	45393
Years of Education	0.174**	0.134	-0.024	0.064	0.265***	0.330**
	(0.076)	(0.099)	(0.068)	(0.170)	(0.079)	(0.150)
p-value	0.023	0.176	0.719	0.705	0.001	0.028
N	42823	45393	42823	45393	45393	45393
Mun-specific trends	No	No	Yes	Yes	Yes	Yes

Notes: This table shows the impact of access to high-intensity of TV-schools on educational outcomes by the presence of services sector at baseline, defined as an indicator for whether the municipality had an above median share of services sector jobs in 1980. The first set of coefficient estimates are the interaction term between the treatment coefficient and the sectoral presence indicator from a fully-interacted single-coefficient two-way fixed effects (TWFE) model, DiD_{TWFE}, which includes municipality and cohort fixed effects (col .1), and additionally includes municipality-specific linear time trends (col. 3). The second set of coefficient estimates present the test of the linear combination of the equally-weighted average treatment on the treated (ATT) of post-treatment estimates by the sectoral presence indicator, coming from the dynamic TWFE event study model, DiD_{ATT}, also including municipality and cohort fixed effects (col. 2), and additionally including municipality-specific linear time trends (col. 4). See Equation (1) and Section IV for details on the econometric specifications. The last set of coefficients are the equally-weighted average treatment on the treated (ATT) of post-treatment estimates from the dynamic TWFE event study model, DiD_{ATT}, including municipality and cohort fixed effects, and municipality-specific linear time trends, ran as split-sample regressions by below-median services sector at baseline (col. 5) and above-median services sector at baseline (col. 6). The treatment is an indicator identifying individuals with access to a high density of TV-schools in the municipality at age 15, defined as being above the sample median density, measured as the number of TV-schools per 50 eligible children. The outcomes are: an indicator for graduating from primary, lower secondary, and upper secondary education, and years of education. Regressions use administrative school construction data from the Mexican Secretariat of Education for 1980–2000, and outcome data from the 10% subsample of the 2010 Mexican census from IPUMS, restricting it to non inter-state migrants. Data is aggregated into birth-year by municipality-of-residence cells, and standard errors are clustered at the municipal level. Statistical significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A10: Heterogenous Impacts of Access to High-Intensity of TV-Schools on Labor Outcomes by Services Sector Presence

	Difference in heterogeneity estimates				DiD_{ATT} estimates	
	DiD_{TWFE}	DiD_{ATT}	DiD_{TWFE}	DiD_{ATT}	Low services	High services
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Labor Outcomes						
Labor Market Participation	0.017***	0.016**	0.006	0.026*	0.005	0.031***
	(0.006)	(0.008)	(0.007)	(0.013)	(0.007)	(0.011)
p-value	0.003	0.033	0.374	0.054	0.506	0.006
N	42823	45393	42823	45393	45393	45393
Earns Positive Income	0.027***	0.026***	0.012*	0.023*	0.011*	0.034***
	(0.006)	(0.009)	(0.007)	(0.014)	(0.006)	(0.012)
p-value	0.000	0.003	0.074	0.095	0.076	0.005
N	42819	45389	42819	45389	45389	45389
Monthly Income	400.694***	402.875***	110.590**	282.069***	121.131***	403.200***
	(66.380)	(75.379)	(44.303)	(89.496)	(36.881)	(81.544)
p-value	0.000	0.000	0.013	0.002	0.001	0.000
N	42819	45389	42819	45389	45389	45389
Hourly Income (log)	0.063***	0.069**	-0.012	0.083	0.018	0.102**
	(0.023)	(0.031)	(0.030)	(0.054)	(0.035)	(0.041)
p-value	0.007	0.024	0.679	0.125	0.601	0.014
N	42824	45394	42824	45394	45394	45394
Panel B: Labor Sectoral Composition						
Agricultural Sector	-0.010***	-0.016***	0.001	-0.003	0.001	-0.002
	(0.003)	(0.005)	(0.004)	(0.009)	(0.006)	(0.007)
p-value	0.002	0.001	0.781	0.772	0.911	0.770
N	42824	45394	42824	45394	45394	45394
Manufacturing Sector	0.006**	0.009**	0.000	0.001	0.001	0.002
	(0.003)	(0.004)	(0.004)	(0.008)	(0.004)	(0.007)
p-value	0.028	0.016	0.965	0.889	0.862	0.791
N	42824	45394	42824	45394	45394	45394
Services Sector	0.024***	0.024***	0.007	0.026**	0.007	0.033***
	(0.005)	(0.007)	(0.006)	(0.012)	(0.005)	(0.011)
p-value	0.000	0.001	0.236	0.032	0.161	0.003
N	42824	45394	42824	45394	45394	45394
Formal Health Insurance	0.005	0.000	0.001	0.002	0.002	0.003
	(0.004)	(0.006)	(0.005)	(0.012)	(0.005)	(0.011)
p-value	0.149	0.958	0.858	0.881	0.718	0.752
N	42823	45393	42823	45393	45393	45393
Mun-specific trends	No	No	Yes	Yes	Yes	Yes

Notes: This table shows the impact of access to high-intensity of TV-schools on labor market outcomes by the presence of services sector at baseline, defined as an indicator for whether the municipality had an above median share of services sector jobs in 1980. The first set of coefficient estimates are the interaction term between the treatment coefficient and the sectoral presence indicator from a fully-interacted single-coefficient two-way fixed effects (TWFE) model, DiD_{TWFE}, which includes municipality and cohort fixed effects (col .1), and additionally includes municipality-specific linear time trends (col. 3). The second set of coefficient estimates present the test of the linear combination of the equally-weighted average treatment on the treated (ATT) of post-treatment estimates by the sectoral presence indicator, coming from the dynamic TWFE event study model, DiD_{ATT}, also including municipality and cohort fixed effects (col. 2), and additionally including municipality-specific linear time trends (col. 4). See Equation (1) and Section IV for details on the econometric specifications. The last set of coefficients are the equally-weighted average treatment on the treated (ATT) of post-treatment estimates from the dynamic TWFE event study model, DiD_{ATT}, including municipality and cohort fixed effects, and municipality-specific linear time trends, ran as split-sample regressions by below-median services sector at baseline (col. 5) and above-median services sector at baseline (col. 6). The treatment is an indicator identifying individuals with access to a high density of TV-schools in the municipality at age 15, defined as being above the sample median density, measured as the number of TV-schools per 50 eligible children. The outcomes are: an indicator for participation in the labor market, unconditional probability of earning positive income, unconditional monthly adult income in pesos, logarithmic transformation of hourly income, and indicators for unconditionally participating in the primary sector, secondary sector, and tertiary sector, and for receiving formal health insurance. Regressions use administrative school construction data from the Mexican Secretariat of Education for 1980–2000, and outcome data from the 10% subsample of the 2010 Mexican census from IPUMS, restricting it to non inter-state migrants. Data is aggregated into birth-year by municipality-of-residence cells, and standard errors are clustered at the municipal level. Statistical significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A11: Heterogenous Impacts of the Density of TV-schools on Educational Outcomes by Services Sector Presence

	Difference in heterogeneity estimates				DiD_{ATT} **estimates**	
	DiD_{TWFE}	DiD_{ATT}	DiD_{TWFE}	DiD_{ATT}	Low services	High services
	(1)	(2)	(3)	(4)	(5)	(6)
Primary Graduation	0.090**	0.092***	-0.043*	0.017	0.035	0.052*
	(0.036)	(0.026)	(0.023)	(0.035)	(0.022)	(0.027)
p-value	0.013	0.000	0.066	0.636	0.110	0.056
N	43026	47511	43026	47511	47511	47511
Lower Sec. Graduation	0.062**	-0.008	-0.006	0.012	0.106***	0.117****
	(0.026)	(0.024)	(0.023)	(0.031)	(0.016)	(0.027)
p-value	0.019	0.739	0.798	0.709	0.000	0.000
N	43026	47511	43026	47511	47511	47511
Higher Sec. Graduation	0.008	-0.013	0.001	0.014	-0.013	0.001
	(0.016)	(0.014)	(0.015)	(0.020)	(0.009)	(0.018)
p-value	0.610	0.346	0.957	0.500	0.163	0.969
N	43026	47511	43026	47511	47511	47511
Years of Education	0.589**	0.470***	-0.224	0.190	0.299**	0.490**
	(0.237)	(0.164)	(0.182)	(0.242)	(0.126)	(0.206)
p-value	0.013	0.004	0.218	0.431	0.018	0.018
N	43026	47511	43026	47511	47511	47511
Mun-specific trends	No	No	Yes	Yes	Yes	Yes

Notes: This table shows the impact of the density of TV-schools on educational outcomes by the presence of services sector at baseline, defined as an indicator for whether the municipality had an above median share of services sector jobs in 1980. The first set of coefficient estimates are the interaction term between the treatment coefficient and the sectoral presence indicator from a fully-interacted single-coefficient two-way fixed effects (TWFE) model, DiD_{TWFE}, which includes municipality and cohort fixed effects (col .1), and additionally includes municipality-specific linear time trends (col. 3). The second set of coefficient estimates present the test of the linear combination of the equally-weighted average treatment on the treated (ATT) of post-treatment estimates by the sectoral presence indicator, coming from the dynamic TWFE event study model, DiD_{ATT}, also including municipality and cohort fixed effects (col. 2), and additionally including municipality-specific linear time trends (col. 4). See Equation (1) and Section IV for details on the econometric specifications. The last set of coefficients are the equally-weighted average treatment on the treated (ATT) of post-treatment estimates from the dynamic TWFE event study model, DiD_{ATT}, including municipality and cohort fixed effects, and municipality-specific linear time trends, ran as split-sample regressions by below-median services sector at baseline (col. 5) and above-median services sector at baseline (col. 6). The continuous treatment is TV-school density in the municipality at age 15, measured as the number of TV-schools per 50 eligible children. The outcomes are: an indicator for graduating from primary, lower secondary, and upper secondary education, and years of education. Regressions use administrative school construction data from the Mexican Secretariat of Education for 1980–2000, and outcome data from the 10% subsample of the 2010 Mexican census from IPUMS, restricting it to non inter-state migrants. Data is aggregated into birth-year by municipality-of-residence cells, and standard errors are clustered at the municipal level. Statistical significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A12: Heterogenous Impacts of the Density of TV-Schools on Labor Market Outcomes by Services Sector Presence

	Difference in heterogeneity estimates				**DiD ATT estimates**	
	DiD_{TWFE}	DiD_{ATT}	DiD_{TWFE}	DiD_{ATT}	Low services	High services
	(1)	(2)	(3)	(4)	(5)	(6)
			Panel A: Labor Outcomes			
Labor Market Participation	0.035***	0.006	0.020	0.018	0.025	0.043*
	(0.012)	(0.013)	(0.017)	(0.029)	(0.016)	(0.024)
p-value	0.004	0.669	0.237	0.528	0.123	0.075
N	43027	47511	43027	47511	47511	47511
Earns Positive Income	0.069***	0.044***	0.026	0.028	0.034**	0.062***
	(0.014)	(0.015)	(0.017)	(0.027)	(0.014)	(0.023)
p-value	0.000	0.003	0.129	0.297	0.013	0.007
N	43026	47509	43026	47509	47509	47509
Monthly Income	1157.376***	744.696***	242.246*	566.161***	169.974**	736.136***
	(167.465)	(131.511)	(142.819)	(216.262)	(76.143)	(202.414)
p-value	0.000	0.000	0.090	0.009	0.026	0.000
N	43026	47509	43026	47509	47509	47509
Hourly Income (log)	0.215***	0.083	0.097	-0.019	0.038	0.019
	(0.057)	(0.067)	(0.096)	(0.155)	(0.086)	(0.129)
p-value	0.000	0.216	0.312	0.902	0.659	0.884
N	43027	47512	43027	47512	47512	47512
			Panel B: Labor Sectoral Composition			
Agricultural Sector	-0.029***	-0.034***	-0.010	-0.029	0.005	-0.024
	(0.009)	(0.010)	(0.013)	(0.022)	(0.014)	(0.017)
p-value	0.002	0.001	0.421	0.198	0.714	0.171
N	43027	47511	43027	47511	47511	47511
Manufacturing Sector	0.004	-0.001	-0.005	0.015	0.003	0.018
	(0.007)	(0.008)	(0.010)	(0.016)	(0.008)	(0.014)
p-value	0.611	0.889	0.630	0.347	0.721	0.189
N	43027	47511	43027	47511	47511	47511
Services Sector	0.066***	0.042***	0.029**	0.032	0.018	0.050***
	(0.012)	(0.011)	(0.014)	(0.021)	(0.011)	(0.018)
p-value	0.000	0.000	0.046	0.125	0.100	0.006
N	43027	47511	43027	47511	47511	47511
Formal Health Insurance	0.016*	0.009	-0.002	-0.005	-0.015	-0.020
	(0.009)	(0.010)	(0.015)	(0.021)	(0.010)	(0.018)
p-value	0.079	0.366	0.886	0.801	0.136	0.265
N	43027	47512	43027	47512	47512	47512
Mun-specific trends	No	No	Yes	Yes	Yes	Yes

Notes: This table shows the impact of the density of TV-schools on labor market outcomes by the presence of services sector at baseline, defined as an indicator for whether the municipality had an above median share of services sector jobs in 1980. The first set of coefficient estimates are the interaction term between the treatment coefficient and the sectoral presence indicator from a fully-interacted single-coefficient two-way fixed effects (TWFE) model, DiD_{TWFE}, which includes municipality and cohort fixed effects (col .1), and additionally includes municipality-specific linear time trends (col. 3). The second set of coefficient estimates present the test of the linear combination of the equally-weighted average treatment on the treated (ATT) of post-treatment estimates by the sectoral presence indicator, coming from the dynamic TWFE event study model, DiD_{ATT}, also including municipality and cohort fixed effects (col. 2), and additionally including municipality-specific linear time trends (col. 4). See Equation (1) and Section IV for details on the econometric specifications. The last set of coefficients are the equally-weighted average treatment on the treated (ATT) of post-treatment estimates from the dynamic TWFE event study model, DiD_{ATT}, including municipality and cohort fixed effects, and municipality-specific linear time trends, ran as split-sample regressions by below-median services sector at baseline (col. 5) and above-median services sector at baseline (col. 6). The continuous treatment is TV-school density in the municipality at age 15, measured as the number of TV-schools per 50 eligible children. The outcomes are: an indicator for participation in the labor market, unconditional probability of earning positive income, unconditional monthly adult income in pesos, logarithmic transformation of hourly income, and indicators for unconditionally participating in the primary sector, secondary sector, and tertiary sector, and for receiving formal health insurance. Regressions use administrative school construction data from the Mexican Secretariat of Education for 1980–2000, and outcome data from the 10% subsample of the 2010 Mexican census from IPUMS, restricting it to non inter-state migrants. Data is aggregated into birth-year by municipality-of-residence cells, and standard errors are clustered at the municipal level. Statistical significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A13: Heterogenous Impacts of the Density of TV-schools by Type of Region

	Full sample	By services jobs		By standard school presence				Services jobs + standard school presence		
		Low	High	None at baseline	Low density	Has at baseline	High density	Low serv. + low density	High serv. + low dens.	High serv. + high dens.
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
					Panel A: Education Outcomes					
Lower Sec. Graduation	0.128***	0.106***	0.117***	0.122***	0.116***	0.075***	0.116***	0.123***	0.087**	0.120***
	(0.014)	(0.016)	(0.027)	(0.016)	(0.016)	(0.025)	(0.030)	(0.017)	(0.035)	(0.044)
Mean	0.43	0.35	0.51	0.38	0.39	0.49	0.52	0.34	0.46	0.57
Observations	47511	24823	22688	26663	33363	20848	14148	20916	12447	10241
Years of Education	0.410***	0.299**	0.490**	0.447***	0.372***	-0.099	0.332	0.432***	0.293	0.470
	(0.108)	(0.126)	(0.206)	(0.119)	(0.114)	(0.235)	(0.259)	(0.135)	(0.217)	(0.408)
Mean	7.16	6.44	7.94	6.73	6.79	7.71	8.02	6.40	7.45	8.54
Observations	47511	24823	22688	26663	33363	20848	14148	20916	12447	10241
					Panel B: Income					
Monthly Income	468.063***	169.974**	736.136***	189.806**	206.294***	722.351***	939.170***	154.573**	337.432	1003.049***
	(87.321)	(76.150)	(202.489)	(81.765)	(80.198)	(207.265)	(237.998)	(75.816)	(216.242)	(378.700)
Mean	1847.28	1256.69	2493.33	1444.48	1542.45	2362.38	2566.05	1209.96	2101.06	2970.14
Observations	47509	24820	22689	26661	33361	20848	14148	20913	12448	10241
Hourly Income (log)	0.041	0.038	0.019	0.042	0.034	-0.070	-0.013	0.051	0.023	-0.063
	(0.070)	(0.086)	(0.129)	(0.084)	(0.078)	(0.130)	(0.160)	(0.088)	(0.171)	(0.162)
Mean	2.24	1.93	2.58	2.00	2.09	2.54	2.60	1.88	2.42	2.77
Observations	47512	24823	22689	26664	33364	20848	14148	20916	12448	10241
					Panel C: Labor market composition					
Labor Market Participation	0.038***	0.025	0.043*	0.022	0.017	0.041	0.078**	0.013	0.031	0.018
	(0.013)	(0.016)	(0.024)	(0.015)	(0.014)	(0.028)	(0.036)	(0.017)	(0.027)	(0.048)
Mean	0.58	0.55	0.63	0.56	0.57	0.61	0.63	0.54	0.60	0.66
Observations	47511	24822	22689	26663	33363	20848	14148	20915	12448	10241
Services Sector	0.038***	0.018	0.050***	0.022**	0.028***	0.043**	0.046*	0.021*	0.043**	0.024
	(0.010)	(0.011)	(0.018)	(0.010)	(0.010)	(0.022)	(0.024)	(0.011)	(0.021)	(0.032)
Mean	0.23	0.16	0.30	0.18	0.19	0.28	0.31	0.16	0.26	0.35
Observations	47511	24822	22689	26663	33363	20848	14148	20915	12448	10241

Notes: This table shows the impact of access to a high density of TV-schools on education and labor market outcomes by different municipality characteristics using split-sample regressions. The samples of analysis are: Full sample (col. 1), below and above the median of the share of services sector in 1980 (col. 2-3), without and with presence of standard schools in 1980 (col. 4,6) below and above the density of standard schools in 2000 (col. 5,7), low share of services sector in 1980 and low density of standard schools in 2000 (col. 8), high share of services sector in 1980 and low density of standard schools in 2000 (col. 9), high share of services sector and high density of standard schools in 2000 (col. 10). The continuous treatment is TV-school density in the municipality at age 15, measured as the number of TV-schools per 50 eligible children. The coefficient estimates are the equally-weighted average treatment on the treated (ATT) of post-treatment estimates from the dynamic TWFE event study model, DiD_{ATT}, which includes municipality and cohort fixed effects, and municipality-specific linear time trends. See Equation (1) and Section IV for details on the econometric specification. The outcomes are: an indicator for graduating from lower secondary education, years of education, the unconditional monthly adult income in pesos, the logarithmic transformation of hourly income, an indicator for participating in the labor market, and an indicator for participating in the services sector. Regressions use administrative school construction data from the Mexican Secretariat of Education for 1980–2000, and outcome data from the 10% subsample of the 2010 Mexican census from IPUMS. Data is aggregated into birth-year by municipality-of-residence cells, and standard errors are clustered at the municipal level. Statistical significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B FOR ONLINE PUBLICATION.

Creation of Secondary School Construction Variables

School construction date. The secondary school data comes from the Mexican Secretariat of Education (*Secretaría de Educación Pública*, SEP). We use two sources for lower secondary school data: the 2015-2016 school directory ("formato 911"), and yearly school records from 1990 to 2000. The 2015-2016 school directory is a database of all lower secondary schools in Mexico, including each school's unique identifier, address, geographical coordinates, school type, foundation date, registration date into the school record, and, if applicable, closing and reopening dates. The registration system was created in 1981, and all schools existing before 1981 have the same registration date, making the distinction between the foundation date and registration date important. The yearly school records are annual databases of all lower secondary schools opened in a given academic year in Mexico, including each school's unique code, address, geographical coordinates, school type, and total number of enrolled students by grade.

We use three sources of information to determine the school construction date: the foundation date and registration date from the 2015-2016 school directory, and the annual records that show when schools were operating. These three dates should match, but they sometimes do not, and the number of date discrepancies varies significantly by state. We impute the school construction date by combining these three data sources using the procedure described below. The earliest year among the foundation date, registration date, and yearly opening is designated as the "probable school construction date". However, the school registry was created in 1981, and in some states, schools built before this year were recorded as being constructed in 1981 or 1982. Similarly, since yearly records began in 1990, some schools built before this year are recorded as having opened in 1990. Therefore, sudden large increases in recorded school constructions at the state level in 1981, 1982, or 1990 are considered as unreliable.[33]

Due to this, states are manually assigned into groups based on the reliability of their probable construction dates. This reliability is determined visually. The classification is as follows: Groups 1

[33]Note that in 1981-1982, there was a TV-school construction boom, with the introduction of this school modality to new states. Hence, this unreliability assignment is potentially overly restrictive, and we are throwing away potentially useful variation for caution.

and 5 do not have any school date replacements. Constructions in Groups 2, 4, and 6 are considered unreliable if the school construction date is prior to 1982. Constructions in Group 3 are considered unreliable if the school construction date is prior to 1983. Additionally, schools constructed in 1990 are deemed unreliable if they came from the yearly files and were not included in the previous exceptions. We create the "first construction year" variable as the earliest year a school is open at the municipality and school-type level (TV-school, standard school).

We combine several sources of school coordinates to have the maximum coverage. We use the school coordinates from the school directory and the yearly school records, if available. If not, we use the locality coordinates if the locality is rural, and the locality centroid coordinates for urban localities. Lastly, we use the average of primary schools coordinates from the same locality.

CONSTRUCTION OF VARIABLES RELATED TO SCHOOL TREATMENTS. All the variables below are defined at the (municipality)-(year)-(school-type) level:

- **Imputed first construction year (minimum):** This variable is marked as missing if no school was constructed during the 1980-2000 period, if the first school was constructed after 2000, or if the first construction year is considered unreliable (see above).

- **Number of schools open (sum)**: If the number of schools is missing or if the first construction occurred post-2000, it is marked as zero. If deemed unreliable, it is marked as missing for all years prior to the probable first construction year.

- **Presence of a school type:** This is defined as having a positive number of schools open, with a zero if the number is zero and not unreliable. If deemed as unreliable, it is marked as missing if the year is prior to the probable first construction year and the start year is not missing

- **Density of a school type (continuous treatment, $_{mc}$):** Defined as the number of schools open at the municipality-year level divided by the population aged 12 to 14 in 1980, multiplied by 50. The density is marked as zero if the number of schools open is zero, and as missing if the population data is missing or deemed unreliable for all the years prior to the probable first construction date. The density variable is winsorized at the 99 percentile.

- **Year relative to the first construction year**: This is the difference between the current year and the municipality's probable first construction year. It is marked as missing if the starting year is missing, either due to no schools being constructed in the relevant period or because it is considered unreliable.

- **Post-treatment average density (used for the continuous event study):** This variable is the average density of schools after the first construction year.

- Above-median density treatment indicator (binary treatment, $AboveTS_{mc}$): This indicator is computed by determining whether the municipality-year has a density above the median of all positive densities across all years. If the density is missing, the indicator is marked as missing if the switching year occured *before* or *at* the unreliable probable date.

- **Year of switching to above median density:** This identifies the first year the municipality switches from not being above median to being above median, excluding missing years. If no school was ever constructed, if the data is unreliable, or if the switch occurred after 2000, it is marked as missing.

- **Relative year to the switching year:** This is the difference between the current year and the year the municipality switches to above-median density. It is marked as missing if the switching year is missing.

We manually correct incorrectly allocated TV-schools in the years prior to 1968, setting their probable construction year to 1968. We create "placebo relative years" for municipalities without school constructions or without switching by assigning a placebo first construction year (switching year). We generate a variable containing all the years of school constructions between 1980 and 2000, separate them by percentiles, and randomly assign the percentile values to all untreated municipalities. This variable is used only for sample restriction purposes before analysis, ensuring balanced samples between treated and control municipalities during the window timing sample restrictions around the switching event.

AGGREGATE ENROLLMENT SHARES. To construct the aggregate enrollment shares, we combine the annual secondary school enrollment data from the Mexican Secretariat of Education for the period 1990-2000, and population counts from the Mexican census. The school records are yearly databases of all lower secondary schools open in a given academic year in Mexico. Among other information, they include the unique school code for each school, address, geographical coordinates, school type and total number of enrolled students per grade. The population counts at the locality level come from the 1990, 2000 and 2010 census and from the 1995 and 2005 population counts, all from the Instituto Nacional de Estadística y Geografía (INEGI).[34] The population counts in each census year are aggregated at the locality-cohort level. For individuals older than 25 years-old, they are also binned in 5-age intervals.

Whenever possible, we split the 5-age population count bins into cohort population counts following the cohort proportions from the 1990 census. If the specific cohorts proportions are not available, and given that there are almost no differences in cohort sizes within a 5-age bin, we divide the population groups into five equally-sized cohorts. We obtain yearly population counts using a cubic spline interpolation across census years.

We aggregate the school-level enrollment data by separately computing the total number of standard school and TV-school students in a given locality and year. Assuming no individuals leave their locality to attend a school, we use the cohort size from the imputed population data to compute the enrollment shares in TV-schools and standard school students, and proportion of individuals not enrolled in secondary education.

[34]Specifically, the population data come from the following datasets: *XI Censo General de Población y Vivienda 1990*, *I Conteo de población y vivienda 1995*, *XII Censo General de Población y Vivienda 2000*, *II Conteo de población y Vivienda 2005*, and *XIII Censo de Población y Vivienda 2010 Cuestionario Básico*.